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Macaluso

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(54) **VEHICLE ENERGY GENERATION SYSTEM**

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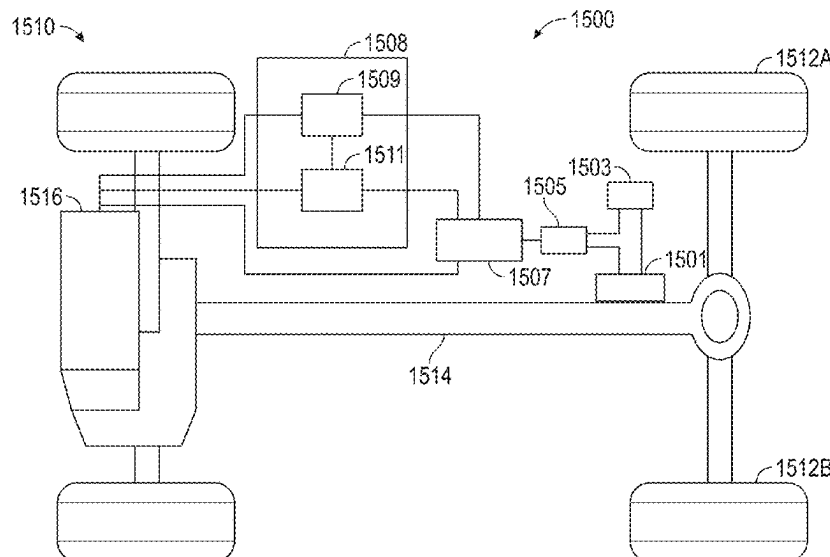
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(57) **ABSTRACT**

A system may generate energy in response to a movement of a vehicle. The system may include an energy recovery mechanism, a generator, an energy storage, and a motor. The energy recovery mechanism may include one or more rollers configured to rotate in response to a rotation of a drive shaft of the vehicle. The generator may be rotatably coupled to the roller and may be configured to generate an electrical output in response to a rotation of the roller. The energy storage may be electrically coupled with the generator and may be configured to receive a portion of the electrical output to store as energy. The motor may be electrically coupled with the energy storage and may be configured to receive a portion of the energy.

30 Claims, 39 Drawing Sheets



US 12,407,219 B2

Page 2

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Page 5

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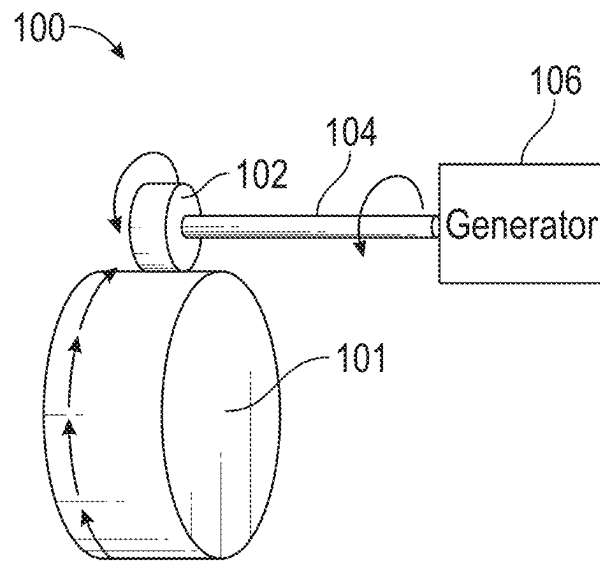


FIG. 1A

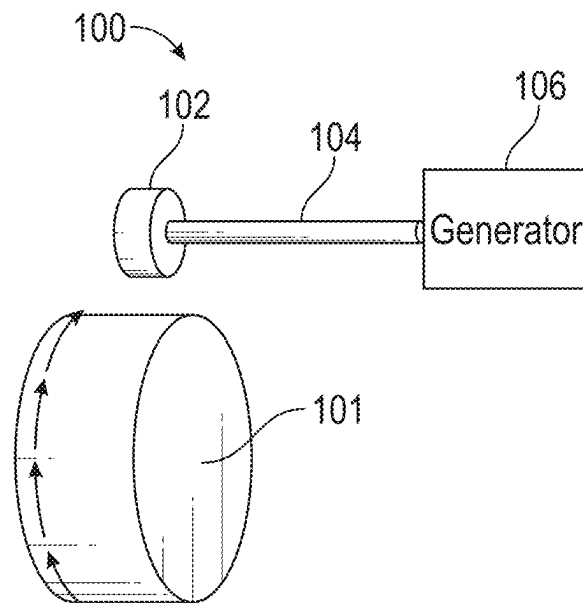


FIG. 1B

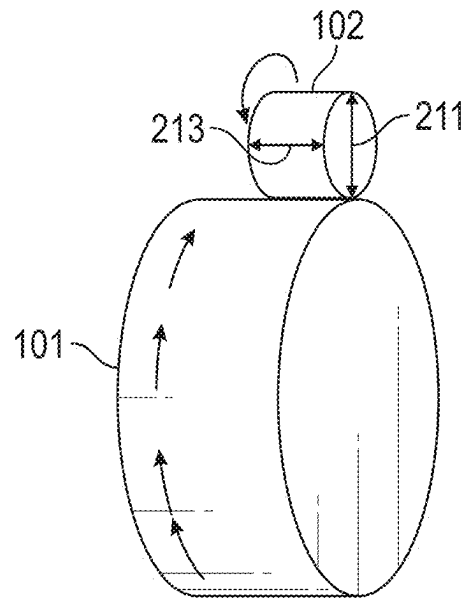


FIG. 2

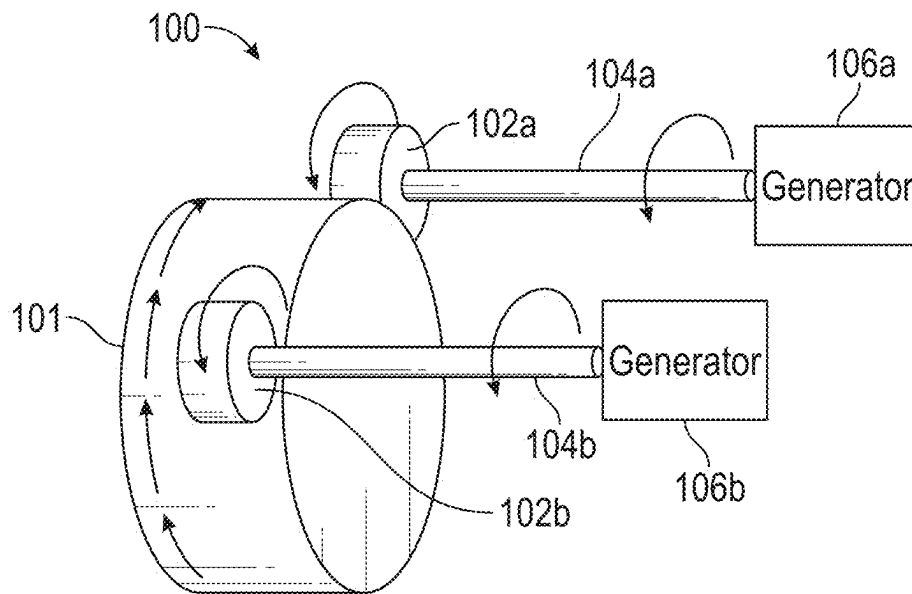


FIG. 3A

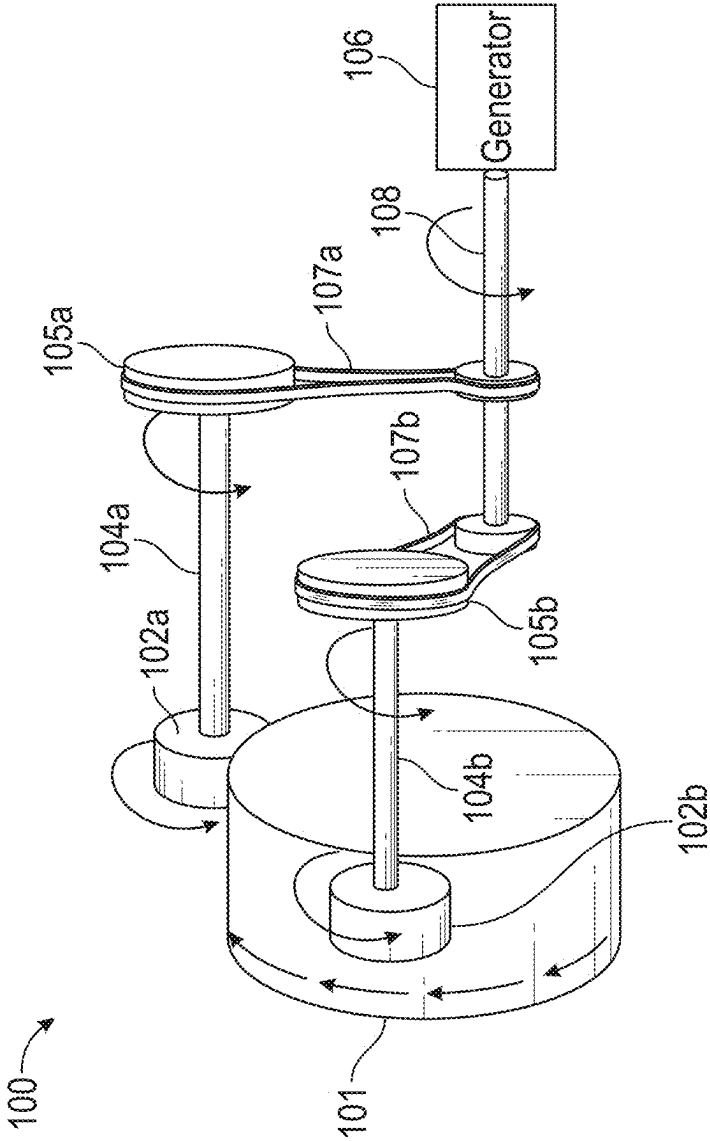
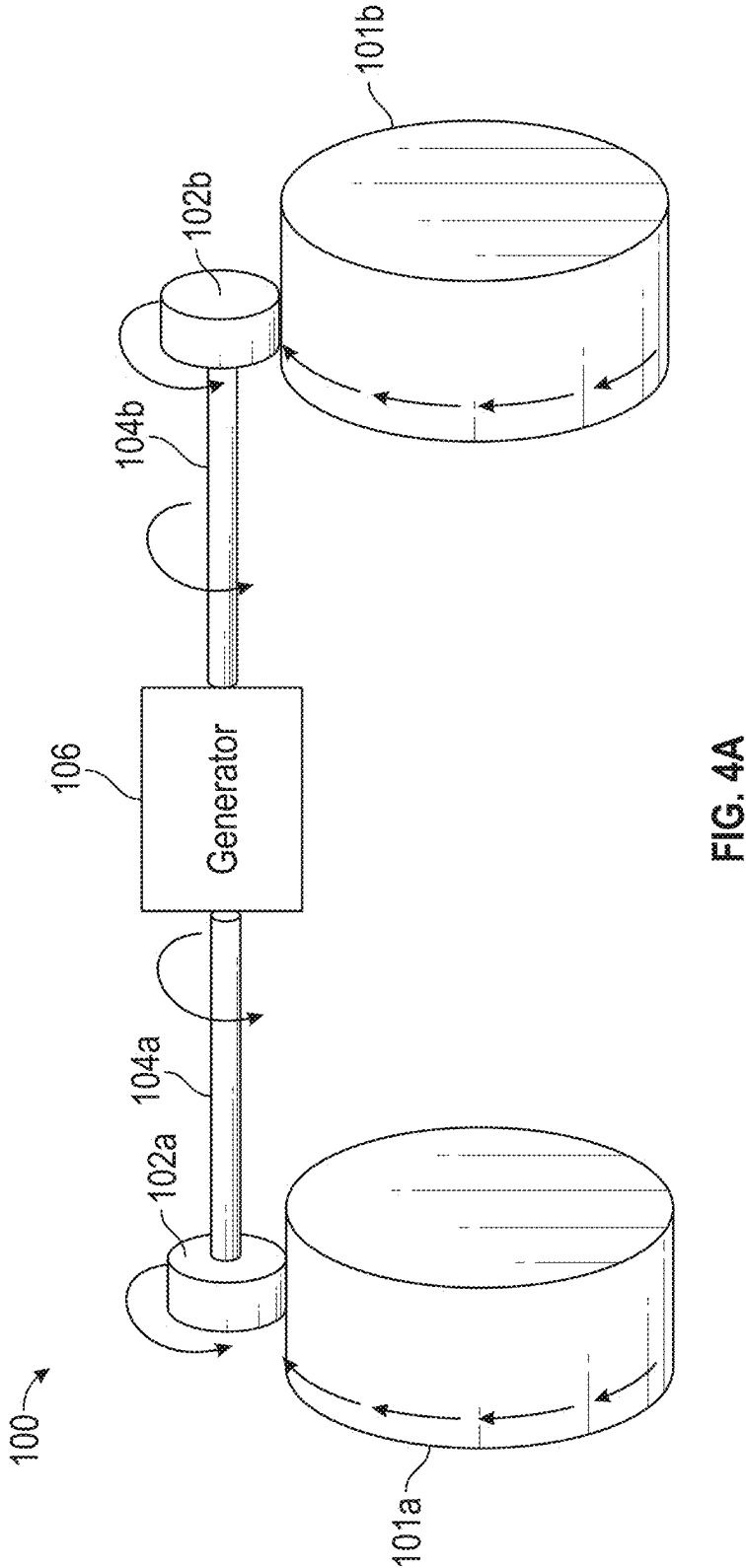


FIG. 3B



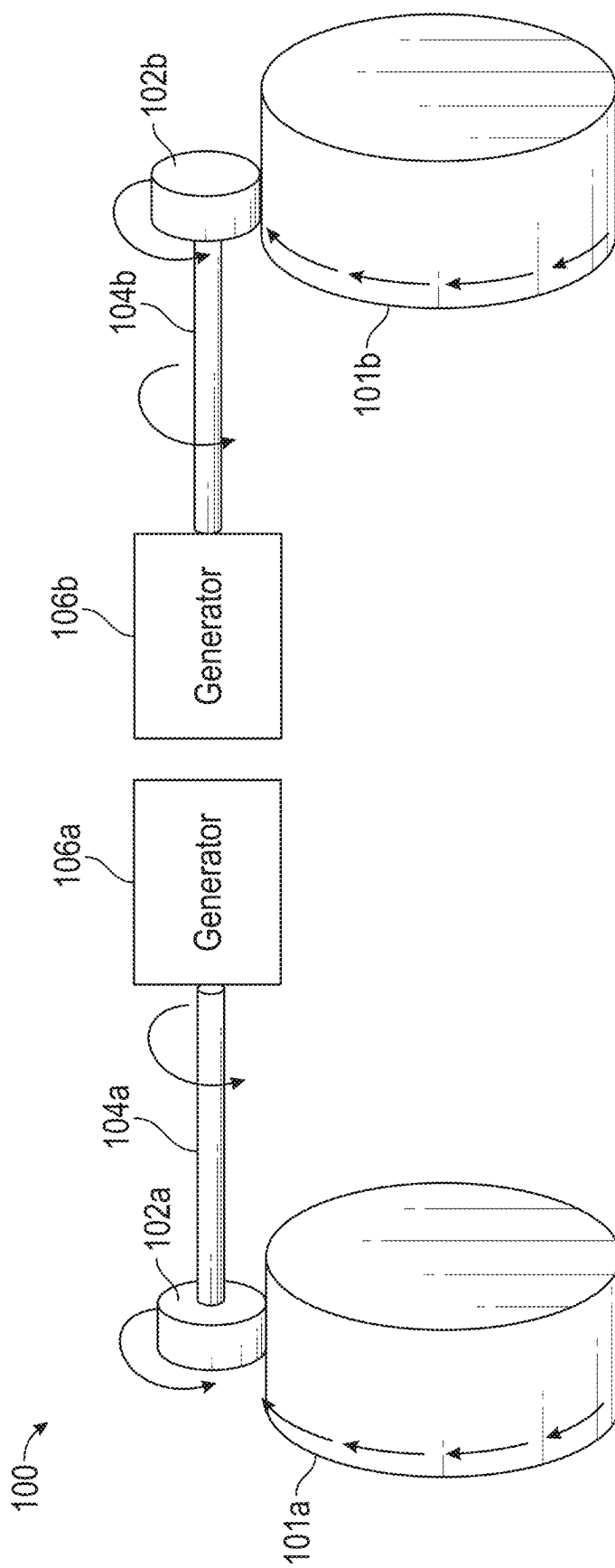


FIG. 4B

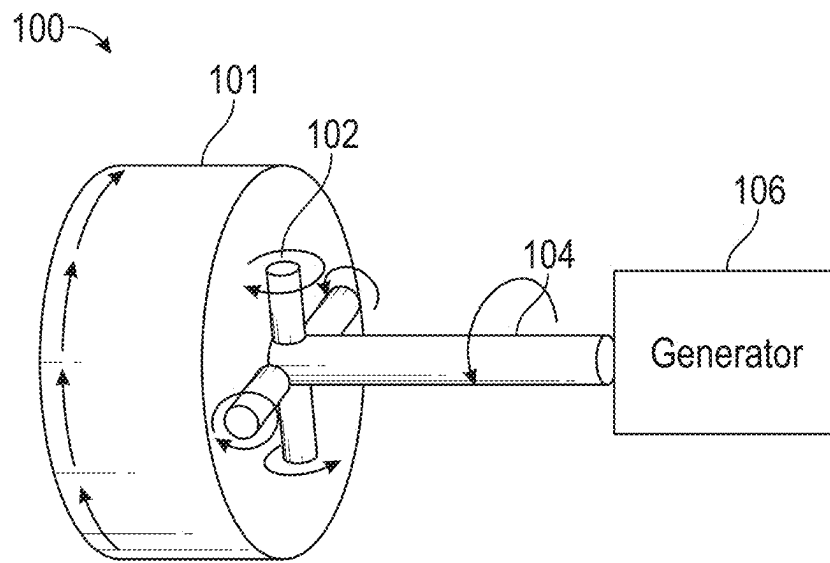


FIG. 5A

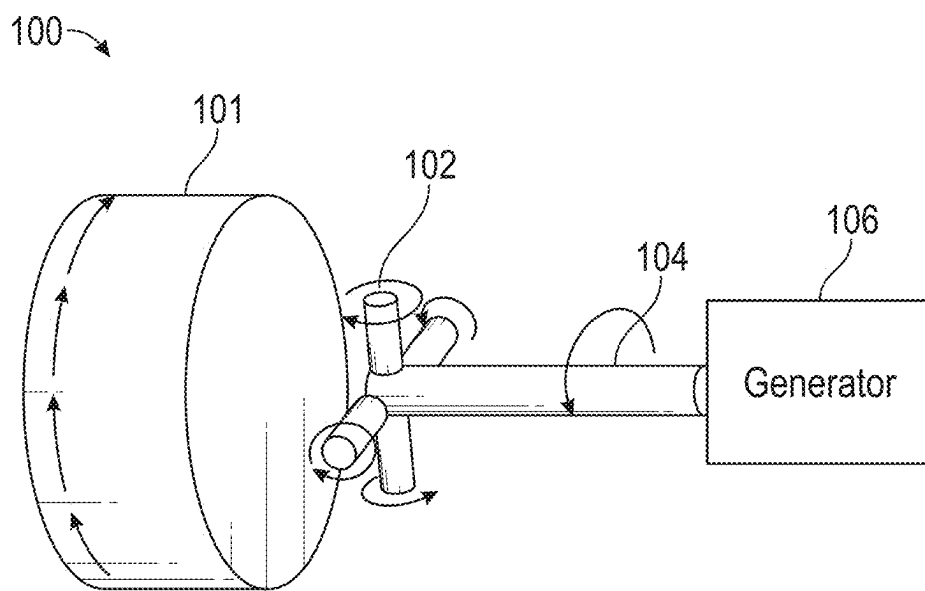


FIG. 5B

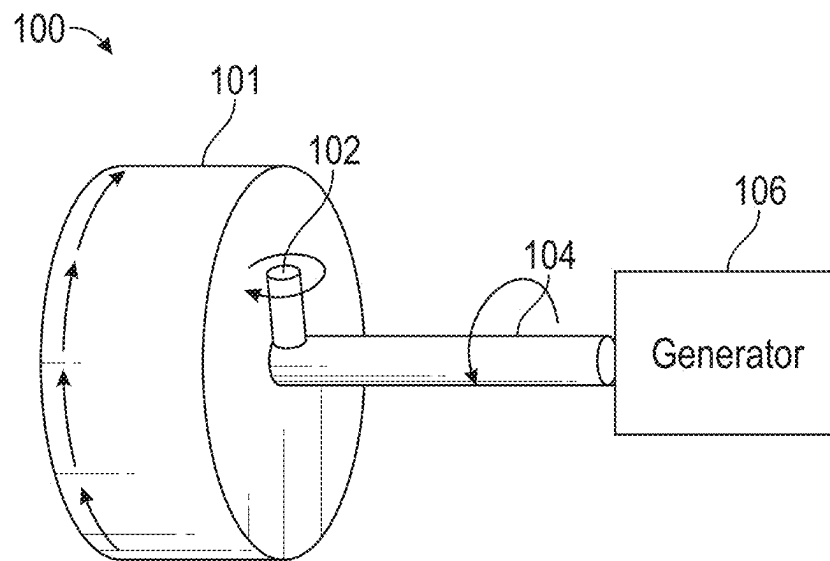


FIG. 6A

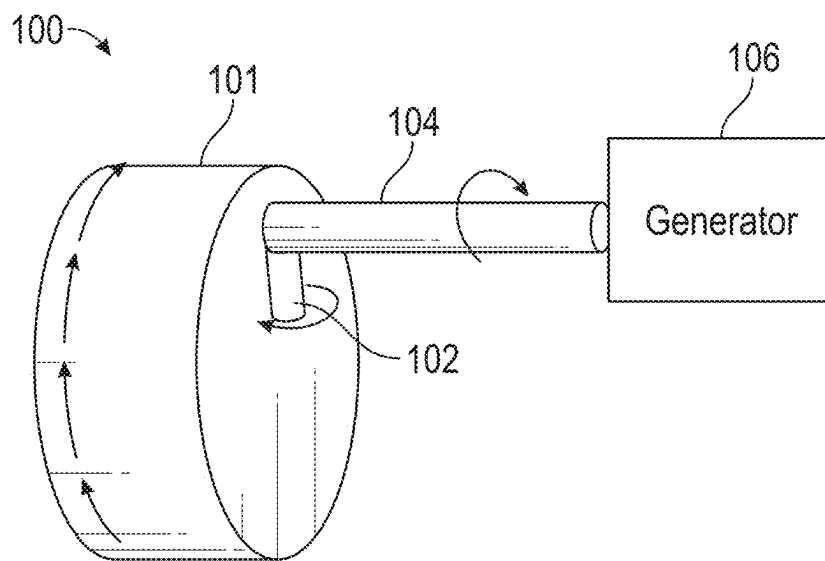


FIG. 6B

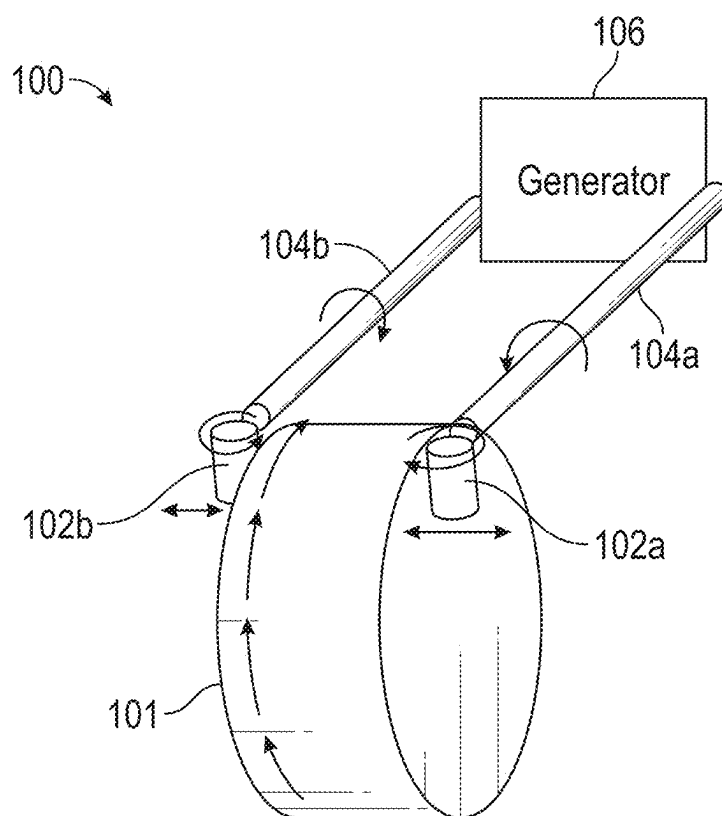


FIG. 6C

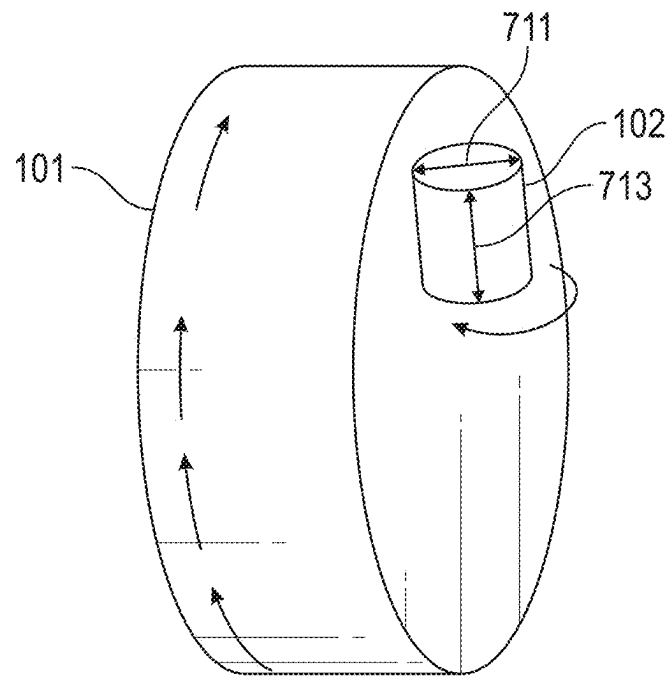


FIG. 7A

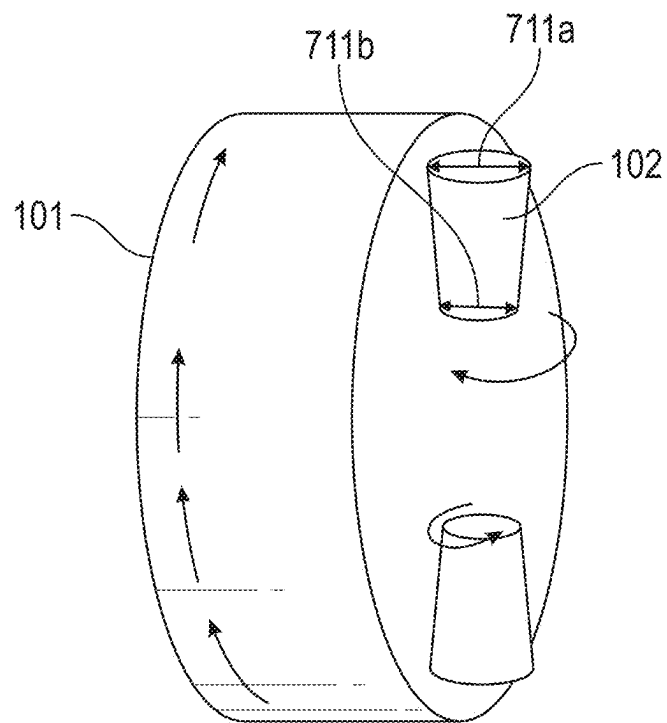


FIG. 7B

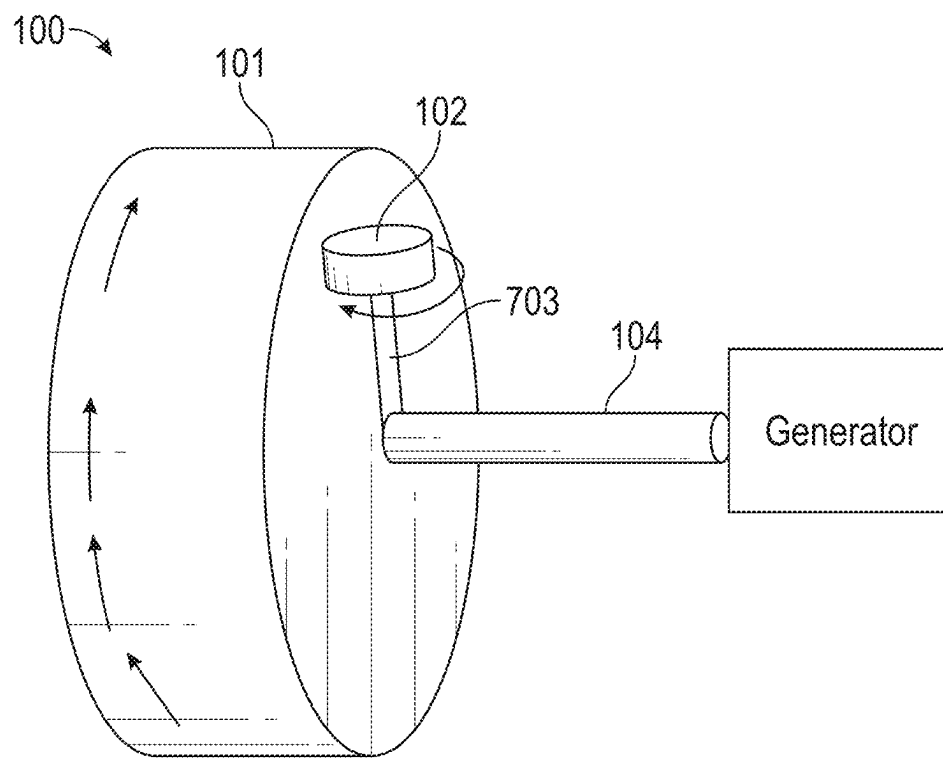


FIG. 7C

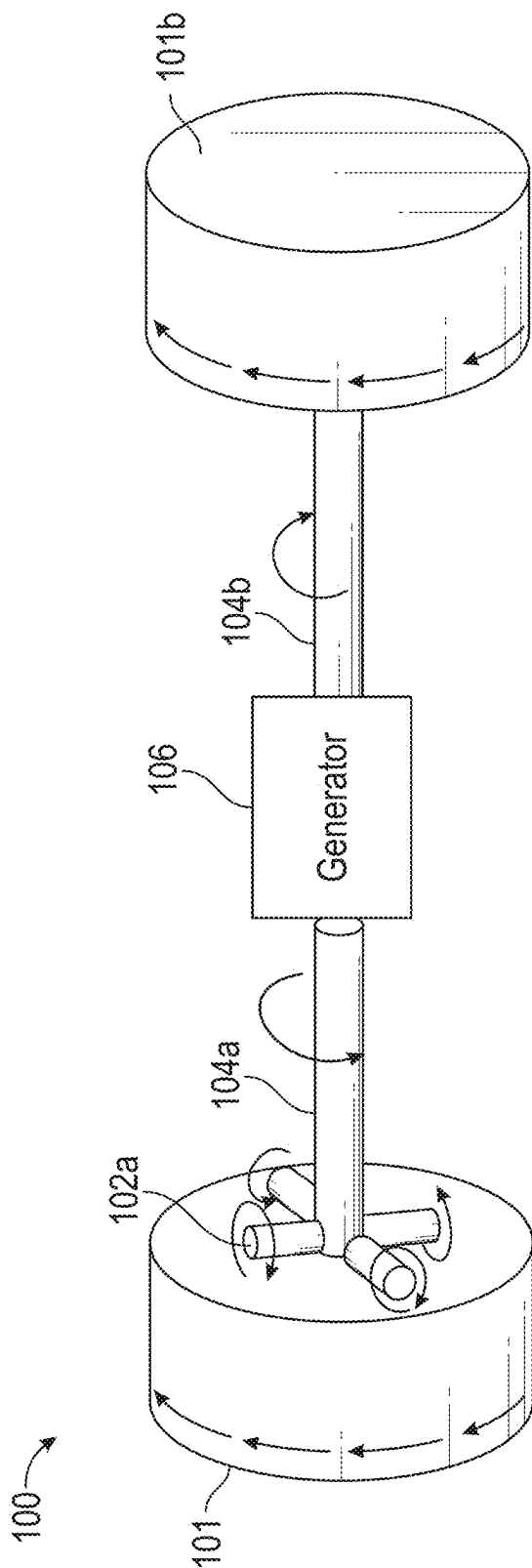


FIG. 8A

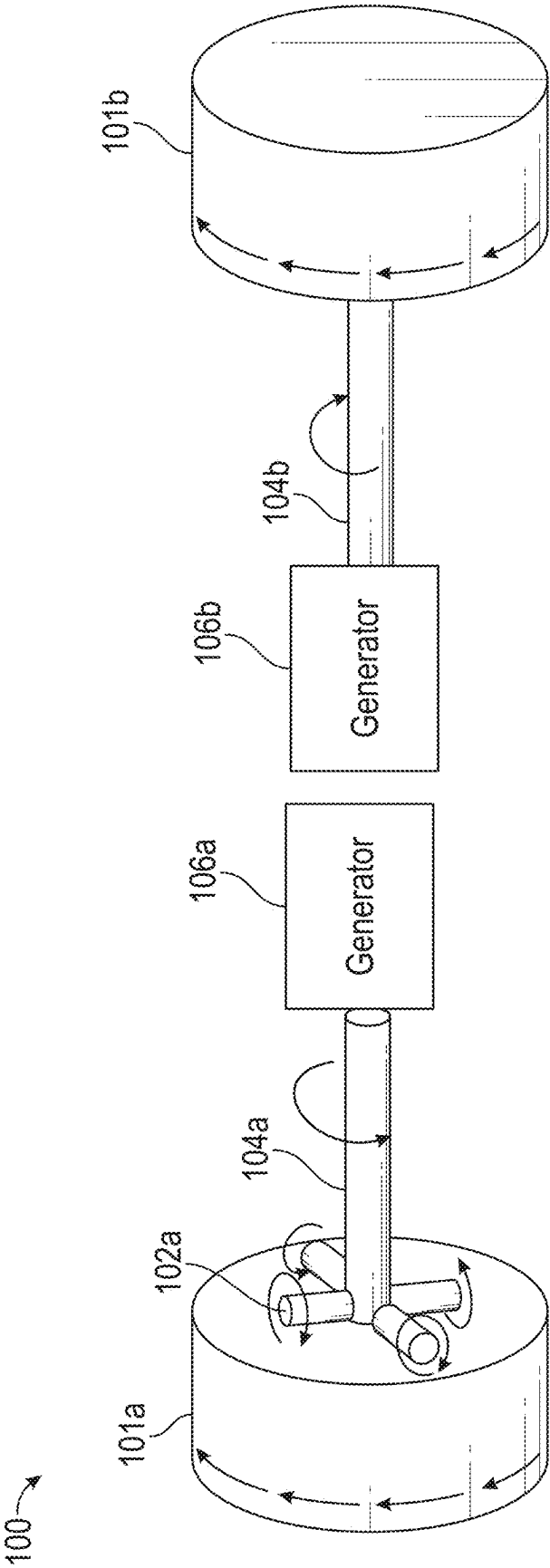
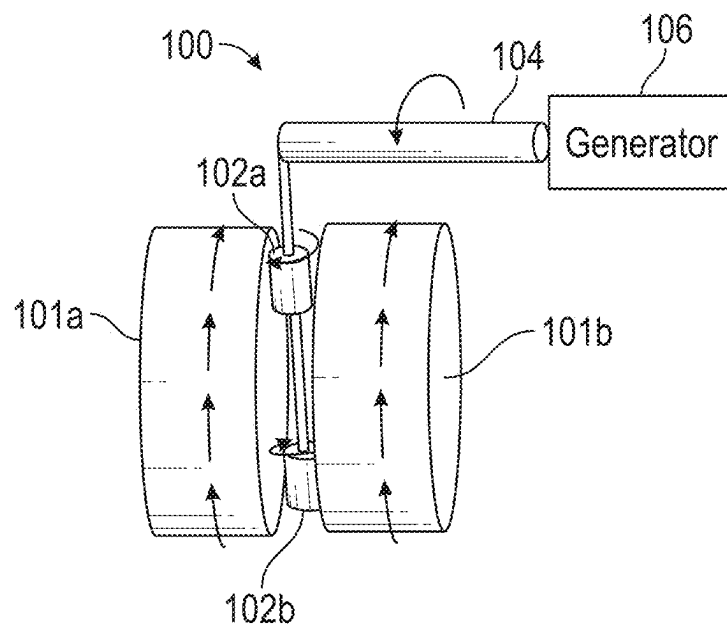


FIG. 8B

**FIG. 9**

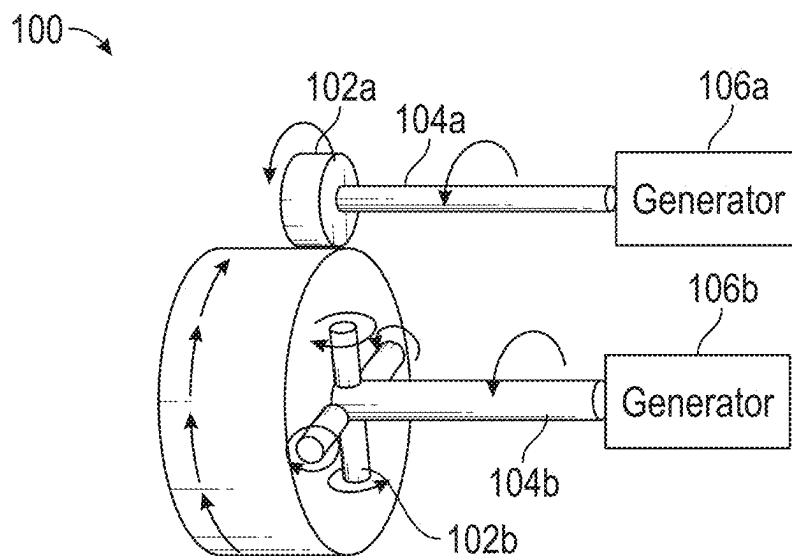


FIG. 10A

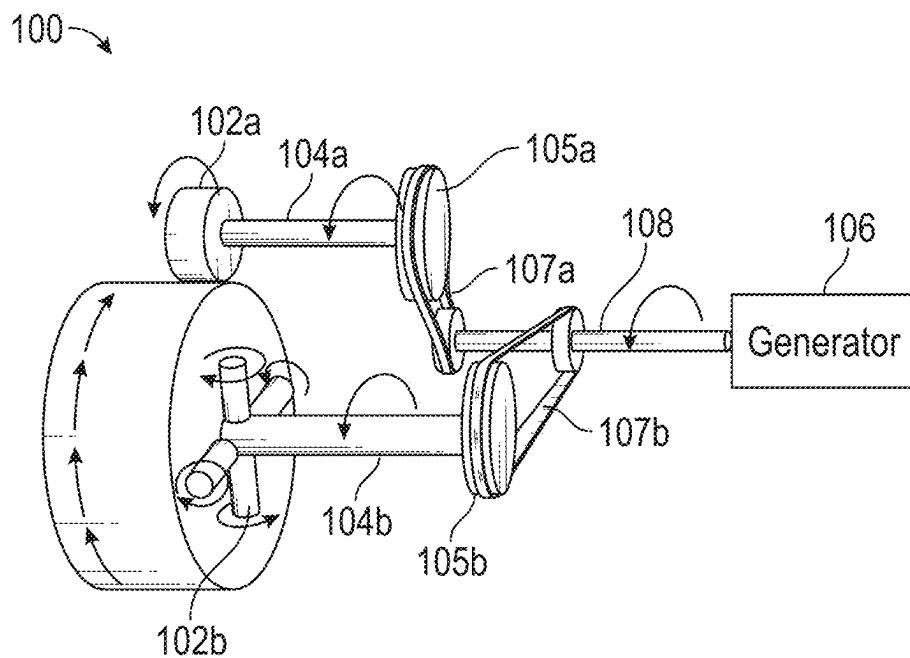


FIG. 10B

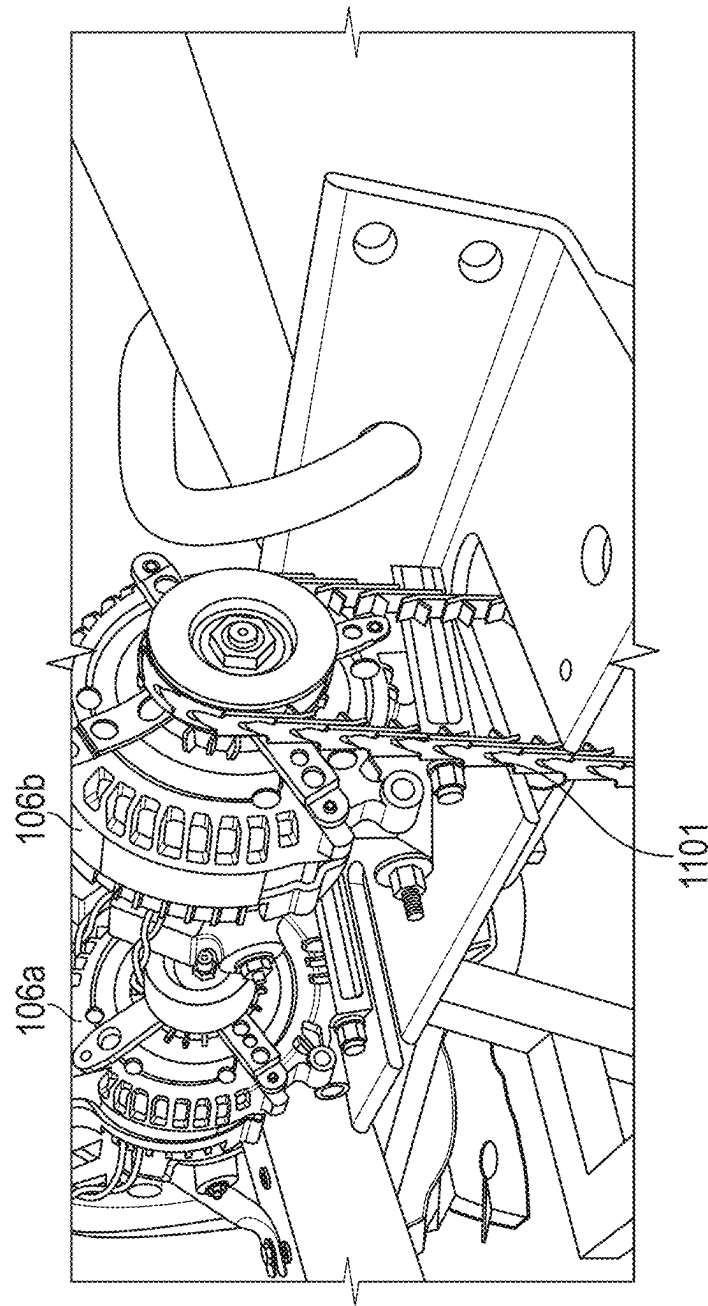


FIG. 11A

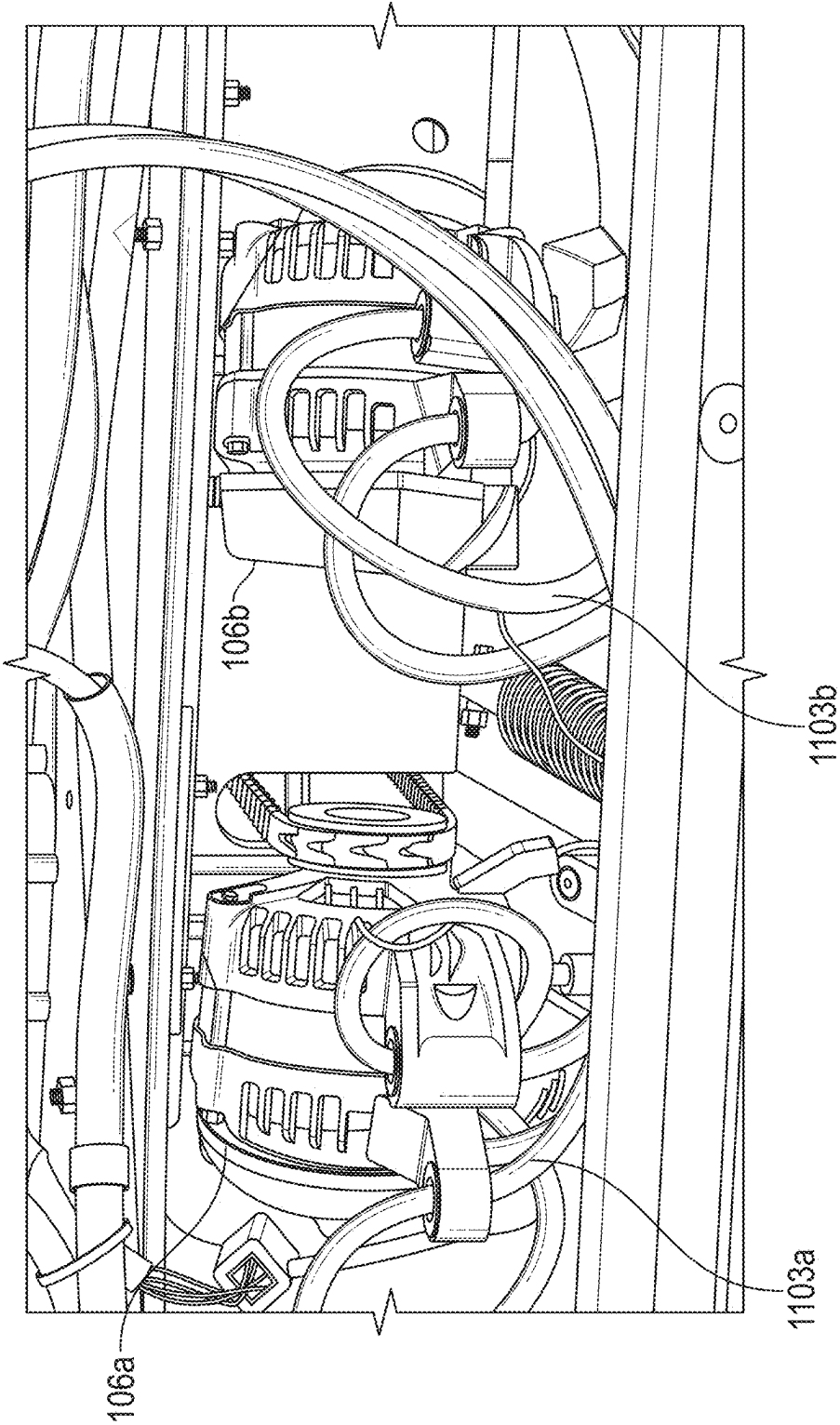


FIG. 11B

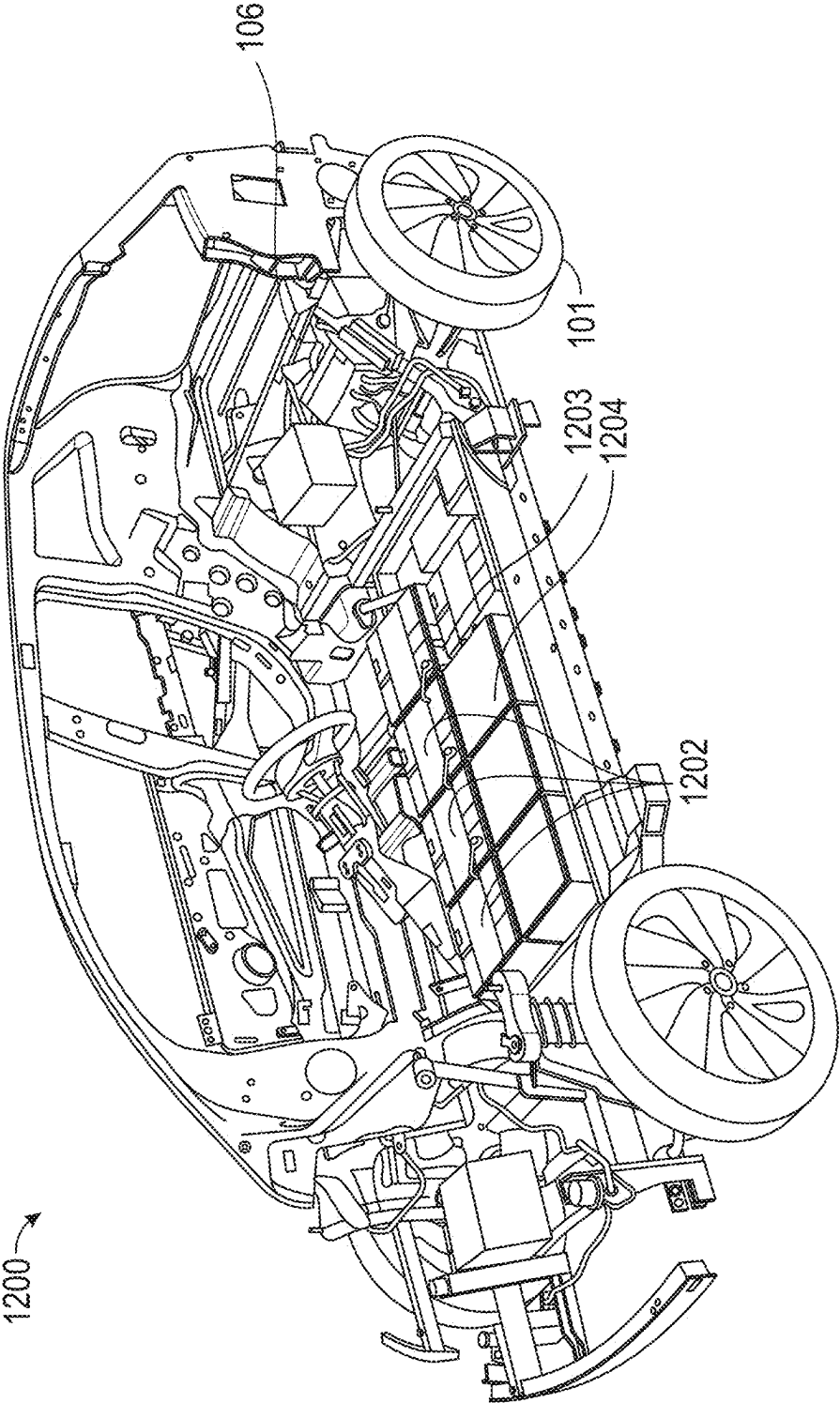


FIG. 12

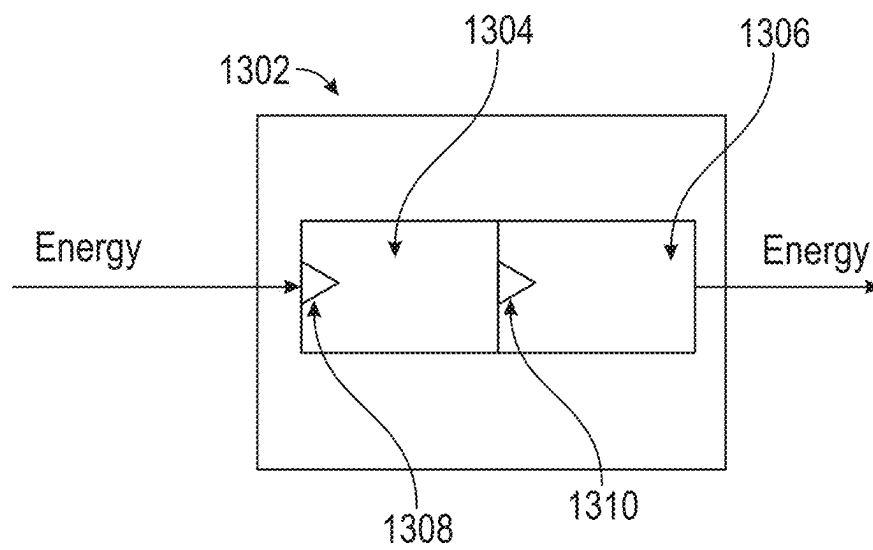


FIG. 13

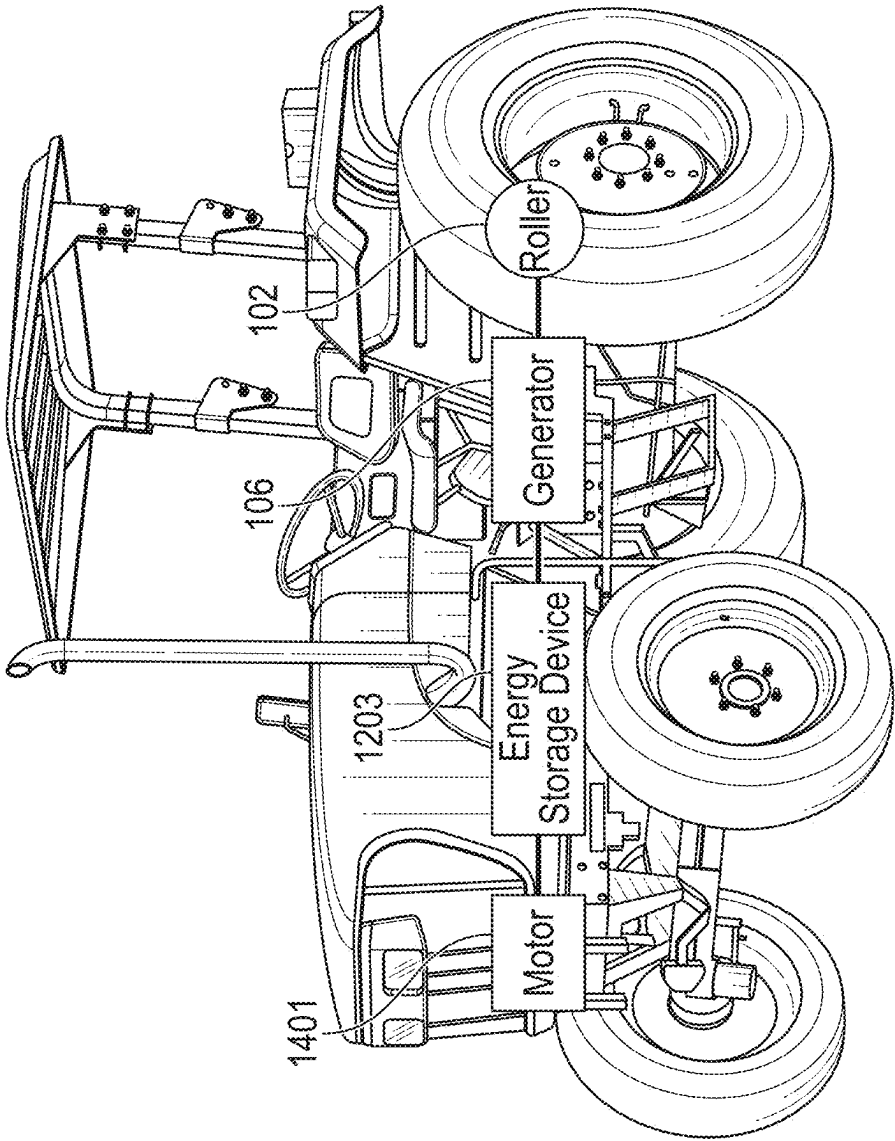


FIG. 14A

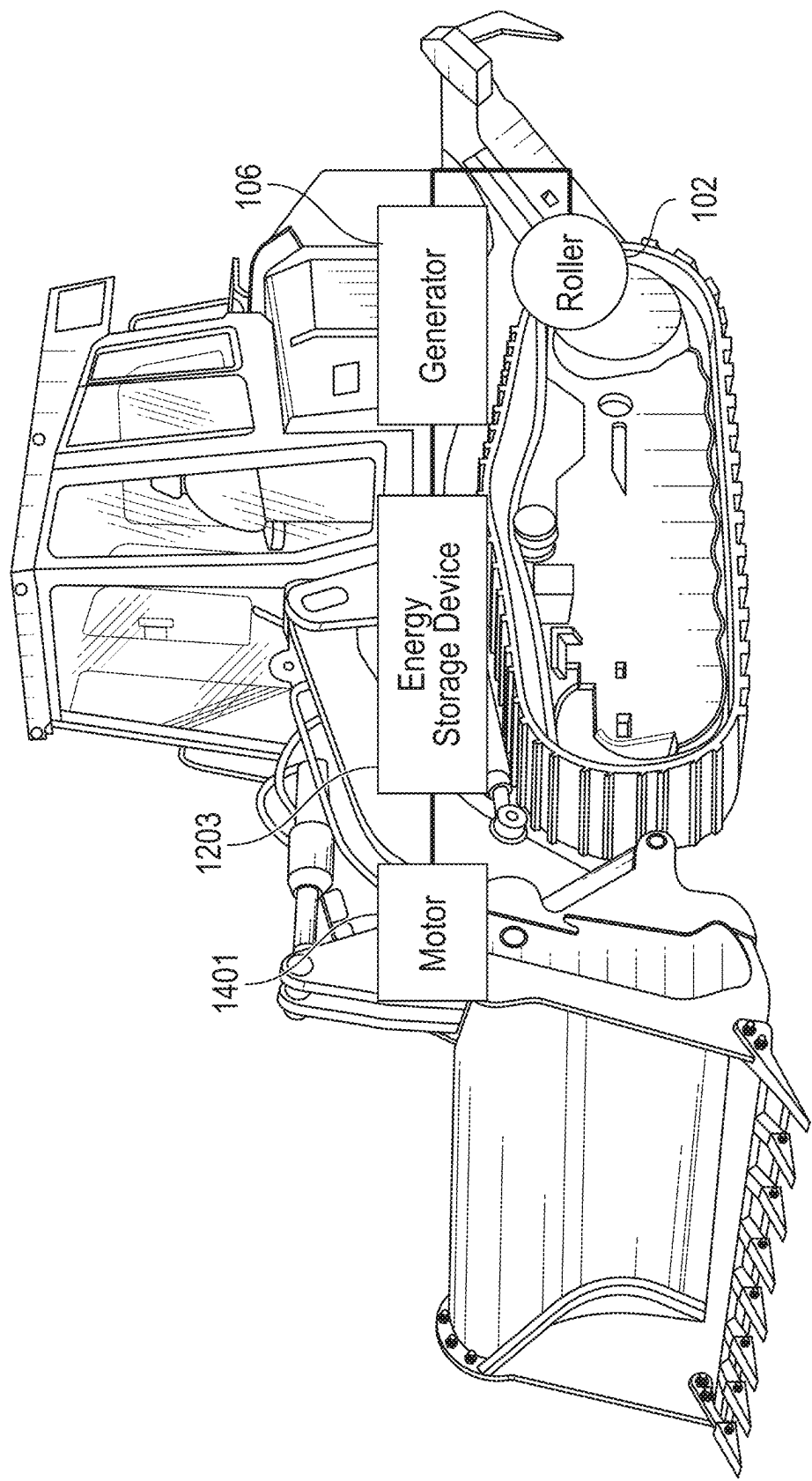


FIG. 14B

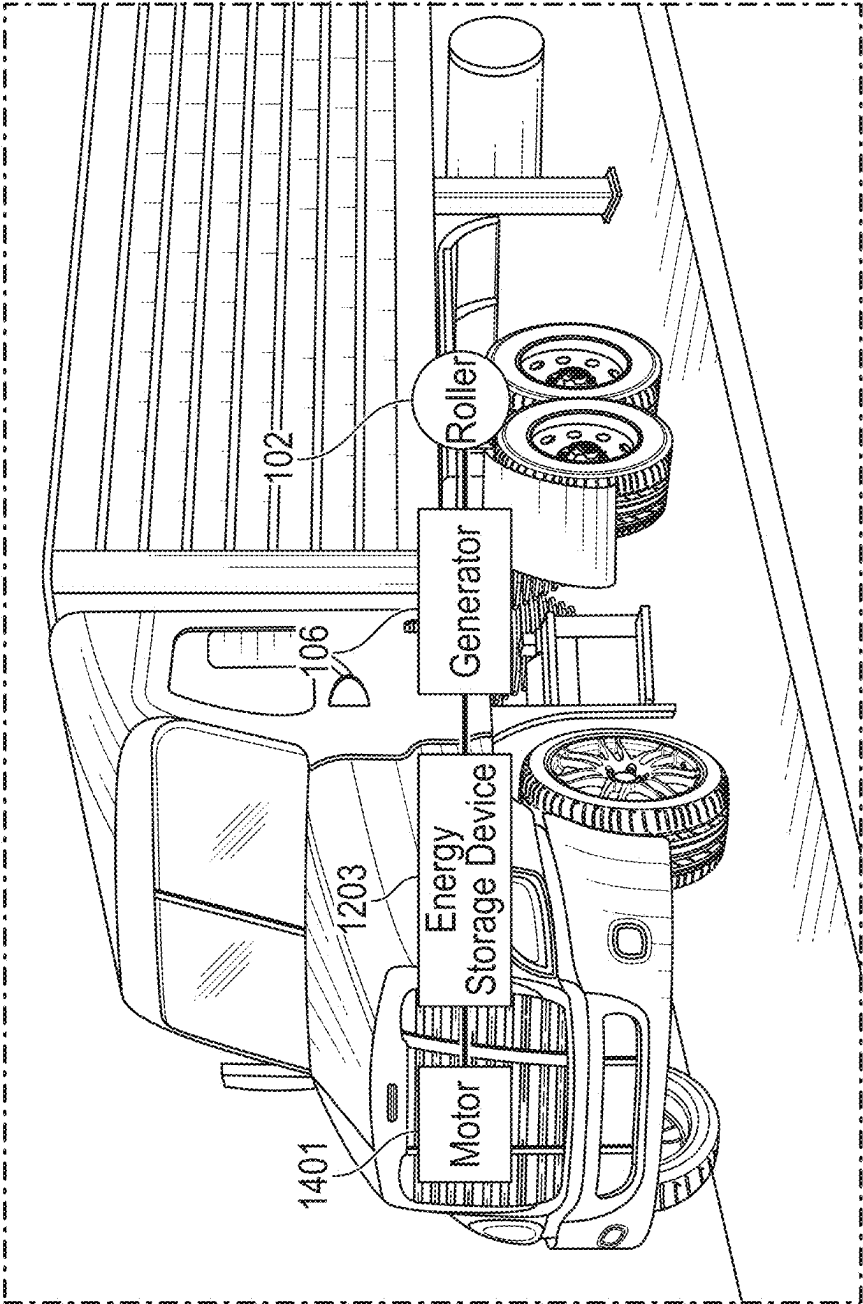


FIG. 14C

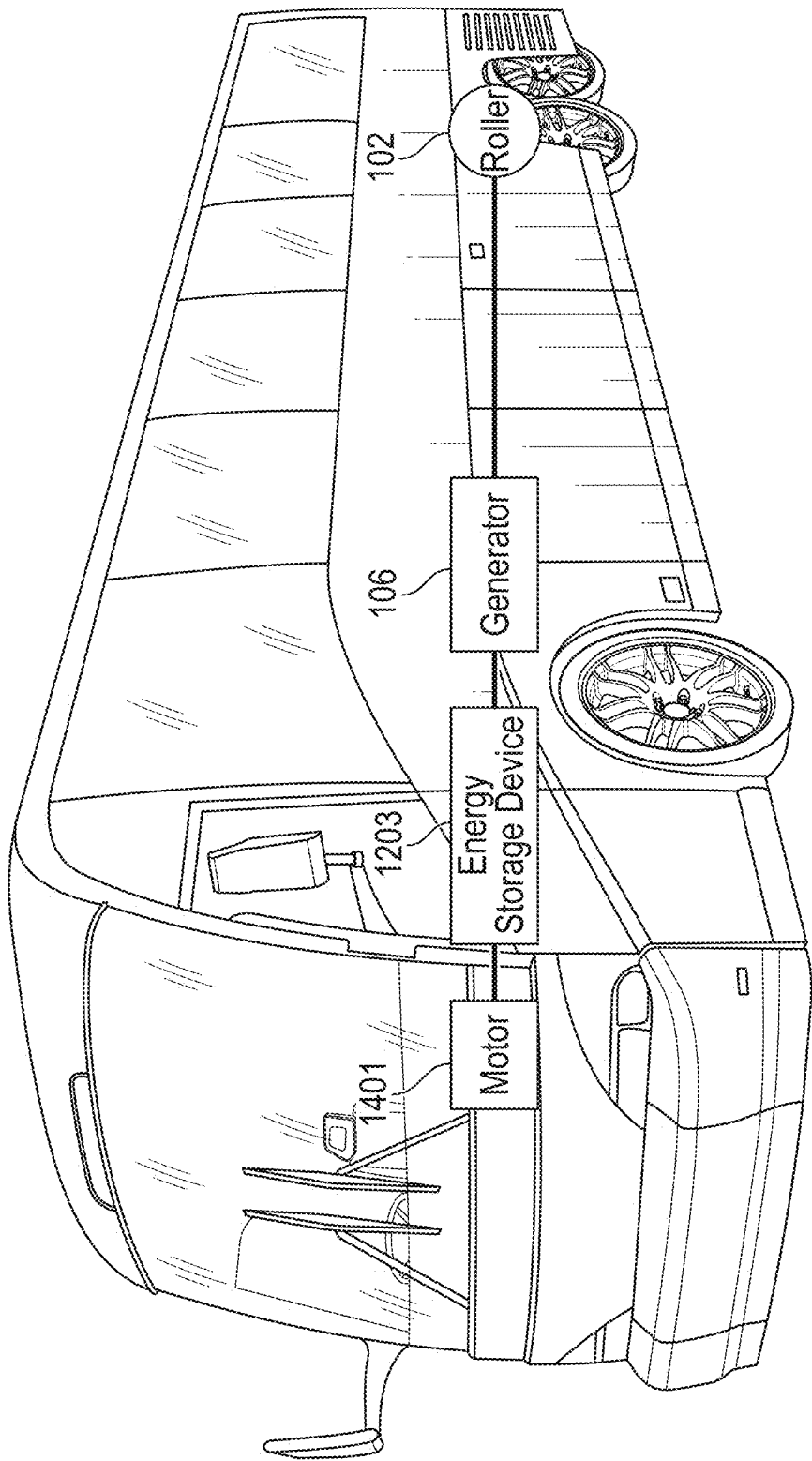
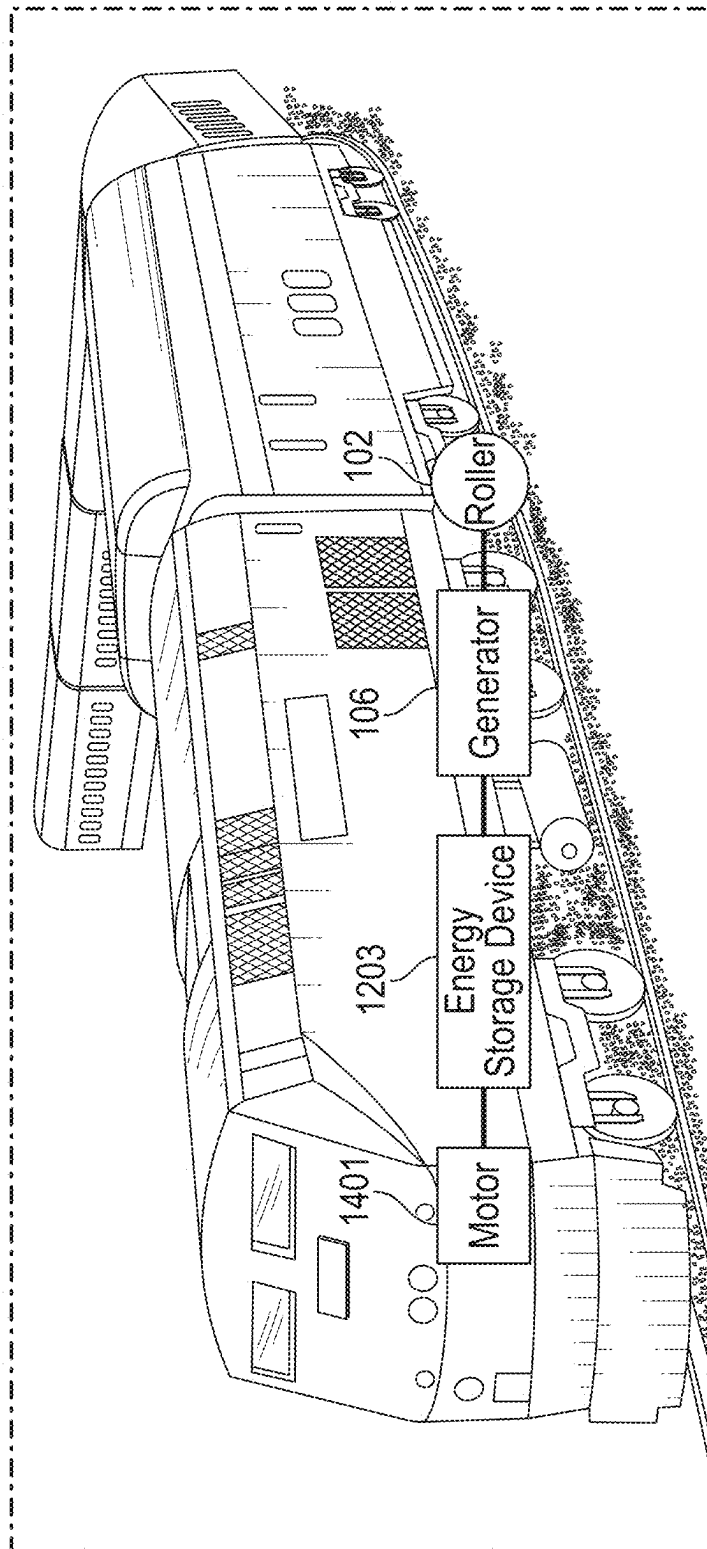


FIG. 14D



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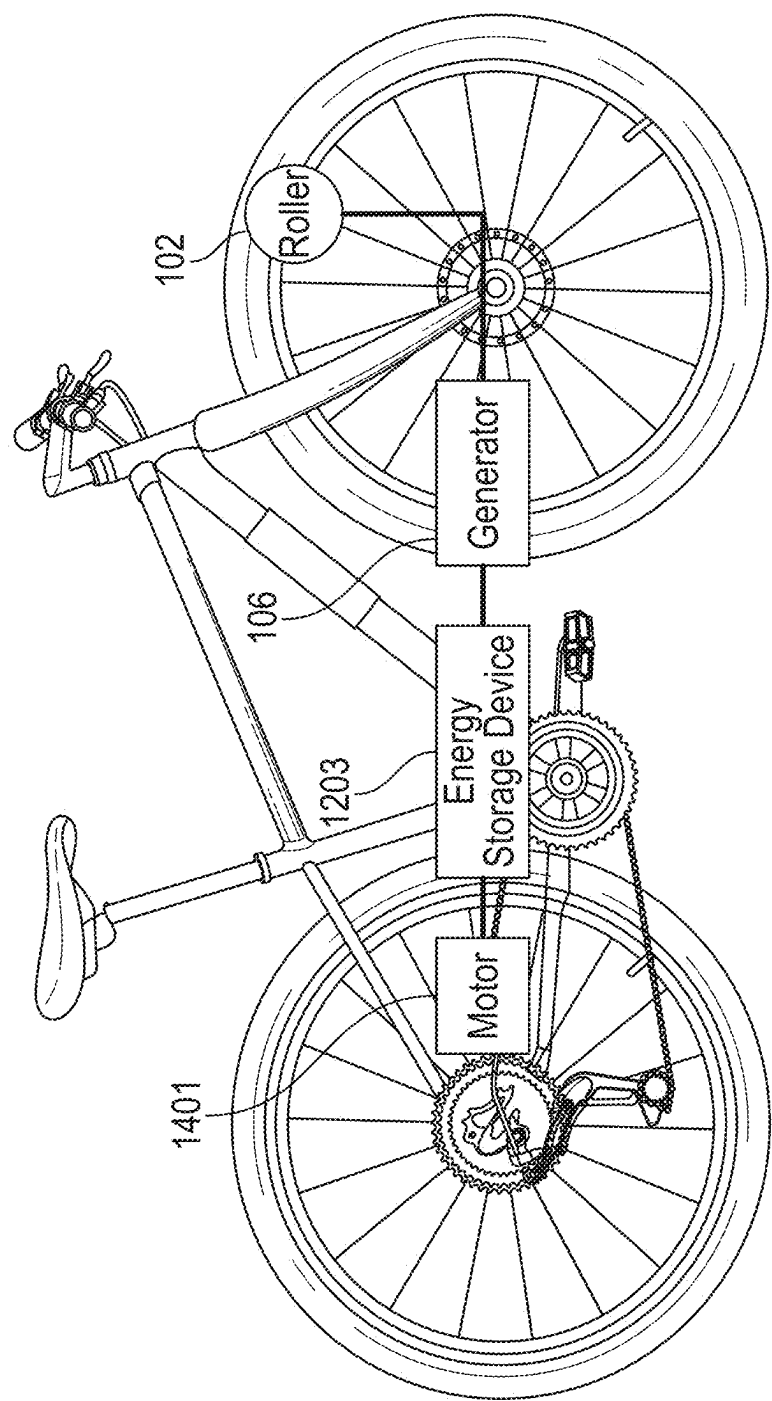


FIG. 14F

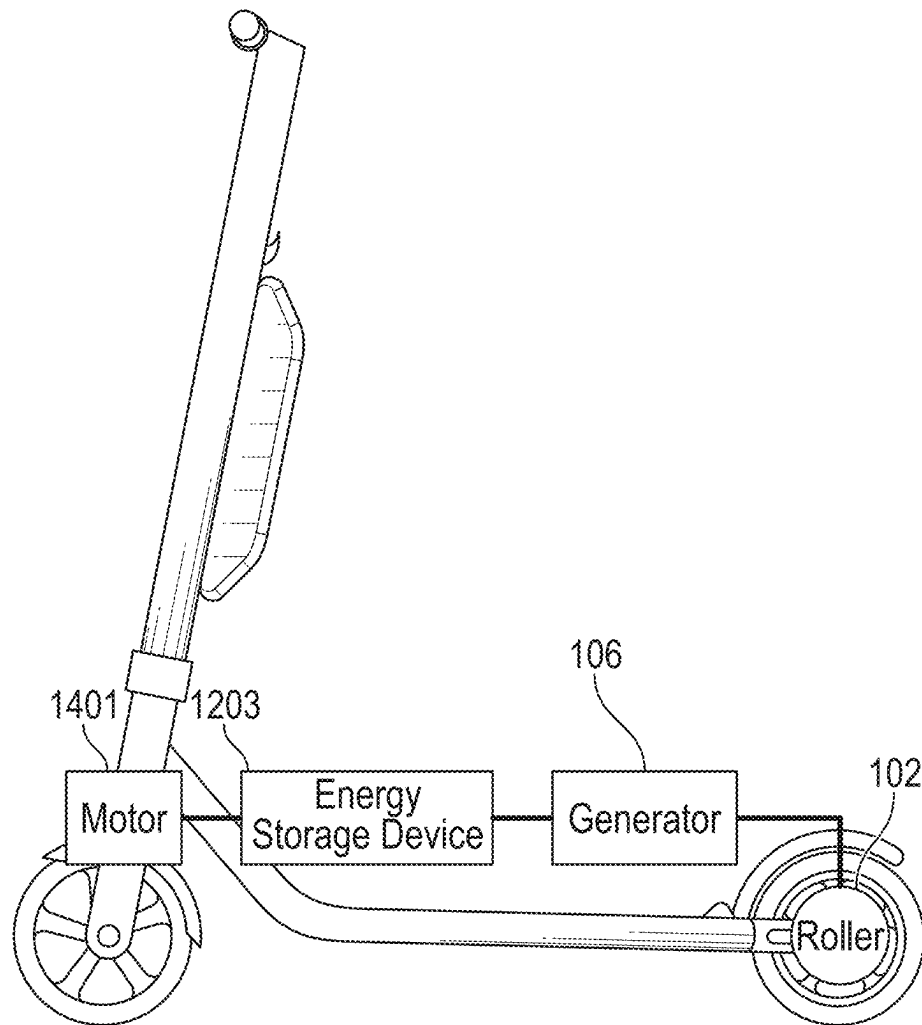


FIG. 14G

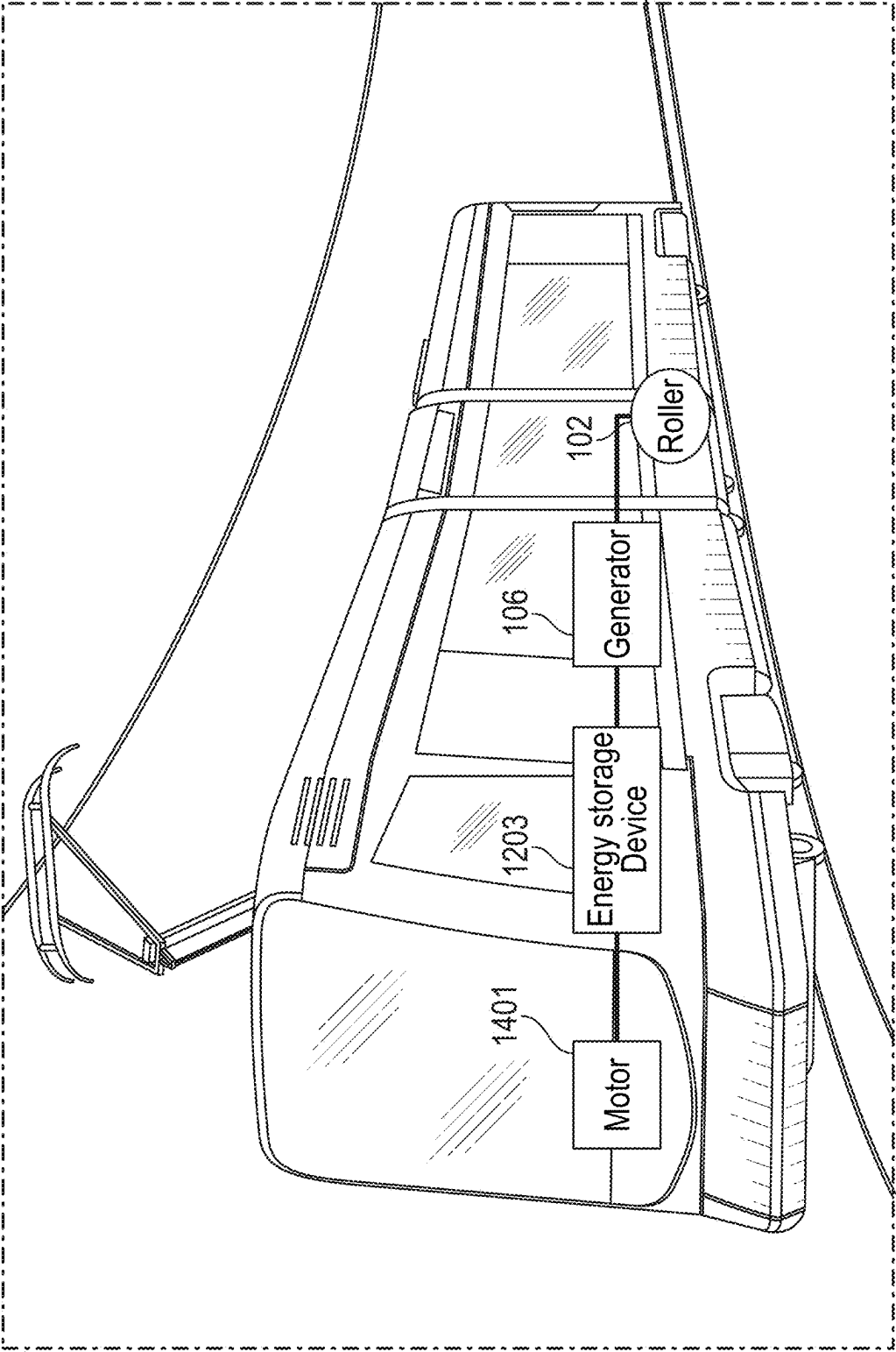


FIG. 14H

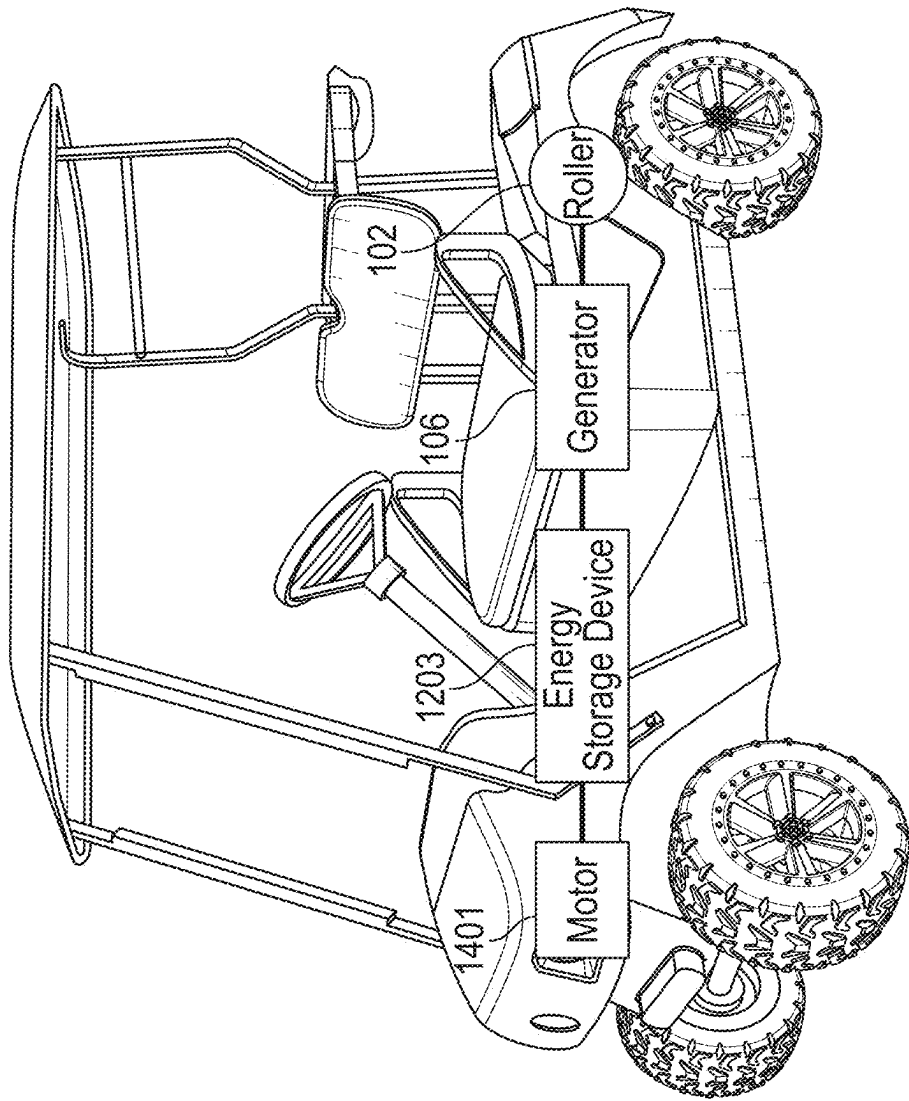


FIG. 14I

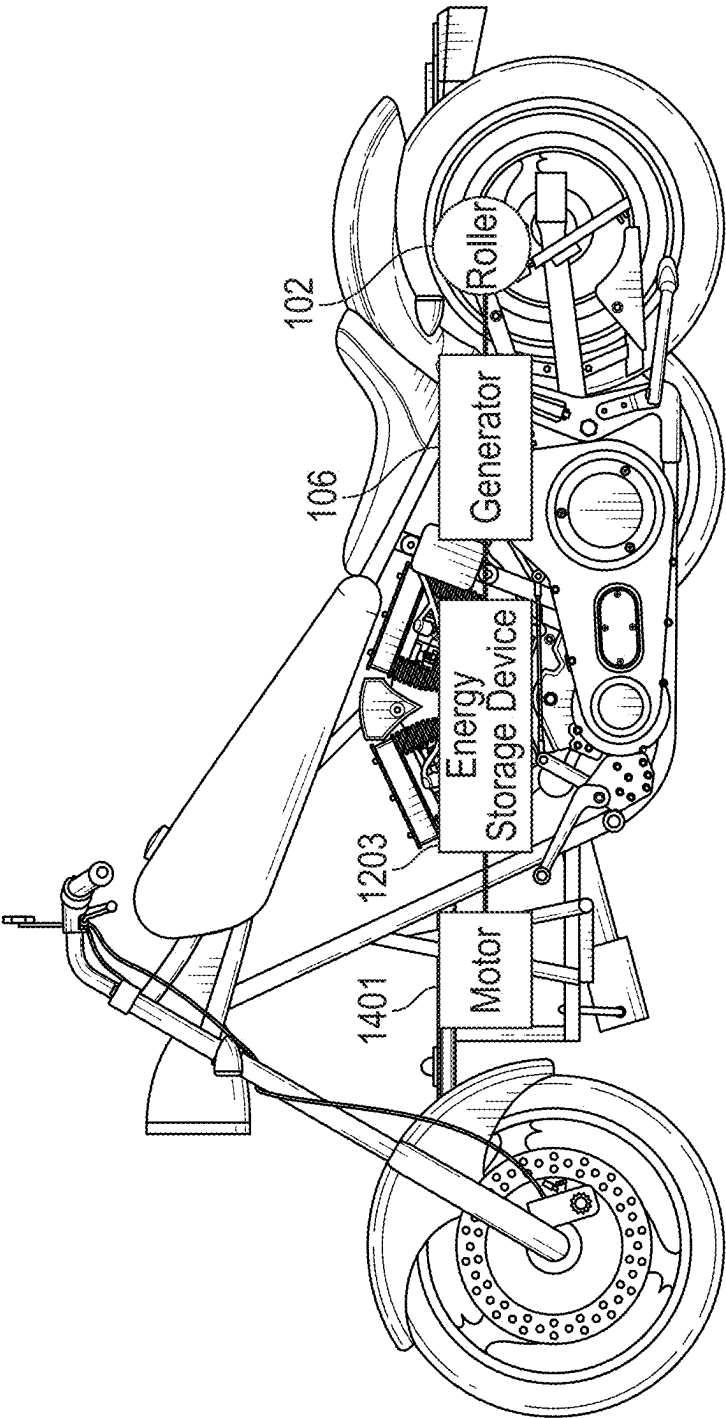


FIG. 14J

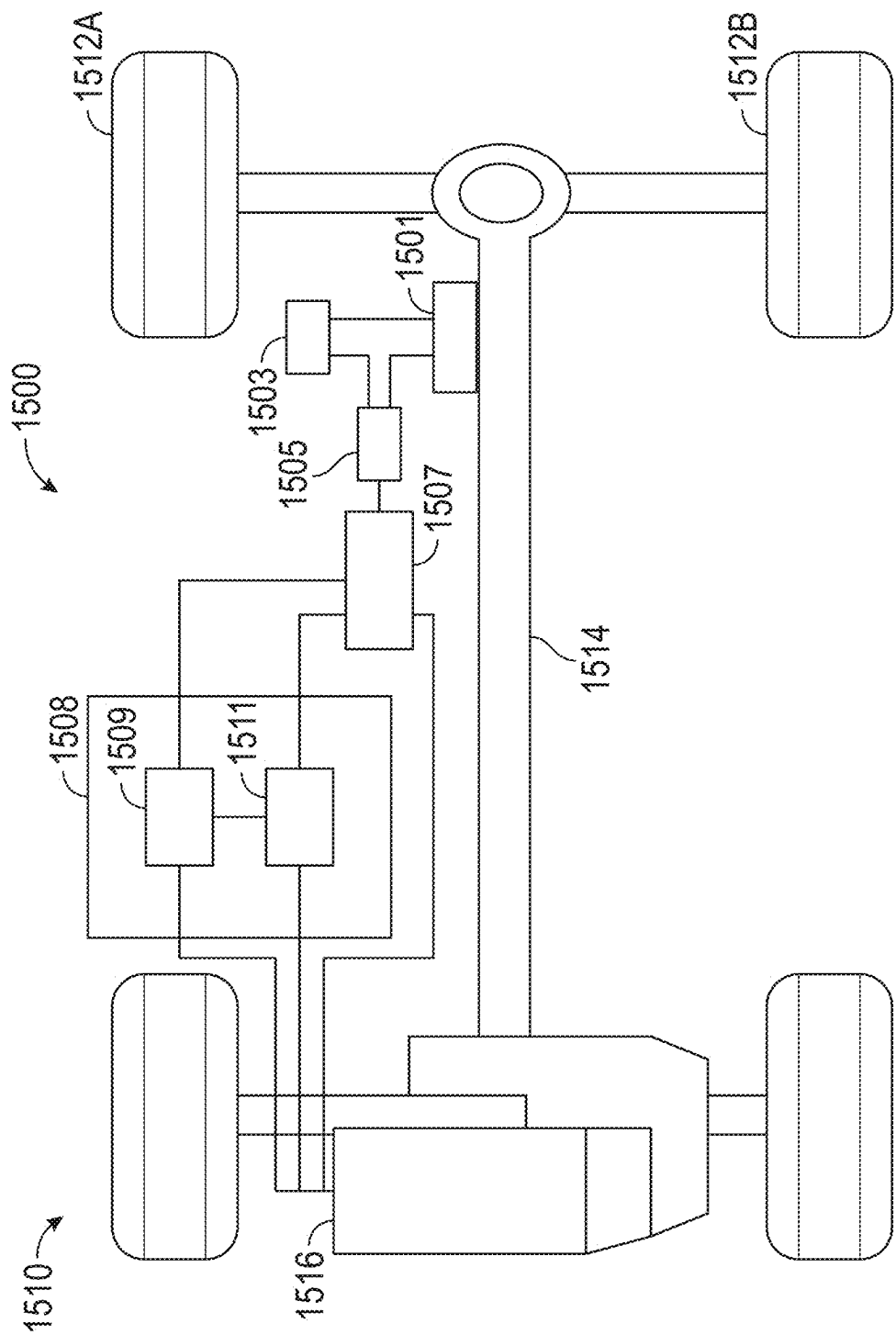


FIG. 15A

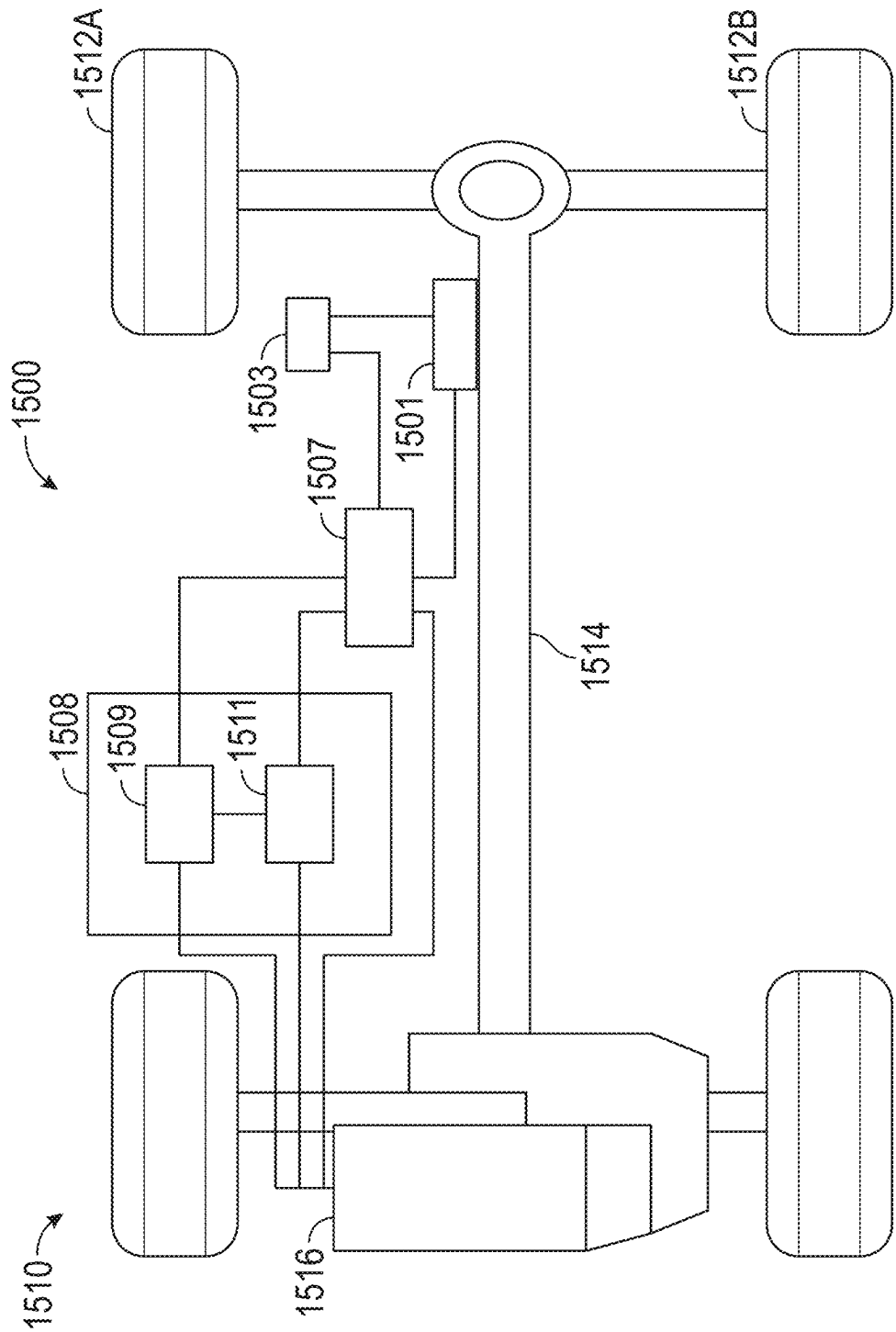


FIG. 15B

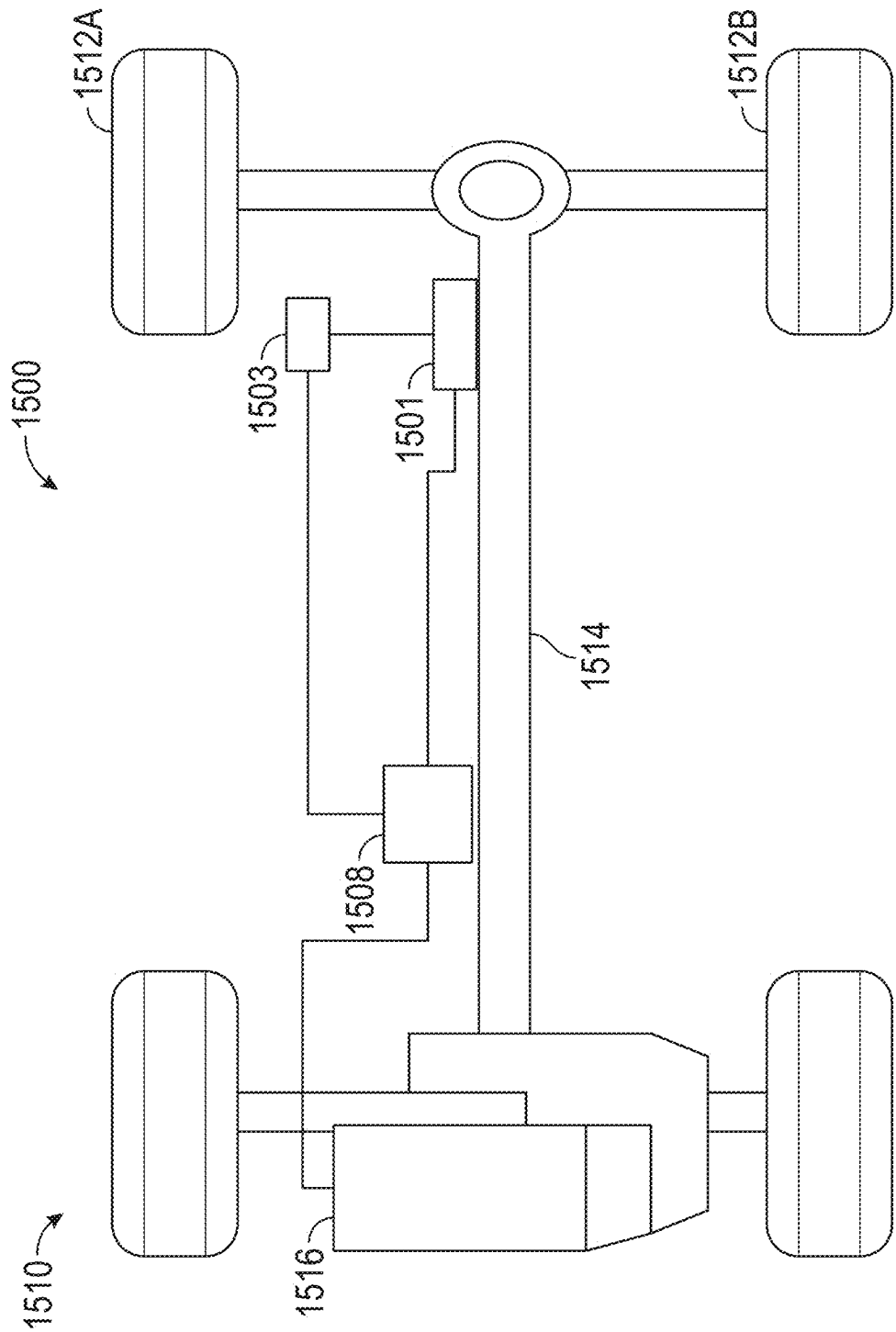


FIG. 15C

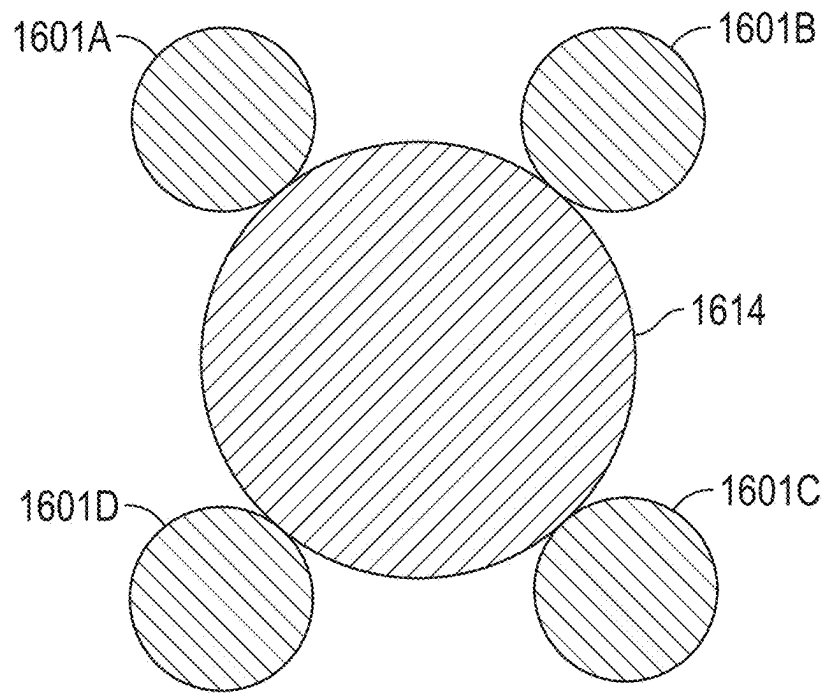


FIG. 16

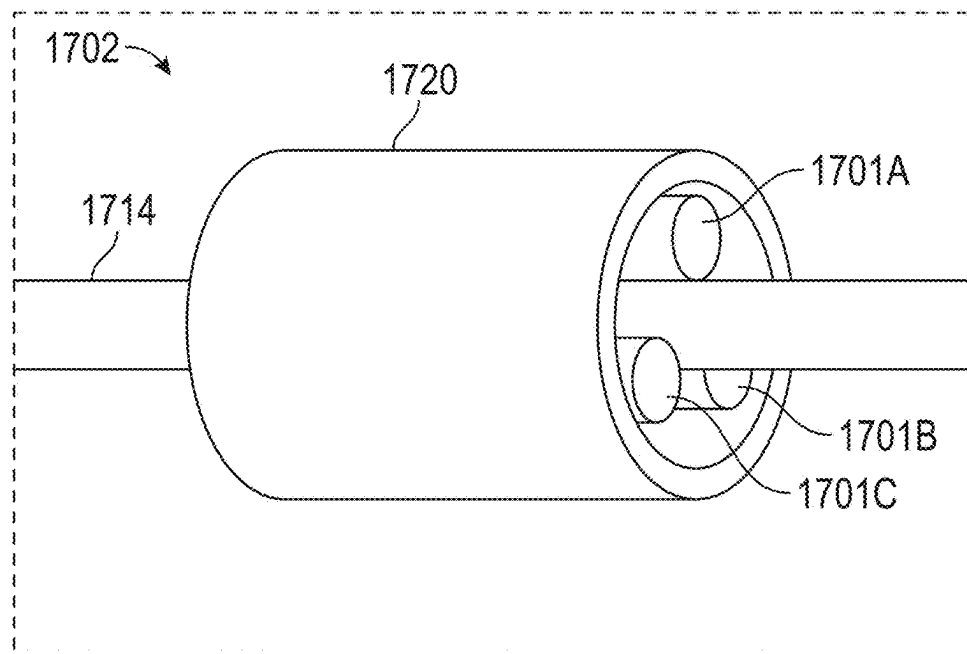


FIG. 17

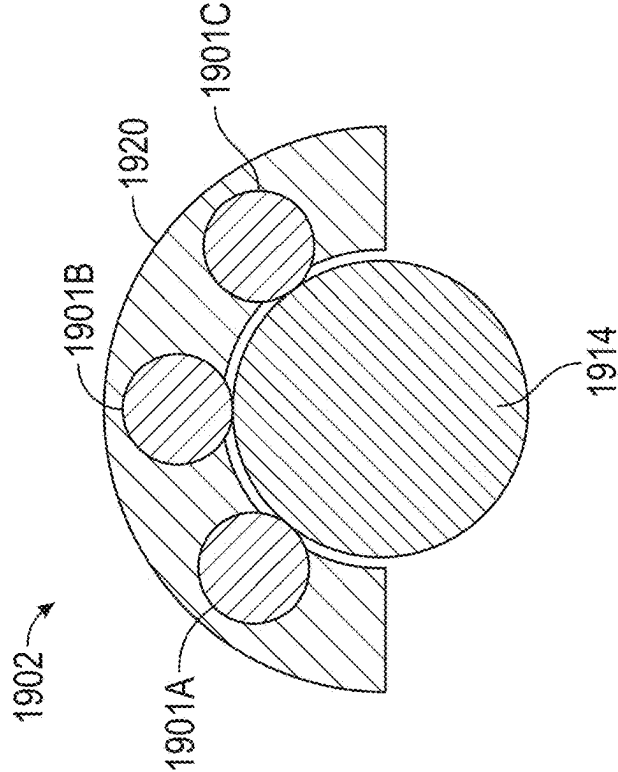


FIG. 18

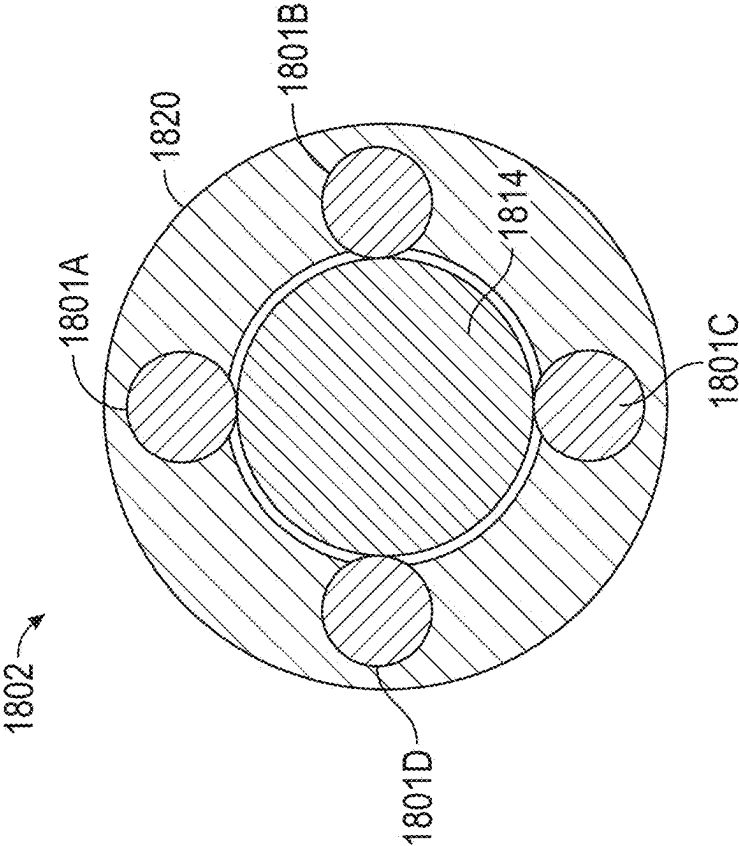


FIG. 19

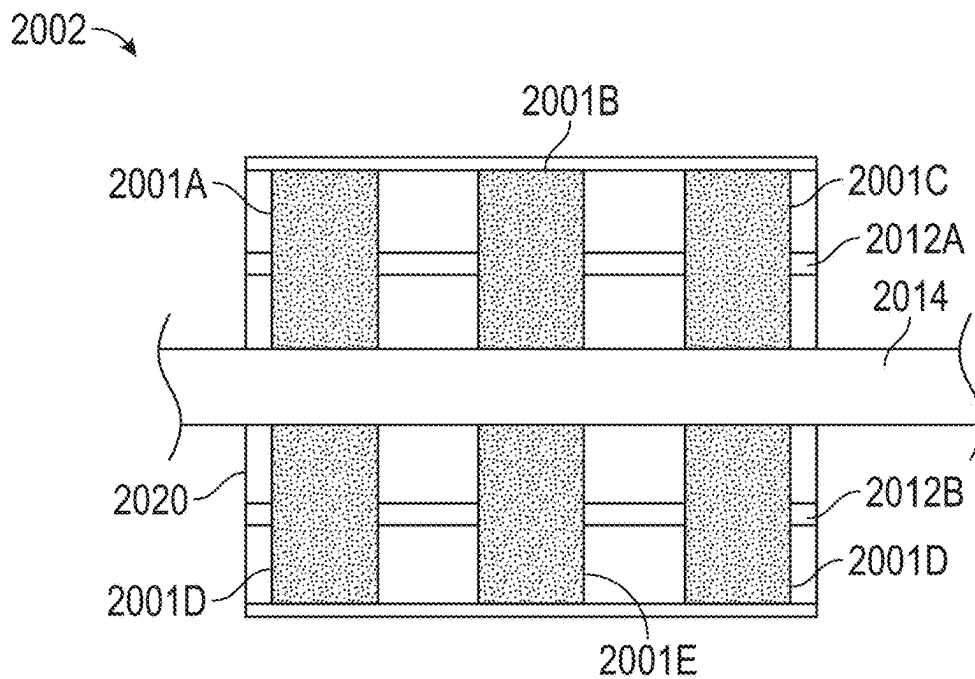


FIG. 20A

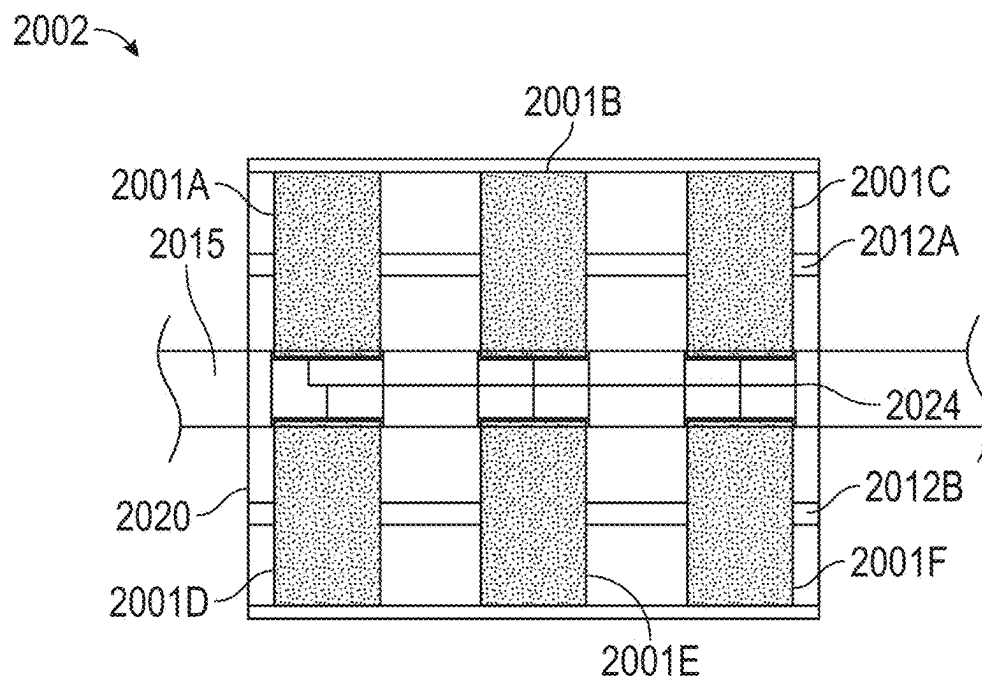


FIG. 20B

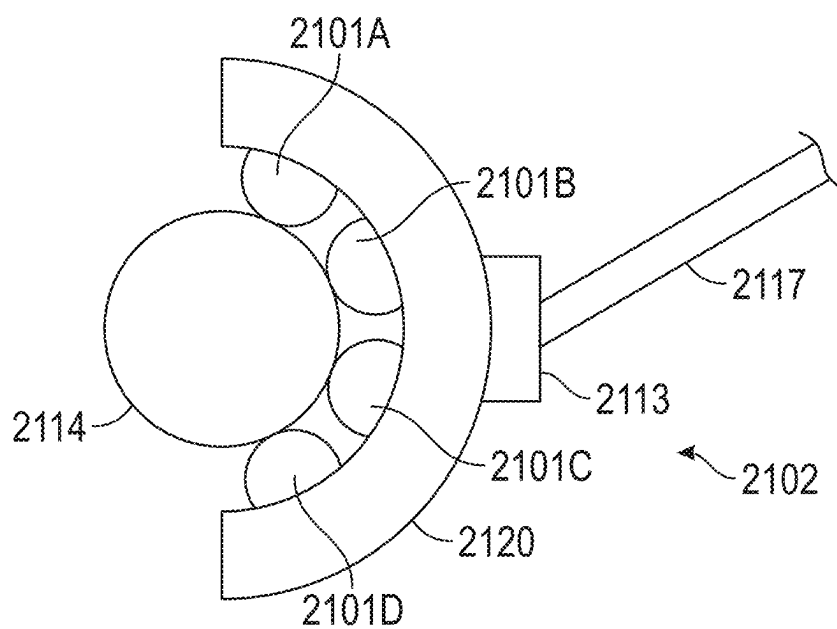


FIG. 21A

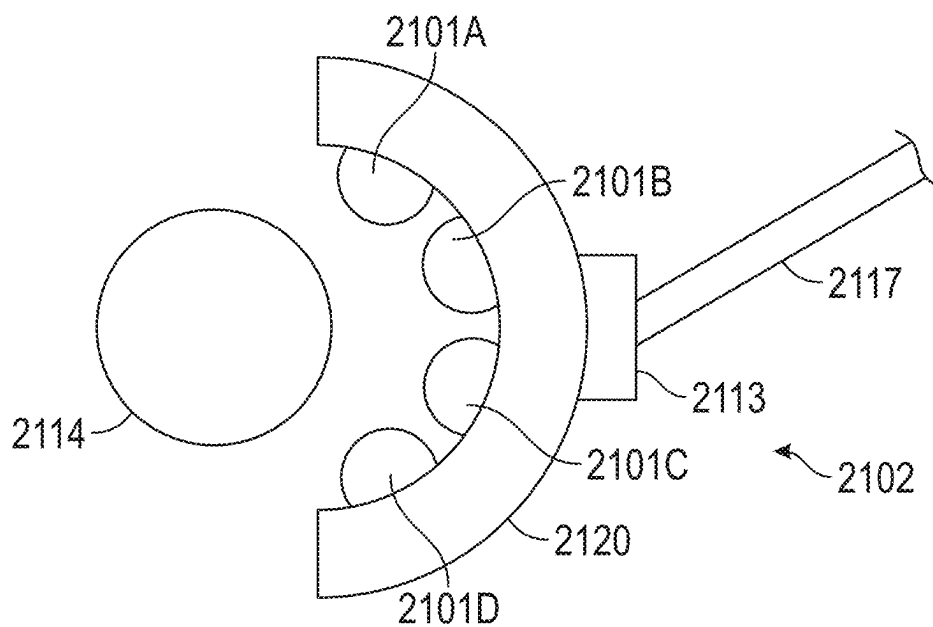


FIG. 21B

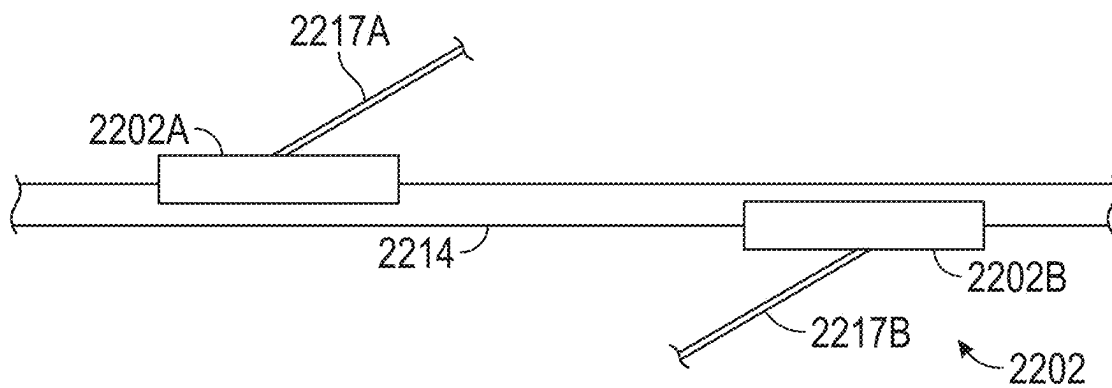


FIG. 22

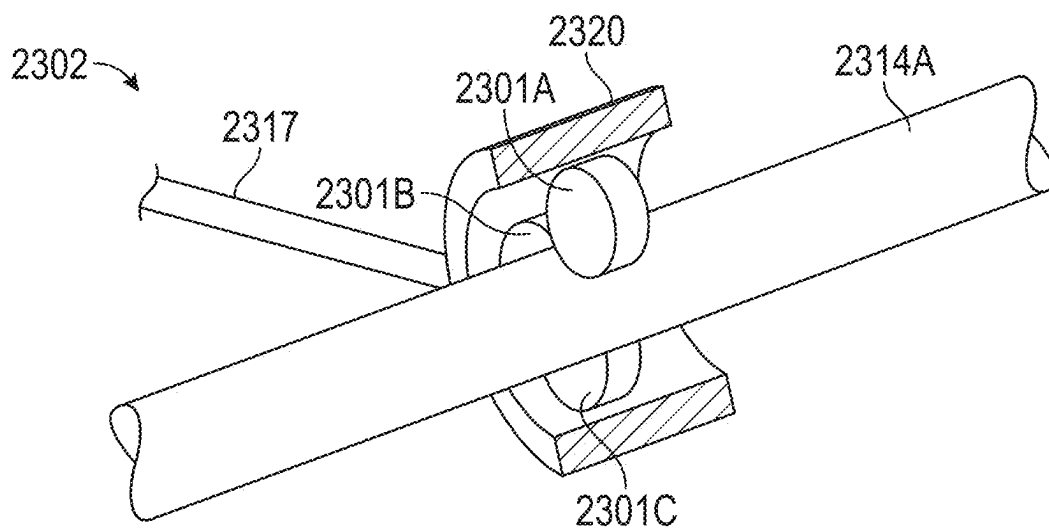


FIG. 23A

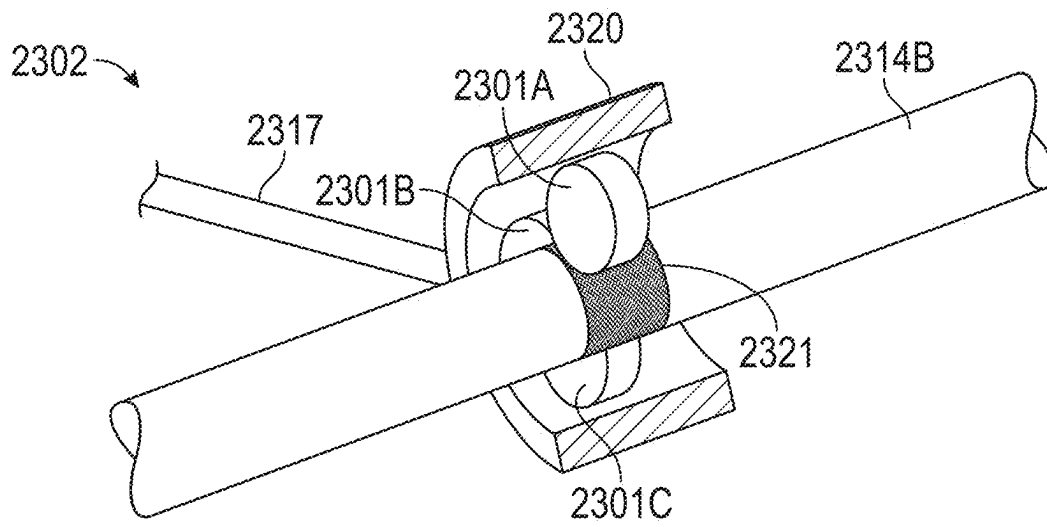


FIG. 23B

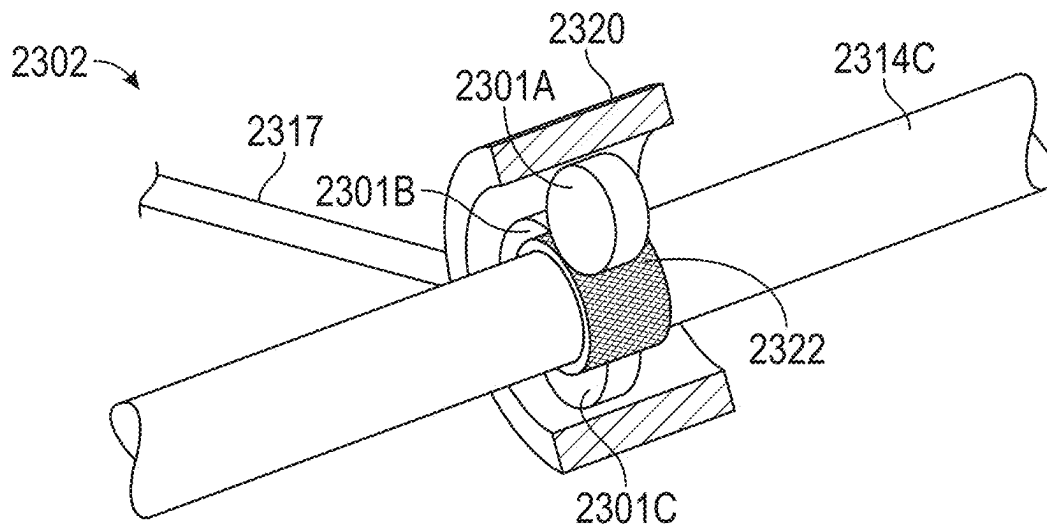


FIG. 23C

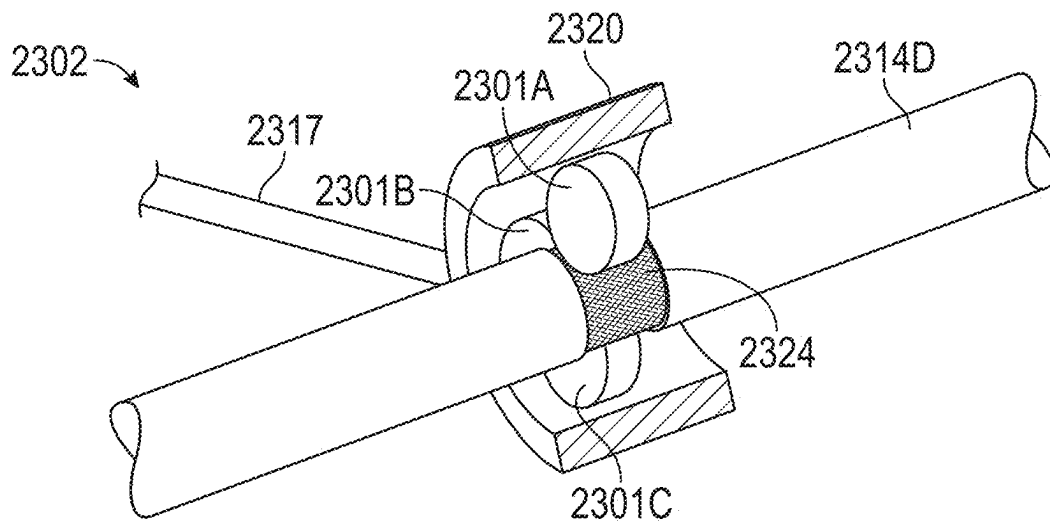


FIG. 23D

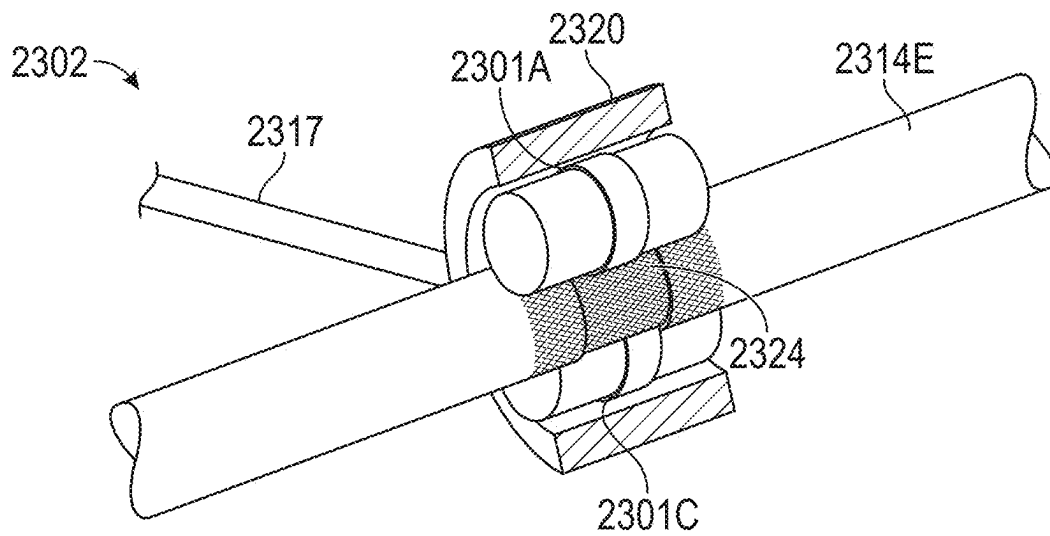


FIG. 23E

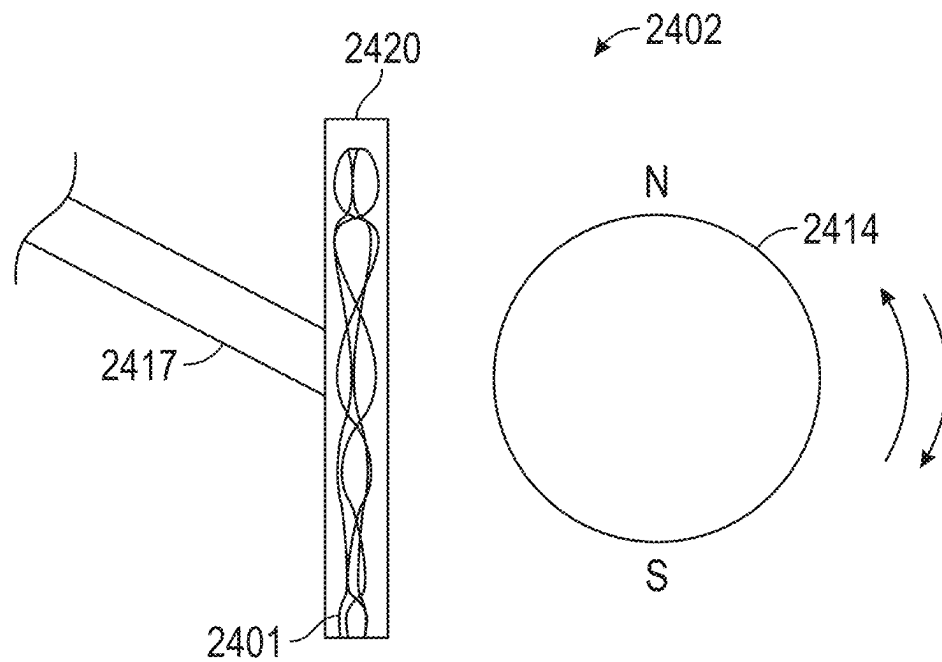


FIG. 24

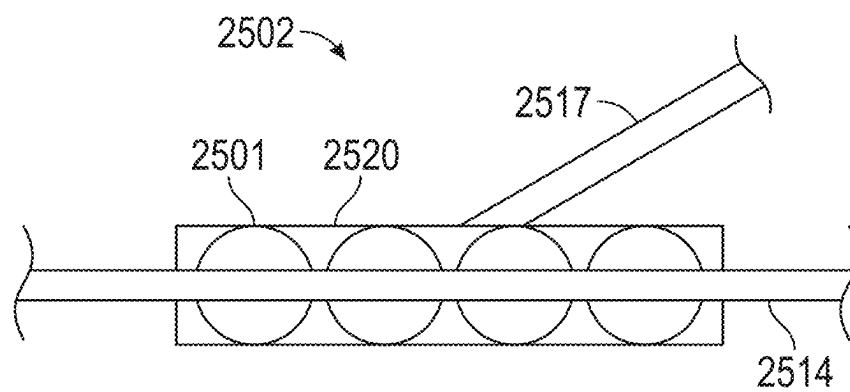


FIG. 25

1

VEHICLE ENERGY GENERATION SYSTEM**INCORPORATION BY REFERENCE TO ANY
PRIORITY APPLICATIONS**

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. This application is a continuation of U.S. application Ser. No. 18/381,120, filed Oct. 17, 2023, which is a continuation of U.S. application Ser. No. 18/243,524, filed Sep. 7, 2023, which claims priority benefit of U.S. Provisional Application No. 63/487,515, filed Feb. 28, 2023, the entirety of each of which is hereby incorporated by reference.

FIELD OF THE DISCLOSURE

The present disclosure relates generally to generating and providing energy for a vehicle powered, at least in part, by electricity, and more specifically, to generating and conveying the energy to the vehicle while the vehicle is mobile.

BACKGROUND

Electric vehicles derive locomotion power from electricity often received from an energy storage device within the electric vehicle. Battery electric vehicles (BEVs) are often proposed to have an energy storage/containment device, such as a battery, charged through some type of wired or wireless connection at one or more stationary locations, for example household or commercial supply sources. The wired charging connections require cables or other similar connectors physically connected to a stationary power supply. The wireless charging connections require one or more antennas or other similar structures wirelessly connected to a power supply that generates a wireless field via its own antenna(s). However, such wired and wireless stationary charging systems may be inconvenient or cumbersome and have other drawbacks, such as degradation during energy transference, inefficiencies or losses, requiring a specific location for charging, and so forth. As such, alternatives for stationary wired or wireless charging systems and methods that efficiently and safely transfer energy for charging electric vehicles are desirable.

SUMMARY

Various embodiments of systems, methods and devices within the scope of the appended claims each have several aspects, no single one of which is solely responsible for the desirable attributes described herein. Without limiting the scope of the appended claims, the description below describes some prominent features.

Details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that relative dimensions of the following figures may not be drawn to scale.

The present disclosure provides a system and apparatus for generating energy in response to a rotating component of a vehicle. In a first aspect, a system for generating energy in response to a movement of a vehicle includes an energy recovery mechanism configured to receive rotational energy from a drive shaft of the vehicle where the energy recovery

2

mechanism is secured to a feature of the vehicle. The system further includes an energy storage electrically coupled with the energy recovery mechanism and configured to receive a portion of an electrical output to store as energy, and a motor electrically coupled with the energy storage and configured to receive a portion of the energy.

In some embodiments, the energy storage includes a battery and/or a capacitor. In some embodiments, the feature of the vehicle is one of a vehicle frame, capacitor, generator, gearbox, or battery. In some embodiments, the system includes a flexible arm coupling the energy recovery mechanism to the feature of the vehicle. In some embodiments, the energy recovery mechanism is immovably fixed to the drive shaft. In some embodiments, the system includes a second energy recovery mechanism.

In another aspect, a system for generating energy in response to a movement of a vehicle includes an energy recovery mechanism configured to receive rotational energy from a drive shaft of the vehicle, where the energy recovery mechanism includes a housing with one or more rollers positioned tangentially adjacent the drive shaft and configured to rotate in response to a rotation of the drive shaft of the vehicle. The system further includes a generator rotatably coupled to the energy recovery mechanism and configured to generate an electrical output in response to the rotation of the roller, a capacitor electrically coupled with the generator and configured to receive a portion of the electrical output to store as capacitor energy, and a motor electrically coupled with the capacitor and configured to receive a portion of the capacitor energy.

In some embodiments, the system includes a flexible arm mechanically connected to the housing and a feature of the vehicle, where the flexible arm is configured to apply a force to the one or more rollers to cause the one or more rollers to contact the drive shaft. In some embodiments, the flexible arm is configured to bend. In some embodiments, the flexible arm is configured to pivot about one or more joints. In some embodiments, the flexible arm is configured to adjust a magnitude of the force. In some embodiments, the flexible arm is configured to adjust a magnitude of the force based on at least a braking of the vehicle. In some embodiments, the flexible arm is configured to adjust a magnitude of the force based on at least an acceleration of the vehicle. In some embodiments, the system further includes a battery electrically coupled with the capacitor and configured to receive a portion of the capacitor energy to store as a battery energy. In some embodiments, the motor is configured to receive a portion of the battery energy. In some embodiments, the drive shaft includes one or more grooves for accommodating the one or more rollers. In some embodiments, the one or more rollers fit substantially within the one or more grooves. In some embodiments, a material on a surface of the one or more grooves is different than a material of the drive shaft. In some embodiments, a surface of the one or more grooves is textured to increase friction with the one or more rollers. In some embodiments, the one or more rollers are positioned circumferentially around the drive shaft. In some embodiments, the one or more rollers are positioned longitudinally along the drive shaft. In some embodiments, the housing partially encompasses the drive shaft. In some embodiments, the housing fully encompasses the drive shaft.

In another aspect, a system for generating energy in response to a movement of a vehicle includes an energy recovery mechanism configured to receive rotational energy from a magnetic drive shaft of the vehicle, the energy recovery mechanism including one or more wires positioned

adjacent the magnetic drive shaft of the vehicle, where a current is induced in the one or more wires as the magnetic drive shaft rotates. The system further includes a capacitor electrically coupled with the energy recovery mechanism and configured to receive a portion of the electrical output to store as capacitor energy, and a motor electrically coupled with the capacitor and configured to receive a portion of the capacitor energy.

In some embodiments, the system includes a battery electrically coupled with the capacitor and configured to receive a portion of the capacitor energy to store as battery energy. In some embodiments, the motor is configured to receive a portion of the battery energy. In some embodiments, the energy recovery mechanism is coupled to a feature of the vehicle via a flexible arm. In some embodiments, the flexible arm is configured to move from an engaged to a disengaged position. In some embodiments, the feature of the vehicle is one of a vehicle frame, capacitor, or battery. In some embodiments, the magnetic drive shaft includes a permanent magnet. In some embodiments, the magnetic drive shaft partially includes a permanent magnet. In some embodiments, the magnetic drive shaft includes a permanent electromagnet positioned between two semicylinders of metal.

In another aspect, a method for generating energy in response to rotational movement of a moving vehicle includes rotating a shaft, the shaft including a textured surface spanning an outer circumference, rotating a roller in response to the rotation of the shaft, where the roller contacts the textured surface of the shaft, applying a force to the roller to cause the roller to maintain contact with the textured surface, generating, via a generator, an electrical output based on the rotation of the roller, and conveying the electrical output to an energy storage device or motor.

In some embodiments, the shaft is a drive shaft. In some embodiments, the textured surface is positioned within a groove of the drive shaft. In some embodiments, the textured surface is positioned on a shoulder of the drive shaft. In some embodiments, the energy storage device is a one or more capacitors. In some embodiments, the method further includes notifying a driver that the one or more capacitors is fully charged. In some embodiments, the force is applied to the roller based on a received signal from the driver. In some embodiments, method further includes transferring the energy from the capacitor to the motor based on a signal from the driver. In some embodiments, the force is reduced or eliminated while the energy is transferred from the capacitor to the motor.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a schematic diagram illustrating an example embodiment of an apparatus for generating energy in response to rotation of a wheel of vehicle.

FIG. 1B is a schematic diagram illustrating an example embodiment of the apparatus of FIG. 1A in a disengaged state.

FIG. 2 is a schematic diagram illustrating that a roller of the apparatus can have various dimensions.

FIGS. 3A-3B are schematic diagrams illustrating example embodiments of the apparatus comprising multiple rollers.

FIGS. 4A-4B are schematic diagrams illustrating example embodiments of the apparatus implemented on multiple wheels of a vehicle.

FIG. 5A is a schematic diagram illustrating an example embodiment of the apparatus comprising rollers implemented on a sidewall of a wheel of a vehicle.

FIG. 5B is a schematic diagram illustrating an example embodiment of the apparatus of FIG. 5A in a disengaged state.

FIGS. 6A-6B are schematic diagrams illustrating example embodiments of the apparatus comprising a shaft coupled to various portions of a roller.

FIG. 6C is a schematic diagram illustrating an example embodiment of the apparatus on both sides of a wheel.

FIG. 7A-7B is a schematic diagram illustrating various sizes of a roller of the apparatus implemented on a sidewall of a wheel.

FIG. 7C is a schematic diagram illustrating an example embodiment of a roller of the apparatus coupled to a shaft of the apparatus.

FIGS. 8A-8B are schematic diagrams illustrating example embodiments of the apparatus implemented on sidewalls of multiple wheels of a vehicle.

FIG. 9 is a schematic diagram illustrating an example embodiment of the apparatus comprising rollers between two adjacent wheels of a vehicle.

FIGS. 10A-10B are schematic diagrams illustrating example embodiments of the apparatus comprising rollers implemented on a sidewall surface and curved surface of a wheel of a vehicle.

FIGS. 11A-11B is a diagram illustrating examples embodiments of generators coupled to roller(s) of the apparatus.

FIG. 12 is a diagram illustrating an example vehicle incorporating the apparatus, a generator and an energy storage device.

FIG. 13 is a diagram illustrating an example embodiment of a hypercapacitor as an energy storage device.

FIGS. 14A-14J illustrate example vehicles incorporating the apparatus, a generator, an energy storage device and a motor.

FIG. 15A is a schematic diagram illustrating an example implementation of an energy generation system, according to some embodiments.

FIG. 15B is another schematic diagram illustrating an example implementation of an energy generation system, according to some embodiments.

FIG. 15C is another schematic diagram illustrating an example implementation of an energy generation system, according to some embodiments.

FIG. 16 is a cross section view of an example drive shaft and rollers, according to some embodiments.

FIG. 17 is a perspective view of an example energy recovery mechanism with a partial housing, according to some embodiments.

FIG. 18 is a cross sectional view of an example energy recovery mechanism with a full housing, according to some embodiments.

FIG. 19 is a cross sectional view of an example energy recovery mechanism, according to some embodiments.

FIG. 20A is a side view of an example energy recovery mechanism and drive shaft, according to some embodiments.

FIG. 20B is a side view of an example energy recovery mechanism and drive shaft with grooves, according to some embodiments.

FIG. 21A is a side view of an example energy recovery mechanism in an engaged position.

FIG. 21B is a side view of an example energy recovery mechanism in a disengaged position.

FIG. 22 is a top down view of a drive shaft and several energy recovery mechanisms.

5

FIG. 23A is a perspective view of an example energy recovery mechanism and drive shaft, according to some embodiments.

FIG. 23B is a perspective view of an example energy recovery mechanism and drive shaft with a textured surface, according to some embodiments.

FIG. 23C is a perspective view of an example energy recovery mechanism and drive shaft with a raised shoulder, according to some embodiments.

FIG. 23D is a perspective view of an example energy recovery mechanism and drive shaft with a groove, according to some embodiments.

FIG. 23E is a perspective view of an example energy recovery mechanism and drive shaft with a groove, according to some embodiments.

FIG. 24 is a sectional view of an example energy recovery mechanism and magnetic drive shaft, according to some embodiments.

FIG. 25 is a side view of an example energy recovery mechanism and magnetic drive shaft, according to some embodiments.

DETAILED DESCRIPTION

Overview

Example embodiments and implementations of an apparatus for generating energy (for example, in response to the rotation of a component of a vehicle) are described herein. The apparatus can be implemented in conjunction with a vehicle, such as an electric vehicle. The vehicle can include a car, a truck, a semi-truck, a tractor-trailer, a tractor, farm equipment, construction equipment, carts, scooters, motorcycles, bicycles, trains, trams, and the like, for example. The apparatus can comprise one or more rollers configured to be rotatably couplable (for example, removably coupled either through direct physical contact or through indirect operable coupling) to one or more components of a vehicle (for example, wheel, drive shaft, axle etc.) such that rotation of the component of the vehicle causes rotation of the one or more rollers. The one or more rollers can be rotatably coupled (either through direct physical contact or through indirect operable coupling) to one or more generators. The generators can be configured to generate energy (for example, an electrical output), in response to rotation of the one or more rollers. In some embodiments, the one or more rollers can be rotatably coupled to the one or more generators via one or more flexible arms configured to rotate in response to a rotation of the one or more rollers. In some embodiments, the one or more rollers can be rotatably coupled to the one or more generators via one or more other mechanical coupling devices such as a chain, belt, gearing, pulley, sprocket and the like. In some embodiments, the one or more generators can provide generated energy (for example, electrical output) to the vehicle. The electrical output that is provided to the vehicle from the generator may be used to power the vehicle. For example, the electrical output may be conveyed to a motor of the vehicle and/or to an energy storage device of the vehicle for later use and/or consumption by the vehicle. In some embodiments, the energy of the rotating component is captured by inducing a current in one or more wires positioned adjacent the rotating component.

Example Apparatus Embodiments and Implementations

Various example embodiments of an apparatus for generating energy are described herein, for example, with reference to the figures. The various embodiments and their

6

implementations are given as examples and are not meant to be limiting of the present disclosure.

Furthermore, the structural and/or operational features described with reference to any of the example embodiments and/or figures are not meant to be limited to that embodiment and/or figure. Rather the structural and/or operation features of the various embodiments and figures may be implemented or otherwise combined in each of the various other embodiments.

In some embodiments, the term “shaft” may refer to a “flexible arm” as described in U.S. Pat. No. 11,577,606 which is hereby incorporated by reference in its entirety.

FIG. 1A is a diagram illustrating an example embodiment of an apparatus 100 comprising a roller rotatably couplable to a wheel of a vehicle. As shown in FIG. 1A, the apparatus 100 may comprise a roller 102, a shaft 104 and a generator 106. The roller 102 may comprise a substantially cylindrical shape comprising a length, a diameter, a curved surface and a center axis as described in greater detail with reference to FIG. 2. A curved surface of the roller 102 may be in substantial physical contact with a curved surface of the wheel 101. The center axis of the roller 102 may be substantially parallel to a center axis of the wheel 101. The roller 102 may be configured to rotate about its center axis. The roller 102 may be rotatably couplable to a wheel 101 of the vehicle such that rotation of the wheel 101 causes rotation of the roller 102. The roller 102 may rotate in an opposite direction than the wheel 101, for example as shown in FIG. 1A. The roller 102 may rotate at a greater rotational velocity than the wheel 101.

With continued reference to FIG. 1A, the roller 102 may be rotatably coupled to a shaft 104 such that rotation of the roller 102 can cause rotation of the shaft 104. The shaft 104 may rotate about an axis that is substantially parallel to the axis of the roller 102 and may rotate in a same direction as the roller 102, for example as shown in FIG. 1A. In some embodiments, the shaft 104 may be fixedly rotatably coupled to the roller 102 such that the shaft 104 can only rotate when the roller 102 rotates. In some embodiments, the shaft 104 may be configured to rotate when the roller 102 is not rotating. For example, after a roller 102 discontinues rotating, the shaft 104 may continue to rotate, for example due to rotational inertia. For example, the roller 102 and/or shaft 104 may comprise a one-way ratchet device that causes the shaft 104 to rotate when the roller 102 rotates and allows the shaft 104 to continue to rotate for a period of time even after the roller 102 stops rotating. In some embodiments, the shaft 104 may be configured to not rotate when the roller 102 is rotating. For example, in a disengaged state, as discussed in greater detail herein, the roller 102 may rotate in response to rotation of a vehicle wheel but may not cause rotation of the shaft 104 to generate energy at the generator 106.

The shaft 104 may be operably coupled to a generator 106. The generator 106 may be configured to generate energy (for example, an electrical output) in response to mechanical movement such as the rotation of the shaft 104. The generator 106 may be electrically coupled to the vehicle and may provide generated energy to the vehicle, for example to a motor of the vehicle and/or to an energy storage device of the vehicle that includes one or more batteries and/or capacitors (for example, ultracapacitors) or one or more hypercapacitors.

FIG. 1B is a diagram illustrating an example embodiment of the apparatus 100 comprising a roller that is removably coupled to a wheel of a vehicle. The apparatus 100 may exist in one of (1) an engaged state or (2) a disengaged state. In the engaged state, the roller 102 may be in physical contact

7

with the wheel **101** (for example, rotatably coupled to the wheel **101**) in which the rotation of the wheel **101** causes the roller **102** to rotate. In some embodiments, in the disengaged state, the roller **102** may not be in physical contact with the wheel **101** such that rotation of the wheel **101** does not cause the roller **102** to rotate. In some embodiments, in the disengaged state, the roller **102** may be in physical contact with the wheel **101** such that rotation of the wheel **101** causes the roller **102** to rotate but the roller **102** may not be rotatably coupled to the shaft **104** such that rotation of the roller **102** does not cause the shaft **104** (or other similar component) to rotate to cause generation of energy at the generator **106**.

FIG. 1B shows a roller **102** in an example disengaged state such that the roller **102** is not in physical contact with the wheel **101** and will not rotate in response to a rotation of the wheel **101**. The roller **102** may transition between the engaged and the disengaged states. In some embodiments, the roller **102** may transition between the engaged and the disengaged states automatically, for example, based at least in part on an energy demand of the vehicle (for example, an energy demand of a motor of the vehicle) and/or a rotational velocity of the wheel **101**. In some embodiments, the roller **102** may transition between the engaged and the disengaged states in response to a user input, such as a driver of the vehicle activating a user input device, such as a button, lever, or switch.

FIG. 2 is a diagram illustrating an example embodiment of a roller **102**. As shown in FIG. 2, the roller **102** may comprise a length **213** and a diameter **211**. The roller **102** may have any length **213** such as is required or desired. The roller **102** may have any diameter **211** such as is required or desired. The diameter **211** of the roller **102** may be less than the diameter of the wheel **101** such that the roller **102** rotates at a greater rotational velocity than the wheel **101** in response to a rotation of the wheel **101**. In some embodiments comprising multiple rollers, one, some or each of the multiple rollers may have a length and/or diameter that is different than the length and/or diameters of the other rollers.

In some embodiments, the roller **102** may be configured to change a size of diameter **211**. In response to changing size of diameter **211**, the roller **102** may rotate at various rotational velocities in response to rotation of the wheel **101** at a single rotational velocity. In some embodiments, the roller **102** may be configured to change size of diameter **211** automatically, for example, based at least in part on an energy demand of the vehicle (for example, an energy demand of a motor of the vehicle) and/or a rotational velocity of the wheel **101**.

FIG. 3A is a diagram illustrating an example embodiment of the apparatus **100** comprising two rollers and two generators. As shown in FIG. 3A, the apparatus **100** may comprise a first roller **102a**, a first shaft **104a**, a first generator **106a**, a second roller **102b**, a second shaft **104b** and a second generator **106b**. The components of the example embodiment shown in FIG. 3A may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiment of FIG. 1A. For example, the rotation of the wheel **101** may cause the rollers **102a/102b** to rotate thereby causing shafts **104a/104b** to rotate thereby causing the generators **106a/106b** to generate energy. FIG. 3A is not meant to be limiting of the present disclosure. The apparatus **100** may comprise any number of rollers, shafts and/or generators as required and/or desired.

FIG. 3B is a diagram illustrating an example embodiment of the apparatus **100** comprising two rollers and a generator.

8

As shown in FIG. 3B, the apparatus **100** may comprise a first roller **102a**, a first shaft **104a**, a first sprocket **105a**, a first coupling device **107a**, a second roller **102b**, a second shaft **104b**, a second sprocket **105b**, a second coupling device **107b**, a third shaft **108** and generator **106**. The components of the example embodiment shown in FIG. 3B may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example FIG. 1A. The sprockets **105a/105b** may be rotatably coupled to the shafts **104a/104b** and may rotate in response to rotation of the shafts **104a/104b**. The sprockets **105a/105b** may be rotatably coupled to a third shaft **108**, for example via coupling devices **107a/107b** as shown in FIG. 3B. The coupling devices **107a/107b** may comprise one or more of a chain, belt, gearing, pulley or the like. The third shaft **108** may be operably coupled to the generator **106** such that rotation of the third shaft **108** causes the generator to generate energy. Thus, the generator **106** may generate energy in response to a rotation of the first and/or second rollers **102a/102b**.

In some embodiments, the third shaft **108** may rotate in response to simultaneous rotations of the first and second rollers **102a/102b**. In some embodiments, the third shaft **108** may rotate in response to rotation of either the first or second rollers **102a/102b**.

In some embodiments, the shafts **104a/104b** may be fixedly rotatably coupled to the sprockets **105a/105b** such that the sprockets **105a/105b** can only rotate when the shafts **104a/104b** rotate. In some embodiments, the sprockets **105a/105b** may be configured to rotate when the shafts **104a/104b** are not rotating, for example, after the shafts **104a/104b** discontinue rotating, the sprockets **105a/105b** may continue to rotate, for example due to rotational inertia. For example, the shafts **104a/104b** and/or sprockets **105a/105b** may comprise a one-way ratchet device that causes the sprockets **105a/105b** to rotate when the shafts **104a/104b** rotate and allows the sprockets **105a/105b** to continue to rotate when the shafts **104a/104b** are not rotating. The sprockets **105a/105b** and the third shaft **108** may comprise similar operational and/or structural features to allow the third shaft **108** to rotate when one or more of the sprockets **105a/105b** are not rotating in some embodiments or to cause the third shaft **108** to rotate only when the sprockets **105a/105b** are rotating in other embodiments.

FIG. 4A is a diagram illustrating an example embodiment of the apparatus **100** implemented with multiple wheels of a vehicle. As shown in FIG. 4A, the apparatus **100** may comprise a first roller **102a** rotatably couplable to a first wheel **101a** of a vehicle, a second roller **102b** rotatably couplable to a second wheel **101b** of a vehicle. The components of the example embodiment shown in FIG. 4A may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiment of FIG. 1A. For example, rotation of the first and/or second rollers **102a/102b** may cause the generator **106** to generate energy.

FIG. 4A is not meant to be limiting of the present disclosure. The apparatus **100** may comprise any number of rollers, shafts and/or generators as required and/or desired and may be implemented on any number of wheels of a vehicle as required or desired, for example on one, two, three or four wheels (for example, with reference to implementation with a car) or 18 wheels (for example, with reference to implementation with a semi-truck).

FIG. 4B is a diagram illustrating an example embodiment of the apparatus **100** implemented with multiple wheels of a vehicle and comprising multiple generators. As shown in

FIG. 4B, the apparatus 100 may comprise a first and second generator 106a/106b. The components of the example embodiment shown in FIG. 4B may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiment of FIG. 1A. For example, rotation of the first roller 102a may cause the first generator 106a to generate energy and rotation of the second roller 102b may cause the generator 106b to generate energy. The generators 106a/106b may be in electrical communication with the vehicle and/or each other.

FIG. 5A is a diagram illustrating an example embodiment of the apparatus 100 comprising one or more rollers rotatably couplable to a sidewall of a wheel of a vehicle. As shown in FIG. 5A, the apparatus 100 may comprise one or more rollers 102, a shaft 104 and a generator 106. Each of the one or more rollers 102 may comprise a substantially cylindrical shape and may further comprise a length, a diameter, a curved surface and a center axis as described in greater detail with reference to FIG. 2 and/or FIGS. 7A-7B. A curved surface of each of the one or more rollers 102 may be in substantial physical contact with a sidewall surface of the wheel 101. The center axis of each of the one or more rollers 102 may be substantially orthogonal to a center axis of the wheel 101. Each of the one or more rollers 102 may be configured to rotate about its center axis. Each of the one or more rollers 102 may be rotatably couplable to the wheel 101 of the vehicle such that rotation of the wheel 101 causes rotation of each of the one or more rollers 102. Each of the one or more rollers 102 may rotate at a greater rotational velocity than the wheel 101.

The roller(s) 102 may be configured to be in physical contact with a sidewall of the wheel 101 at any distance away from a center axis of the wheel. For example, the roller(s) 102 may be in physical contact with a sidewall of the wheel 101 close to the center axis of the wheel or far from a center axis of the wheel. The roller(s) 102 may rotate at a greater rotational velocity when in physical contact with the sidewall of the wheel 101 far from a center axis of the wheel 101 than when in physical contact with the sidewall of the wheel 101 near a center axis of the wheel 101.

With continued reference to FIG. 5A, the roller(s) 102 may be rotatably coupled to a shaft 104 such that rotation of the roller(s) 102 causes rotation of the shaft 104. The roller 102 may be coupled (for example, rotatably coupled) to the shaft 104 for example via one or more coupling devices as required or desired such as gears, sprockets, chains, belts, pulleys and the like. The shaft 104 may rotate about an axis that is substantially orthogonal to the axes of the roller(s) 102. In some embodiments, the shaft 104 may be fixedly rotatably coupled to the roller(s) 102 such that the shaft 104 can only rotate when the roller(s) 102 rotate. In some embodiments, the shaft 104 may be configured to rotate when one or more of the roller(s) 102 is not rotating, for example, after a roller 102 discontinues rotating, the shaft 104 may continue to rotate, for example due to rotational inertia. For example, the roller(s) 102 and/or shaft 104 may comprise a one-way ratchet device that causes the shaft 104 to rotate when the roller(s) 102 rotate and allows the shaft 104 to continue to rotate even when one of the roller(s) 102 is not rotating (for example, has stopped rotating). In some embodiments, the shaft 104 may be configured to not rotate when one or more of the roller(s) 102 are rotating. For example, in a disengaged state, as discussed in greater detail herein, the roller(s) 102 may rotate in response to rotation of a vehicle wheel but may not cause rotation of the shaft 104 to generate energy at the generator 106.

The shaft 104 may be operably coupled to a generator 106. The generator 106 may be configured to generate energy (for example, an electrical output) in response to mechanical movement such as the rotation of the shaft 104. The generator 106 may be electrically coupled to the vehicle and may provide generated energy to the vehicle, for example to a motor of the vehicle and/or to an energy storage device of the vehicle that includes one or more batteries and/or capacitors (for example, ultracapacitors) or one or more hypercapacitors.

FIG. 5B is a diagram illustrating an example embodiment of the apparatus 100 comprising one or more rollers that are removably coupled to a sidewall of a wheel of a vehicle. The apparatus 100 may exist in one of (1) an engaged state or (2) a disengaged state. In the engaged state, the roller(s) 102 may be in physical contact with the wheel 101 (for example, rotatably coupled to a sidewall of the wheel 101) in which the rotation of the wheel 101 causes the roller(s) 102 to rotate. In some embodiments, in the disengaged state, the roller(s) 102 may not be in physical contact with the wheel 101 such that rotation of the wheel 101 does not cause the roller(s) 102 to rotate. In some embodiments, in the disengaged state, the roller(s) 102 may be in physical contact with the wheel 101 such that rotation of the wheel 101 causes the roller(s) 102 to rotate but the roller(s) 102 may not be rotatably coupled to the shaft 104 such that rotation of the roller(s) 102 does not cause the shaft 104 (or other similar component) to rotate to cause generation of energy at the generator 106.

FIG. 5B shows roller(s) 102 in an example disengaged state such that the roller(s) 102 are not in physical contact with the wheel 101 and will not rotate in response to a rotation of the wheel 101. The roller(s) 102 may transition between the engaged and the disengaged states. In some embodiments, the roller(s) 102 may transition between the engaged and the disengaged states automatically, for example, based at least in part on an energy demand of the vehicle (for example, an energy demand of a motor of the vehicle) and/or a rotational velocity of the wheel 101. In some embodiments, the roller(s) 102 may transition between the engaged and the disengaged states in response to a user input, such as a driver of the vehicle toggling a user input device such as a button, switch or lever.

FIG. 6A is a diagram illustrating an example embodiment of the apparatus 100 comprising a roller rotatably couplable to a sidewall of a wheel of a vehicle. As shown in FIG. 6A, the apparatus 100 may comprise a single roller 102 which may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiments of FIG. 1 and/or FIG. 5A.

FIG. 6B is a diagram illustrating an example embodiment of the apparatus 100 comprising a roller. As shown in FIG. 6B, the apparatus 100 may comprise a single roller 102 which may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiments of FIG. 1 and/or FIG. 5A. As shown in FIG. 6B, the shaft 104 may be rotatably coupled to either end of the roller 102. In some embodiments, a shaft may be rotatably coupled to both ends of a roller 102.

FIG. 6C is a diagram illustrating an example embodiment of the apparatus 100. As shown in FIG. 6C, the apparatus 100 may comprise a roller 102a on one side of a wheel 101 and a roller 102b on an opposite side of the wheel 101. The rollers 102a, 102b can be configured to be rotatably couplable to a sidewall portion of the wheel 101 and rotate in

11

response to a rotation of the wheel **101** as described herein. The apparatus **100** may exist in one of (1) an engaged state or (2) a disengaged state and transition between the two states as discussed herein, for example by changing a physical location of the rollers **102a**, **102b** to physically contact or not physically contact the wheel **101**.

The example embodiment of FIG. 6C can be implemented in a vehicle, for example, in conjunction with a braking system of the vehicle. For example, a braking system of a vehicle may comprise a brake pad on one side of wheel and a brake pad on an opposite side of the wheel. The brake pads may normally exist in a state wherein the brake pads are not in physical contact with the wheel. An operator of the vehicle may cause each brake pad to physically contact their respective sides of the wheel to cause friction on the sidewall of the wheel to decelerate the rotation of the wheel. As an example, the apparatus may be implemented with a brake system such that the apparatus may normally exist in a disengaged state wherein the rollers **102a**, **102b** are not in physical contact with the wheel **101** and the brake pads are not in physical contact with the wheel **101**. When braking is desired, and the brake pads are caused to contact the wheel **101**, the rollers **102a**, **102b** may also contact the wheel **101** in an engaged state and thereby rotate in response to a rotation of the wheel **101**.

In some implementations, in the engaged state, the rollers **102a**, **102b** may apply a friction force to the wheel **101** to decelerate the wheel **101**. In some implementations, the apparatus **100** may replace a braking system otherwise employed by the vehicle, such that when braking is desired, the rollers **102a**, **102b** of the apparatus transition to an engaged state thereby applying friction to the wheel **101** to decelerate the rotation of wheel **101** while simultaneously rotating in response to a rotation of the wheel **101** to generate energy at the generator **106** until the wheel **101** stops rotating.

The rotational inertia of the rollers **102** in the example embodiment of FIG. 6C and other examples herein can be changed for example increased or decreased. Increasing the rotational inertia of the rollers can cause more or less friction to be applied to the wheel **101** and also cause more or less energy to be generated at the generator **106**. For example, more energy would be required to rotate a roller **102** with a high rotational inertia than would be required to rotate a roller **102** with less rotational inertia. Thus, a roller **102** with high rotational inertia could more quickly decelerate the rotation of the wheel **101** while simultaneously causing more energy to be generated at the generator **106** than a roller with lower rotational inertia. For example, when acceleration or a constant speed of the vehicle is desired, the rotational inertia of the roller(s) **102** may be low to apply less friction to the wheel **101** (which may thereby cause less energy to be generated at the generator **106**) and when deceleration of the vehicle is desired (for example, stopping), the rotational inertia of the roller(s) **102** may be high to apply more friction to the wheel **101** (which may thereby cause more energy to be generated at the generator **106**). Thus, for any given desired mode of operation of the vehicle (for example, acceleration, deceleration) a maximum energy may be generated at the generator **106** by changing a rotational inertia of the rollers **102**.

In some implementations, the rotational inertia of the rollers **102** can change automatically for example in response to an energy demand of the motor of the vehicle, a rotational velocity of the wheel, and/or desired braking etc. In some implementations, the rotational inertia of the rollers can change in response to a manual user input. The rotational

12

inertia of the roller **102** can be changed by changing a state of the roller **102**, the shaft **104** (or other coupling device), and/or changing a state of the generator **106**.

FIG. 7A is a diagram illustrating an example embodiment of a roller. As shown in FIG. 2, the roller **102** may comprise a length **713** and a diameter **711**. The roller **102** may have any length **713** such as is required or desired. The roller **102** may have any diameter **711** such as is required or desired. The diameter **711** of the roller **102** may be less than the diameter of the wheel **101** such that the roller **102** rotates at a greater rotational velocity than the wheel **101** in response to a rotation of the wheel **101**. In some embodiments comprising multiple rollers, one, some or each of the multiple rollers may have a length and/or diameter that is different than the lengths and/or diameters of the other rollers.

FIG. 7B is a diagram illustrating an example embodiment of a roller. As shown in FIG. 7B, the roller **102** may comprise a diameter that varies in size along a length of the roller **102**. For example, one end of the roller **102** may comprise a diameter **711a** of a first size and the other end of the roller **102** may comprise a diameter **711b** of a second size that is different than the diameter **711a**. A roller **102** having a diameter that varies in size along a length of the roller **102** may facilitate the rotation of the roller **102** in response to a rotation of the wheel **101**.

FIG. 7C is a diagram illustrating an example embodiment of the apparatus **100**. As shown in FIG. 7C, the apparatus **100** may comprise a roller shaft **703** rotatably coupled to the roller **102** and the shaft **104**. The roller shaft **703** may not be in physical contact with the wheel **103**. The roller shaft **703** may be any length to allow the roller **102** to be in contact with a sidewall of the wheel **101** at any distance away from a center axis of the wheel **101**.

FIG. 8A is a diagram illustrating an example embodiment of the apparatus **100** implemented with multiple wheels of a vehicle. As shown in FIG. 8A, the apparatus **100** may comprise one or more first rollers **102a** rotatably couplable to a first wheel **101a** of a vehicle and one or more second rollers (not shown) rotatably couplable to a second wheel **101b** of the vehicle. The components of the example embodiment shown in FIG. 4A may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiment of FIG. 5A. For example, rotation of the one or more first rollers **102a** and/or rotation of the one or more second rollers (not shown) may cause the generator **106** to generate energy.

FIG. 8A is not meant to be limiting of the present disclosure. The apparatus **100** may comprise any number of rollers, shafts and/or generators as required and/or desired and may be implemented on any number of wheels of a vehicle as required or desired, for example on one, two, three or four wheels (for example with reference to implementation with a car) or 18 wheels (for example with reference to implementation with a semi-truck).

FIG. 8B is a diagram illustrating an example embodiment of the apparatus **100** implemented with multiple wheels of a vehicle and comprising multiple generators. As shown in FIG. 8B, the apparatus **100** may comprise a first and second generator **106a/106b**. The components of the example embodiment shown in FIG. 8B may comprise similar structural and/or operational features as described with reference to other embodiments described herein, for example, the example embodiment of FIG. 5A. For example, rotation of the one or more first rollers **102a** may cause the first generator **106a** to generate energy and rotation of the one or

13

more second rollers (not shown) may cause the generator **106b** to generate energy. The generators **106a/106b** may be in electrical communication with the vehicle and/or each other.

FIG. 9 is a diagram illustrating an example embodiment of the apparatus **100** implemented between two wheels. As shown in FIG. 9, the apparatus **100** may comprise a first roller **102a** and a second roller **102b** located between two wheels **101a/101b** such as two adjacent wheels on a truck, van, semi-truck, tractor-trailer and the like. The first roller **102a** may be in physical contact with a sidewall surface of the first wheel **101a** and configured to rotate in response to a rotation of the first wheel **101a**. The second roller **102b** may be in physical contact with a sidewall surface of the second wheel **101b** and configured to rotate in response to a rotation of the second wheel **101b**. The first and second rollers **102a/102b** may be coupled to each other (for example, rotatably coupled) via one or more coupling devices such as a shaft as shown in FIG. 9 and/or any other coupling device as required or desired such as gears, sprockets, chains, belts, pulleys and the like.

The first roller **102a** and/or second roller **102b** may be coupled (for example, rotatably coupled) to a shaft **104** for example via one or more coupling device such as a shaft as shown in FIG. 9 and/or any other coupling device as required or desired such as gears, sprockets, chains, belts, pulleys and the like. The shaft **104** may be configured to rotate in response to a rotation of the first roller **102a** and/or second roller **102b**. The shaft **104** may be operably coupled to a generator **106** and the generator **106** may be configured to generate energy (for example, electrical output) in response to a rotation of the shaft **104** as described elsewhere herein.

FIG. 9 is given as an example and is not meant to be limiting. In some embodiments, the apparatus **100** may comprise any number of rollers, for example one roller or more than two rollers. Furthermore, the rollers **102a/102b** shown in example FIG. 9 may be arranged with any orientation between their respective axes of rotation, as required or desired. For example, the respective axes of rotation of rollers **102a/102b** may be substantially parallel, as shown in FIG. 9, or may be substantially orthogonal or may be oriented in any other way as required or desired. Additionally, the rollers **102a/102b** may be configured to rotate independently of each other such that one roller may rotate while the other does not or may be configured to be fixedly rotatably coupled to each other such that one roller may not rotate without the other roller also rotating.

FIG. 10A is a diagram of an example embodiment of the apparatus **100** comprising various configurations of rollers and multiple generators. As shown in FIG. 10A, the apparatus **100** may comprise one or more first rollers **102a**, a first shaft **104a**, a first generator **106a** and one or more second rollers **102b**, a second shaft **104b** and a second generator **106b**. The example apparatus **100** of FIG. 10A and its various components may operate in a manner similar to that described in other example embodiments herein such as with reference to FIG. 1A and/or FIG. 5A, for example.

FIG. 10B is a diagram of an example embodiment of the apparatus **100** comprising various configurations of rollers and a single generator. As shown in FIG. 10B, the apparatus **100** may comprise one or more first rollers **102a**, a first shaft **104a**, a first sprocket **105a**, a first coupling device **107a**, a third shaft **108** and a generator **106**. The apparatus **100** may further comprise one or more second rollers **102b**, a second shaft **104b**, a second sprocket **105b**, and a second coupling device **107b**. The example apparatus **100** of FIG. 10B and its

14

various components may operate in a manner similar to that described in other example embodiments herein such as with reference to FIG. 1A, FIG. 5A, and/or FIG. 3B, for example. Example Energy Generation and Storage Systems

FIG. 11A is a diagram of two generators **106a** and **106b** configured to be mechanically coupled to roller(s) and that convert mechanical rotation of roller(s) **102** into electrical energy outputs, in accordance with an exemplary embodiment. In some embodiments, the generators **106a** and **106b** may be replaced with alternators or similar electricity generating devices. The generators **106a** and **106b** can be mechanically coupled to roller(s) via one or more of a shaft, linkage, gear, pulley, chain, belt, sprocket or other similar mechanism or device. The example embodiment of FIG. 11A illustrates the generator **106b** as mechanically coupled to roller(s) **102** via at least a chain **1101**. The chain **1101** may rotate, in response to rotation of the roller(s) **102**, causing a corresponding rotor of the generator **106b** to rotate and causing the generator **106b** to generate an electrical energy output via a cable (not shown in this figure). In some embodiments, the two generators **106a** and **106b** may be replaced by any number of generators **106**, from a single generator to many generators. In some embodiments, the generators **106** may generate AC electricity or DC electricity, depending on the application. When the generators **106** generate AC power, an AC-to-DC converter may be used to condition and convert the generated electricity for storage. When the generators **106** generate DC power, a DC-to-DC converter may be used to condition the generated electricity for storage.

FIG. 11B is an alternate view of the two generators **106a** and **106b** of FIG. 11A and cabling **1103a** and **1103b** that couples the generators **106a** and **106b** to a charger (for example, a battery and/or capacitor charger) and/or to an energy storage device such as a battery and/or capacitor. The charger may comprise one or more other components or circuits used to rectify or otherwise condition the electricity generated by the generators **106a** and **106b**. For example, the one or more other components or circuits may comprise one or more of a matching circuit, an inverter circuit, a conditioning circuit, a rectifying circuit, a conversion circuit, and so forth. The matching circuit may match conditions of a load to the source (for example, impedance matching, and so forth). The conversion circuit may comprise a circuit that converts an alternating current (AC) signal to a direct current (DC) signal, a DC/DC conversion circuit, a DC/AC conversion circuit and so forth. The conditioning circuit may condition a signal input into the conditioning circuit, and the rectifying circuit may rectify signals.

Additional details regarding FIGS. 11A-11B can be found in at least paragraphs [0080]-[0099] of U.S. patent application Ser. No. 17/332,824, which is hereby incorporated by reference.

FIG. 12 is a diagram of an example vehicle **1200** incorporating an apparatus **100** comprising roller(s) **102**, a generator **106**, as well as an energy storage device **1203** electrically coupled with the generator **106**. Energy generated at the generator **106**, in response to a rotation of the roller(s) **102** can be provided to the energy storage device **1203**. The energy storage device **1203** can comprise one or more batteries **1202** and/or one or more capacitor modules **1204**. The energy storage device **1203** may comprise the one or more capacitor modules **1204** as a supplemental and/or intermediate energy storage device. In some embodiments, the capacitor modules **1204** are disposed alongside the one or more batteries **1202**. The capacitor modules **1204** and the

15

battery 1202 can be electrically coupled to at least a motor of the vehicle, such as an electric motor.

In some embodiments, the capacitor modules 1204 may be used in combination with the battery 1202. For example, as shown in FIG. 12, the vehicle 1200 may include one or more the capacitor modules 1204 installed alongside the battery 1202. In some embodiments, the vehicle 1200 includes a plurality of capacitor modules 1204. In some embodiments, one or more batteries 1202 are replaced with one or more capacitor modules 1204. As shown, the capacitor modules 1204 may be connected in series or in parallel with the battery 1202, dependent on the use case. For example, the capacitor modules 1204 may be connected in series or parallel with the battery 1202 when supplementing the voltage in the battery 1202 or when charging the battery 1202 and/or the capacitor modules 1204. Therefore, the battery 1202 and the capacitor modules 1204 may provide voltage support to each other. As such, the capacitor modules 1204 may provide supplemental energy when the battery 1202 are discharged or be used in place of the battery 1202 altogether.

In some embodiments, the energy storage device 1203 may comprise one or more hypercapacitors. FIG. 13 schematically illustrates a diagram of an example embodiment of a hypercapacitor 1302 for storing energy (for example, such as may be used in an electric vehicle), which may also be referred to as a hypercapacitor energy storage system or device. As shown, the hypercapacitor 1302 may comprise or consist essentially of an ultracapacitor portion 1304, an energy retainer portion 1306, one or more inbound diodes 1308, and one or more outbound diodes 1310. In some embodiments, the hypercapacitor 1302 may not comprise the inbound diode 1308 and/or the outbound diode 1310.

The ultracapacitor portion 1304 may be electrically coupled to the energy retainer portion 1306 and in some embodiments, together may comprise a single integrated unit or package (for example, the hypercapacitor 1302). The ultracapacitor portion 1304 may provide energy to the energy retainer portion 1306 as the energy in the energy retainer portion 1306 is depleted (for example resulting from an energy demand at a load).

The electrical connection between the ultracapacitor portion 1304 and the energy retainer portion 1306 may stabilize the voltage levels of the ultracapacitor portion 1304 and prevent self-discharge as the energy retainer portion 1306 retains energy provided from the ultracapacitor portion 1304 via their electrical connection. Advantageously, stabilizing the voltage levels in the ultracapacitor portion 1304 by reducing and/or substantially eliminating self-discharge provides a superior energy device capable of storing energy (for example, maintaining high voltage levels) for much longer than existing energy devices in widespread use today.

The ultracapacitor portion 1304 of the hypercapacitor 1302 may comprise one or more ultracapacitors and/or supercapacitors. The ultracapacitor portion 1304 may incorporate structural and operational features described in connection with any of the embodiments of the capacitor module 1204 described herein.

The energy retainer portion 1306 may comprise a device or multiple devices capable of storing energy such as a battery, a battery field and/or a capacitor. For example, in some embodiments the energy retainer portion 1306 may include a battery such as the battery 1202 described herein and may incorporate structural and operational features of the battery 1202. In some embodiments, the energy retainer portion 1306 may include a battery field such as a battery field comprising batteries 1202 such as shown in FIG. 12. In

16

some embodiments, the energy retainer portion 1306 may comprise one or more capacitors, such as the capacitor module 1204 described herein.

Additional details regarding FIG. 13 can be found in at least paragraphs [0211]-[0246] of U.S. patent application Ser. No. 17/332,824, which is hereby incorporated by reference.

FIG. 14A illustrates an example farm equipment such as a tractor that may incorporate the various components and systems discussed herein such as the apparatus 100, which may comprise a generator 106 and one or more rollers 102 rotatably couplable to a wheel of the vehicle, as well as a motor 1401, such as an electric motor, and an energy storage device 1203 which may comprise a capacitor 1204, a battery 1202 and/or a hypercapacitor 1302, as discussed herein. The generator 106 may be electrically coupled to the energy storage device 1203 and may be capable of providing energy to the energy storage device 1203, as discussed herein. The energy storage device 1203 may be electrically coupled to the motor 1401 and may be capable of providing energy to the motor 1401.

FIG. 14B illustrates an example construction equipment that may incorporate the various components and systems discussed herein such as the apparatus 100, which may comprise a generator 106 and one or more rollers 102 rotatably couplable to a wheel of the vehicle, as well as a motor 1401, such as an electric motor, and an energy storage device 1203 which may comprise a capacitor 1204, a battery 1202 and/or a hypercapacitor 1302, as discussed herein. The generator 106 may be electrically coupled to the energy storage device 1203 and may be capable of providing energy to the energy storage device 1203, as discussed herein. The energy storage device 1203 may be electrically coupled to the motor 1401 and may be capable of providing energy to the motor 1401.

FIG. 14C illustrates an example commercial vehicle such as a tractor-trailer or semi-truck that may incorporate the various components and systems discussed herein such as the apparatus 100, which may comprise a generator 106 and one or more rollers 102 rotatably couplable to a wheel of the vehicle, as well as a motor 1401, such as an electric motor, and an energy storage device 1203 which may comprise a capacitor 1204, a battery 1202 and/or a hypercapacitor 1302, as discussed herein. The generator 106 may be electrically coupled to the energy storage device 1203 and may be capable of providing energy to the energy storage device 1203, as discussed herein. The energy storage device 1203 may be electrically coupled to the motor 1401 and may be capable of providing energy to the motor 1401.

FIG. 14D illustrates an example bus that may incorporate the various components and systems discussed herein such as the apparatus 100, which may comprise a generator 106 and one or more rollers 102 rotatably couplable to a wheel of the vehicle, as well as a motor 1401, such as an electric motor, and an energy storage device 1203 which may comprise a capacitor 1204, a battery 1202 and/or a hypercapacitor 1302, as discussed herein. The generator 106 may be electrically coupled to the energy storage device 1203 and may be capable of providing energy to the energy storage device 1203, as discussed herein. The energy storage device 1203 may be electrically coupled to the motor 1401 and may be capable of providing energy to the motor 1401.

FIG. 14E illustrates an example train that may incorporate the various components and systems discussed herein such as the apparatus 100, which may comprise a generator 106 and one or more rollers 102 rotatably couplable to a wheel of the vehicle, as well as a motor 1401, such as an electric

17

motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **14F** illustrates an example bicycle that may incorporate the various components and systems discussed herein such as the apparatus **100**, which may comprise a generator **106** and one or more rollers **102** rotatably couplable to a wheel of the vehicle, as well as a motor **1401**, such as an electric motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **14G** illustrates an example scooter that may incorporate the various components and systems discussed herein such as the apparatus **100**, which may comprise a generator **106** and one or more rollers **102** rotatably couplable to a wheel of the vehicle, as well as a motor **1401**, such as an electric motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **14H** illustrates an example tram that may incorporate the various components and systems discussed herein such as the apparatus **100**, which may comprise a generator **106** and one or more rollers **102** rotatably couplable to a wheel of the vehicle, as well as a motor **1401**, such as an electric motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **14I** illustrates an example cart such as a golf cart that may incorporate the various components and systems discussed herein such as the apparatus **100**, which may comprise a generator **106** and one or more rollers **102** rotatably couplable to a wheel of the vehicle, as well as a motor **1401**, such as an electric motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **14J** illustrates an example motorcycle that may incorporate the various components and systems discussed herein such as the apparatus **100**, which may comprise a generator **106** and one or more rollers **102** rotatably couplable to a wheel of the vehicle, as well as a motor **1401**, such as an electric motor, and an energy storage device **1203** which may comprise a capacitor **1204**, a battery **1202** and/or

18

a hypercapacitor **1302**, as discussed herein. The generator **106** may be electrically coupled to the energy storage device **1203** and may be capable of providing energy to the energy storage device **1203**, as discussed herein. The energy storage device **1203** may be electrically coupled to the motor **1401** and may be capable of providing energy to the motor **1401**.

FIG. **15A** is a schematic diagram illustrating an example implementation of an energy generation system **1500**, also referred to herein as “system”. The system **1500** can include one or more of an energy recovery mechanism **1501**, a controller **1503**, a gearbox **1505**, a generator **1507**, and an energy storage **1508** such as a capacitor **1509** and/or a battery **1511**. Figure FIGS. **15A-C** are provided as an examples and are not intended to be limiting. In some implementations, the system **1500** may include less than each of the components shown and/or described in FIG. **15A**. For example, FIG. **15B** shows an energy generation system **1500** including one or more of a energy recovery mechanism **1501**, a controller **1503**, a generator **1507**, and an energy storage **1508**. Further, FIG. **15C** shows an energy regeneration system **1500** including one or more of the energy recovery mechanism **1501**, the controller **1503**, and the energy storage **1508**. In some implementations, the system **1500** may include a plurality of energy recovery mechanisms **1501**, a plurality of generators **1507**, a plurality of capacitors **1509**, and/or a plurality of batteries **1511**.

The system **1500** may be disposed within a vehicle **1510**. The system **1500** may be mobile with the vehicle **1510**. The system **1500** may generate energy in response to a movement or motion of the vehicle **1510** and/or a movement or motion of components of the vehicle **1510** (for example, a drive shaft, axle, rotor). The system **1500** may generate energy when the vehicle **1510** is stationary. The system **1500** may generate energy to provide to the vehicle **1510**. The system **1500** may provide energy to the vehicle **1510** to cause the vehicle **1510** to move. The vehicle **1510** may be configured to travel on a ground surface. The vehicle **1510** may be configured to travel on a water surface. The vehicle **1510** may be configured to travel through air. The vehicle **1510** may be, and/or incorporate features of, any of the example vehicle shown and/or discussed herein, such as with references to FIGS. **14A-15J**.

The vehicle **1510** can include one or more wheels **1512** (for example, wheel **1512A**, wheel **1512B**). The vehicle **1510** can include a drive shaft **1514**. The vehicle can include a motor **1516**. The motor **1516** may be a traction motor. The motor **1516** may provide locomotive power to the vehicle **1510** to cause the vehicle **1510** to move. The motor **1516** may be an electric motor. The motor **1516** may generate mechanical energy in response to electrical energy. The motor **1516** may be coupled with the drive shaft **1514**. The motor **1516** may cause the drive shaft **1514** to rotate. The drive shaft **1514** may be coupled with the wheels **1512**. The drive shaft **1514** may cause the wheels **1512** to rotate.

The energy recovery mechanism **1501** may be rotatably coupled with the drive shaft **1514** or another rotating component of the vehicle. The energy recovery mechanism **1501** may include one or more rollers that rotates in response to a rotation of the drive shaft **1514**. The generator **1507** may be rotatably coupled to the energy recovery mechanism **1501**. The generator **1507** may generate an electrical output in response to a rotation of the one or more rollers of the energy recovery mechanism **1501**. The generator **1507** may be coupled to the energy recovery mechanism **1501** via one or more of a flexible arm, shaft, rod, axle, gear, pulley, chain, or the like. The one or more rollers of the energy recovery mechanism **1501** may include similar structural and/or

19

operational features as any of the other example rollers shown and/or discussed herein (for example, FIGS. 1-10B).

The energy recovery mechanism 1501 may be positioned adjacent to, but not in contact with, the drive shaft 1514. In some embodiments, the drive shaft 1514 may be magnetic such that it includes a magnetic field. In some embodiments, the drive shaft 1514 may induce a current to flow in the energy recovery mechanism 1501. In some embodiments, the current may be converted to voltage and stored in the energy storage 1508.

The energy recovery mechanism 1501 may be optionally coupled with the generator 1507 via a gearbox 1505 (for example, FIG. 15C). The gearbox 1505 may include one or more gears which may be one or more sizes. The gearbox 1505 can be coupled to the energy recovery mechanism 1501 and to generator 1507. The gearbox 1505 can adjust a rotational velocity of a rotatable component of the generator 1507 relative to a rotational velocity of a rotatable component of the one or more rollers. For example, the one or more rollers may be rotatably coupled to a first gear of the gearbox 1505 and the generator 1507 may be rotatably coupled to a second gear of the gearbox 1505. The first and second gears of the gearbox 1505 may be rotatably coupled. The first and second gears may be different sizes, including having different diameters, such that rotation of the first gear at a first angular velocity causes rotation of the second gear at a second angular velocity. The gearbox 1505 can change a ratio of angular velocity between the one or more rollers and a rotational component of the generator 1507 by changing a gear to which the one or more rollers and/or generator 1507 is rotatably coupled. Changing a ratio of angular velocity between the one or more rollers and a rotatable component of the generator 1507 may change a rate at which the generator 1507 generates an electrical output such as for a given angular velocity of the one or more rollers. The gearbox 1505 can adjust the ratio of rotation, such as by changing the gear to which the one or more rollers and the generator 1507 is rotatably coupled, according to user input and/or according to operational settings, according to any of the examples discussed herein.

In some implementations, the gearbox 1505 may rotatably decouple the one or more rollers from the generator 1507 such that rotation of the one or more rollers does not cause the generator 1507 to generate energy. For example, the gearbox 1505 may transition the one or more rollers between an engaged state and a disengaged state by rotatably decoupling the one or more rollers from the generator 1507. The generator 1507 may not generate energy in response to a rotation of the one or more rollers in the disengaged state. In some implementations, the term “engaged state” and the term “extended state” may be used interchangeably. In some implementations, the term “disengaged state” and the term “retracted state” may be used interchangeably.

The generator 1507 may be electrically coupled with the capacitor 1509. The generator 1507 may be electrically coupled with the capacitor 1509 via one or more switches or relays. The generator 1507 may be electrically coupled with the battery 1511 via one or more diodes biased toward the capacitor 1509. The generator 1507 may disperse energy to the capacitor 1509. The capacitor 1509 may include one or more capacitors. The capacitor 1509 may include one or more ultracapacitors, supercapacitors, or the like. The capacitor 1509 may be configured to store up to 10V, 50V, 100V, 200V, 300V, 400V, or the like.

The generator 1507 may be electrically coupled with the battery 1511. The generator 1507 may be electrically coupled with the battery 1511 via one or more switches or

20

relays. The generator 1507 may be electrically coupled with the battery 1511 via one or more diodes biased toward the battery 1511. The generator 1507 may disperse energy to the battery 1511. The battery 1511 may include one or more batteries. The battery 1511 may include one or more lithium based batteries. The battery 1511 may not include lithium. The battery 1511 may be configured to store up to 10V, 50V, 100V, 200V, 300V, 400V, or the like.

The generator 1507 may be electrically coupled with the motor 1516. The generator 1507 may be electrically coupled with the motor 1516 via one or more switches or relays. The generator 1507 may be electrically coupled with the motor 1516 via one or more diodes biased toward the motor 1516. The generator 1507 may disperse energy to the motor 1516.

The capacitor 1509 may be electrically coupled with the battery 1511. The capacitor 1509 may be electrically coupled with the battery 1511 via one or more diodes. The diodes may be biased toward the battery 1511 and may prevent a flow of energy from the battery 1511 to the capacitor 1509. The one or more diodes may allow energy to flow from the capacitor 1509 to the battery 1511. The capacitor 1509 may be electrically coupled with the battery 1511 via one or more switches or relays. The capacitor 1509 may disperse energy to the battery 1511. The battery 1511 may receive energy from the capacitor 1509. In some implementations, the capacitor 1509 and/or the battery 1511 may include similar structural and/or operational features as the hypercapacitor 1302 shown and/or discussed herein.

The capacitor 1509 may be electrically coupled with the motor 1516. The capacitor 1509 may be electrically coupled with the motor 1516 via one or more switches or relays. The capacitor 1509 may be electrically coupled with the motor 1516 via one or more diodes biased toward the motor 1516. The capacitor 1509 may disperse energy to the motor 1516. The motor 1516 may receive energy from the capacitor 1509.

The battery 1511 may be electrically coupled with the motor 1516. The battery 1511 may be electrically coupled with the motor 1516 via one or more switches or relays. The battery 1511 may be electrically coupled with the motor 1516 via one or more diodes biased toward the motor 1516. The battery 1511 may disperse energy to the motor 1516. The motor 1516 may receive energy from the battery 1511.

The controller 1503 may include one or more hardware processors configured to execute program instructions to cause the controller 1503 or other components of the system 1500 and/or vehicle 1510 to perform one or more operations. The controller 1503 may be in electrical communication with one or more components. The controller 1503 may receive data from one or more components. The controller 1503 may transmit instructions to one or more components. The controller 1503 may be in electrical communication with the energy recovery mechanism 1501. The controller 1503 may receive data relating to an operation of the energy recovery mechanism 1501. Such data may include a rotational velocity and/or acceleration of the roller or a magnitude of a voltage or current induced in the energy recovery mechanism 1501. The controller 1503 may issue instructions, such as to an actuator, to cause the energy recovery mechanism 1501 to transition between the engaged state and the disengaged state. The controller 1503 may be in communication with the gearbox 1505. The controller 1503 may receive data relating to an operation of the gearbox 1505. The controller 1503 may issue instructions to cause the gearbox 1505 to perform one or more operations such as changing a gear to which the roller and/or the generator 1507 is coupled to change a ratio of rotational velocity between

21

the roller and the generator 1507. The controller 1503 may be in communication with the generator 1507. The controller 1503 may receive data relating to an operation of the generator 1507 such as an amount of energy generated by the generator 1507 and/or a rate of energy generated by the generator 1507. The controller 1503 may issue instructions to cause the generator 1507 to perform one or more operations. The controller 1503 may be in communication with the capacitor 1509. The controller 1503 may receive data relating to an operation of the capacitor 1509 such as an amount of energy stored in the capacitor 1509, a voltage of the capacitor 1509, a rate of charge or discharge of the capacitor 1509, or the like. The controller 1503 may be in communication with the battery 1511. The controller 1503 may receive data relating to an operation of the battery 1511 such as an amount of energy stored in the battery 1511, a voltage of the battery 1511, a rate of charge or discharge of the battery 1511, or the like. The controller 1503 may be in communication with the motor 1516. The controller 1503 may receive data relating to an operation of the motor 1516 such as an energy consumed by the motor 1516 or the like. The controller 1503 may be in communication with the wheels 1512. The controller 1503 may receive data relating to one or more operations of the wheels 1512 such as an angular velocity or acceleration of the wheels 1512.

The controller 1503 may control whether the energy recovery mechanism 1501 is the engaged state or in the disengaged state. According to some embodiments, in the engaged state, the energy recovery mechanism 1501 may be in physical contact with the drive shaft 1514. According to some embodiments, in the engaged state, the energy recovery mechanism 1501 is not in physical contact with the drive shaft 1514. According to some embodiments, in the engaged state, the roller of the energy recovery mechanism 1501 may be rotatably coupled to the drive shaft 1514 and may rotate in response to a rotation of the drive shaft 1514. In some implementations, in the disengaged state, the roller of the energy recovery mechanism 1501 may not be in physical contact with the drive shaft 1514 such that rotation of the drive shaft 1514 does not cause the roller to rotate. In some implementations, in the disengaged state, the roller may be in physical contact with the drive shaft 1514 such that rotation of the drive shaft 1514 causes the roller to rotate but the energy recovery mechanism 1501 may not be rotatably coupled to the generator 1507 such that rotation of the roller does not cause the generator 1507 to generate an electrical output. In some embodiments, the energy recovery mechanism 1501 may switch from an engaged state to a disengaged state rapidly by moving the energy recovery mechanism 1501 to and from contact with the drive shaft 1514. The controller 1503 may cause the energy recovery mechanism 1501 to transition between the engaged and disengaged states in response to one or more of a velocity or acceleration of the vehicle 1510, angular velocity or acceleration of the wheels 1512, an amount of energy or rate of energy generated by the generator 1507, an energy demand required by the motor 1516, an amount of energy stored in the capacitor 1509, an amount of energy stored in the battery 1511, a rate of discharge of the capacitor 1509, a rate of discharge of the battery 1511, a user input, or the like.

The controller 1503 may control the ratio of angular velocity between the energy recovery mechanism 1501 and a rotational member of the generator 1507. The controller 1503 may adjust the ratio of angular velocity between the energy recovery mechanism 1501 and the generator 1507 to change a rate at which the generator 1507 generates energy. The controller 1503 may adjust the ratio of angular velocity

22

between the energy recovery mechanism 1501 and the generator 1507 to change a torque applied by the roller of the energy recovery mechanism 1501 on the drive shaft 1514. As an example, increasing a torque applied by the roller on the drive shaft 1514 may cause the drive shaft 1514 to decrease angular velocity which may be desirable such as when decelerating the vehicle. The controller 1503 may control the ratio of angular velocity between the roller and the generator 1507 in response to one or more of a velocity or acceleration of the vehicle 1510, angular velocity or acceleration of the wheels 1512, an amount of energy or rate of energy generated by the generator 1507, an energy demand required by the motor 1516, an amount of energy stored in the capacitor 1509, an amount of energy stored in the battery 1511, a rate of discharge of the capacitor 1509, a rate of discharge of the battery 1511, a user input, or the like.

The controller 1503 may control the force exerted by the energy recovery mechanism 1501 on the drive shaft 1514. The controller 1503 may adjust the force exerted by the energy recovery mechanism 1501 on the drive shaft 1514 to change a rate at which the generator 1507 generates energy. The controller 1503 may adjust the force exerted by the energy recovery mechanism 1501 on the drive shaft 1514 to change a torque applied by the roller on the drive shaft 1514. As an example, increasing a torque applied by the roller on the drive shaft 1514 may cause the drive shaft 1514 to decrease angular velocity which may be desirable such as when decelerating the vehicle. The controller 1503 may control the force exerted by the energy recovery mechanism 1501 on the drive shaft 1514 in response to one or more of a velocity or acceleration of the vehicle 1510, angular velocity or acceleration of the wheels 1512, an amount of energy or rate of energy generated by the generator 1507, an energy demand required by the motor 1516, an amount of energy stored in the capacitor 1509, an amount of energy stored in the battery 1511, a rate of discharge of the capacitor 1509, a rate of discharge of the battery 1511, a user input, or the like.

The controller 1503 may control a flow of energy in the system 1500. The controller 1503 can control whether energy generated at the generator 1507 or the energy recovery mechanism 1501 is dispersed to the capacitor 1509, to the battery 1511, and/or to motor 1516. As an example, the controller 1503 may electrically couple and/or decouple the generator 1507 or energy recovery mechanism 1501 from any of the capacitor 1509, the battery 1511, and/or the motor 1516. As another example, the controller 1503 may electrically couple and/or decouple the capacitor 1509 from the motor 1516, may electrically couple and/or decouple the capacitor 1509 from the battery 1511, and/or may electrically couple and/or decouple the battery 1511 from the motor 1516. In some implementations, the controller 1503 may electrically couple and decouple various electrical components by controlling whether switches, relays, or the like that are disposed between electrical components are in an open or a closed state to control whether energy may flow between the electrical components. As an example, the controller 1503 may control whether one or more switches electrically coupled with the generator 1507 conducts energy or prevents energy from passing from the generator 1507 to the capacitor 1509, the battery 1511, and/or the motor 1516.

In an example embodiment, a driver of a vehicle may receive a signal from the controller 1503 when at least a portion of the energy storage 1508 is full. The driver may send a signal to the controller 1503 to transfer the energy

within the energy storage **1508** to the motor **1516**. As the energy is transferred to the motor **1516**, the energy recovery mechanism **1501** is in a disengaged state. The driver may send a signal to the controller **1503** to stop the transfer of energy within the energy storage **1508** to the motor **1516**. When the energy transfer is stopped, the energy recovery mechanism **1501** may automatically switch to an engaged state.

In some embodiments, a flexible arm attaches the energy recovery mechanism **1501** to a feature of the vehicle. In some embodiments, a shaft, rod, actuator or other connecting mechanical feature may be used instead of the flexible arm as described herein. The feature of the vehicle may include the vehicle housing, a structural component located within the vehicle, the gearbox **1505**, generator **1507**, or the energy storage **1508** (for example, one or both of the capacitor **1509** or battery **1511**). In some embodiments, the flexible arm may be rotatably coupled with the energy recovery mechanism **1501** and any of the above-mentioned features. In some embodiments, the flexible arm is attached to the **1501** via a joint. In some embodiments the joint may be rigidly fixed. In other embodiments, the joint may be a rotatable coupling. In other embodiments, the joint may be a ball and socket joint that allows for 360-degree rotation of the roller housing relative to the flexible arm. In other embodiments, the joint may be a universal joint that transmits motion and power from the roller of the energy recovery mechanism **1501**.

In some embodiments, the flexible arm exerts a downward force on the energy recovery mechanism **1501**. In some embodiments, the flexible arm exerts a downward force only in an engaged state. In some embodiments, the flexible arm exerts a downward force in both the engaged state and the disengaged state. In some embodiments, the downward force ensures that the energy recovery mechanism **1501** maintains constant contact with the drive shaft **1514**. Further, the flexible arm may be configured to accommodate any oscillation of the drive shaft **1514** including vertical or horizontal movement. This allows the flexible arm to flex, bend, or move while maintaining the downward force. In some embodiments, the flexible arm is configured to pivot about one or more joints.

The term “downward force,” is used to signify a force vector, comprising both magnitude and direction, that is normal to, or tangentially adjacent to, a plane of contact between the roller and the drive shaft **1514**. Further, the magnitude of the force vector may be adjustable while the system or apparatus **1500** is in the engaged state. For example, the controller **1503** may control the amount of force exerted on the drive shaft **1514** by the flexible arm. This may be done automatically, or via user command. The controller **1503** may control the magnitude of the force in response to one or more of a velocity or acceleration of the vehicle **1510**, angular velocity or acceleration of the wheels **1512**, an amount of energy or rate of energy generated by the generator **1507**, an energy demand required by the motor **1516**, an amount of energy stored in the capacitor **1509**, an amount of energy stored in the battery **1511**, a rate of discharge of the capacitor **1509**, a rate of discharge of the battery **1511**, a user input, or the like.

Frictional force is a function of the normal force between two objects. Thus, an increase in the downward force on the roller results in an increase in the amount of friction between the roller and the drive shaft **1514**. An increase in friction between the roller and the drive shaft **1514** results in greater contact and a decrease in slippage which in turn results in a greater conversion of rotational movement from the drive

shaft **1514** to the energy recovery mechanism **1501** via the roller. Thus, the downward force serves to increase the overall energy efficiency of the system.

The flexible arm may be made from a variety of materials, such as but not limited to, metals, polymers, or fiber-based materials. The flexible arm may be comprised of a single component or a combination thereof. These components may bend or flex to exert a downward force and accommodate any movement between the feature of the vehicle and the drive shaft **1514**. Additionally, the flexible arm may be made up of a combination of components made of rigid material that are coupled together via joints or hinges such that the arm, as a whole, flexes. In some embodiments, the flexible arm is positioned parallel to the rotational axis of the drive shaft **1514**. In some embodiments, the shape of the flexible arm changes as the force exerted on the drive shaft **1514** increases. In some embodiments, a spring or actuator is coupled to the flexible arm and applies to cause or increase the downward force.

In some embodiments, a surface of the one or more rollers and/or a surface of the drive shaft **1514** which contacts the one or more rollers, may include a material and/or texture designed to increase friction. In some embodiments, the material may include silicon, rubber, polymer or other composite with a high coefficient of friction. In some embodiments, this material may have a coefficient of friction that is higher than the surface of the drive shaft **1514** which does not contact the one or more rollers. Frictional force is a function of the coefficient of friction of the surface of an object. Thus, an increased coefficient of friction of either the surface of the one or more rollers or the drive shaft **1514** results in an increase in the amount of friction between those surfaces, minimize the amount of slippage that occurs between those surfaces and results in an increase to the overall energy efficiency of the system. Further, the material may be high in hardness or resistant to erosion and thus prolong the life of the drive shaft and/or rollers. In some embodiments, the texture includes bumps, ridges, and other irregularities to increase the roughness and friction. In some embodiments, the surface of the one or more rollers and drive shaft include interlocking teeth similar to a gear. In some embodiments, the flexible arm, energy recovery mechanism **1501**, or parts of the energy recovery mechanism **1501** are configured to be rapidly replaceable such as during a pit stop of a racing event.

FIG. **16** illustrates a cross section view of an example drive shaft **1614** and rollers **1601** (for example, rollers **1601A**, **1601B**, **1601C**, **1601D**). The rollers **1601** may have a smaller diameter than the drive shaft **1614**. In some implementations, rollers **1601** may have a larger diameter than the drive shaft **1614**. In some implementations, one or more of the rollers **1601** may have a different diameter than one or more of the other rollers **1601**. In some implementations, each of the rollers **1601** may have the same diameter. The drive shaft **1614** may be encompassed by a plurality of rollers **1601**. In the example shown, four rollers **1601** contact the drive shaft **1614** at various locations around a circumference of the drive shaft **1614**. In some implementations, less than four rollers **1601** may encompass the drive shaft **1614**. In some implementations, more than four rollers **1601** may encompass the drive shaft **1614**. In some implementations, the rollers **1601** may encompass the drive shaft **1614** at non-symmetrical locations around the drive shaft **1614**. In some implementations, the rollers **1601** may encompass the drive shaft **1614** at symmetrical locations around the drive shaft **1614**. The rollers **1601** may rotate at

a different angular velocity than the drive shaft **1614**. The rollers **1601** may rotate as a same angular velocity as the drive shaft **1614**.

As shown in FIG. **16**, the rollers **1601** can include a smooth outer surface. In some implementations, a surface of the rollers **1601** may include texture such as gears, notches, grooves, or the like which may increase a friction between the drive shaft **1614** and the rollers **1601** to facilitate a rotational response of the rollers **1601** due to rotation of the drive shaft **1614**.

FIG. **17** is a perspective view of an example drive shaft **1714** and energy recovery mechanism **1702**. The energy recovery mechanism **1702** includes one or more rollers **1701** (for example, roller **1701A**, **1701B**, **1701C**) disposed within a housing **1720**. The one or more rollers **1701** may each resemble a cylinder with a length that is substantially a length of the housing **1720**. In some embodiments, the length of the one or more rollers **1701** may be less than the length of the housing **1720**. The housing **1720** may encompass the drive shaft **1714**. In some implementations, the housing **1720** may encompass an entire circumference of the drive shaft **1714**. In some implementations, the housing **1720** may encompass less than an entire circumference of the drive shaft **1714**. The housing **1720** may house three rollers **1701**. The housing **1720** may house less than three rollers **1701**. The housing **1720** may house more than three rollers **1701**. A portion of the rollers **1701** may be exposed to an exterior surface of the housing **1720** to be coupled with a generator. The rollers **1701** may rotate relative to the housing **1720**. The drive shaft **1714** may rotate relative to the housing **1720**. The one or more rollers **1701** may be rotatable coupled to the housing **1720**.

In some implementations, the housing **1720** may house a generator rotatably coupled to one or more of the rollers **1701** and configured to generate energy in response to a rotation of the roller(s) **1701**. In some implementations, a cable may electrically connect the energy recovery mechanism **1702** to an energy storage of the system. In some implementations, a shaft or flexible arm may connect the energy recovery mechanism **1702** to a feature of the vehicle.

FIG. **18** illustrates a cross section view of an example drive shaft **1814** and energy recovery mechanism **1802**. The energy recovery mechanism **1802** includes a housing **1820** and one or more rollers **1801** (for example, roller **1801A**, **1801B**, **1801C**, **1801D**). The housing **1820** may ensure that the rollers **1801** remain in contact with the drive shaft **1814** as desired. The housing **1820** may apply a force to the rollers **1801** to ensure that the rollers **1801** remain in contact with the drive shaft **1814**. The housing **1820** may not contact the drive shaft **1814**. The housing **1820** may not rotate. In some implementations, the housing **1820** may change diameter to change a force applied to the rollers **1801** on the drive shaft **1814**. In some implementations, the housing **1820** may change a diameter to remove the rollers **1801** from contacting the drive shaft **1814**, such as in a disengaged state. In some implementations, the rollers **1801** may remain in contact with the drive shaft **1814** in a disengaged state.

FIG. **19** illustrates a cross section view of an example drive shaft **1914** and energy recover mechanism **1902**. The energy recovery mechanism **1902** includes an housing **1920** and one or more rollers **1901** (for example, roller **1901A**, **1901B**, **1901C**, **1901D**). The housing **1920** may cover a portion of the drive shaft **1914**. The housing **1920** may encompass less than an entire circumference of the drive shaft **1914**. In some implementations, multiple energy recover mechanisms **1902** may encompass the drive shaft **1914**. For example, multiple energy recover mechanism

1902 may encompass various portions around a circumference of the drive shaft **1914** at a same length of the drive shaft **1914**.

The housing **1920** may cause the rollers **1901** to contact the drive shaft **1914**. The housing **1920** may cause the rollers **1901** to remove from contacting the drive shaft **1914**. The housing **1920** may transition between causing the rollers **1901** to contact the drive shaft **1914** and causing the rollers **1901** to not contact the drive shaft **1914**. For example, a flexible arm, lever, or other actuator may physically move the energy recover mechanism **1902** such that the rollers **1901** contact the drive shaft **1914** or do not contact the drive shaft **1914**. In some implementations, a flexible arm, lever, or other actuator may apply a force to the energy recover mechanism **1902** to change a force applied by the rollers **1901** on the drive shaft **1914**.

FIG. **20A** illustrates a sideview of an energy recovery mechanism **2002** and a drive shaft **2014**. FIG. **20B** illustrates a sideview of the energy recovery mechanism **2002** and a drive shaft **2015** with a groove.

The energy recovery mechanism **2002** includes a housing **2020** and one or more rollers **2001** (for example **2001A**, **2001B**, **2001C**, **2001D**, **2001E**, **2001F**). The one or more rollers **2001** are rotationally aligned and positioned tangentially adjacent to the drive shaft **2014**. The one or more rollers **2001A-C** are coupled to a roller axel **2012A**. The one or more rollers **2001D-F** are coupled to a roller axel **2012B**. Each roller axel **2012A, B** is attached to the housing **2020**. The roller axels **2012A, B** help to keep the one or more rollers aligned. In some embodiments, the roller axels **2012A, B** are rotationally coupled to a flexible arm, shaft, gear, pulley etc. to transmit the rotational energy received from the drive shaft **2014** to a generator, gear box, or energy storage.

Drive shaft **2015** of FIG. **20B** includes one or more grooves **2024** positioned circumferentially along the shaft. The one or more grooves **2024** are spaced apart to match a spacing of the one or more rollers **2001**. The one or more grooves **2024** are positioned to accommodate each of the one or more rollers **2001**. The point of contact between each one or more rollers **2001** may be at the one or more grooves **2024**. Each of the one or more grooves **2024** may have a width that is substantially the same as the width of the one or more rollers **2001**. The cross-sectional shape of the one or more grooves **2024** may mirror that of the one or more rollers **2001** to maximize contact. The depth of the one or more grooves **2024** may vary depending on the diameter of the one or more rollers **2001** and the thickness of the drive shaft **2015**. The one or more grooves **2024** may operate to guide the one or more rollers **2001** as the drive shaft **2015** rotates. Further, the one or more grooves **2024** provides an increase in contact area between the one or more rollers **2001** and the drive shaft **2015**.

FIG. **21A** illustrates a sideview of an energy recovery mechanism **2102** in contact with a drive shaft **2114**. FIG. **21B** illustrates a sideview of the energy recovery mechanism **2102** not in contact with the drive shaft **2114**. The energy recovery mechanism **2102** includes one or more rollers **2101** (for example **2101A**, **2101B**, **2101C**, **2101D**) and a housing **2120**. A joint **2113** is attached to a backside of the housing **2120** and couples the energy recovery mechanism **2102** to a flexible arm **2117**. The flexible arm **2117** moves the energy recovery mechanism **2102** between the contact and noncontact positions. In some embodiments, the contact position is an engaged state. In some embodiments, the noncontact position is the disengaged state. In some embodiments, the

27

flexible arm **2117** is rotatably and/or pivotally coupled to the joint **2113** which is rotatably coupled to the one or more rollers **2101**.

FIG. **22** is a top-down illustration of a system **2202** with multiple energy recovery mechanisms positioned adjacent a drive shaft **2214**. A first energy recovery mechanism **2202A** is positioned on one side of the drive shaft **2214** and attached to a first flexible arm **2217**. A second energy recovery mechanism **2202B** is positioned on another side of the drive shaft **2214** and attached to a second flexible arm **2217B**. The first energy recovery mechanism **2202A** is laterally spaced apart from the second energy recovery mechanism **2202B**. Both energy recovery mechanisms are in an engaged state with the drive shaft **2214**. In some embodiments, the energy recovery mechanisms may be positioned such that they are not laterally spaced apart. In some embodiments, more than two energy recovery mechanisms may engage the drive shaft **2214**.

FIG. **23A** illustrates an embodiment with a cut-away view of an energy recovery mechanism **2302** engaged with a drive shaft **2314A**. The energy recovery mechanism **2302** includes one or more rollers **2301** (for example **2301A**, **2301B**, **2301C**) and a housing **2320**. A flexible arm **2317** is attached to the energy recovery mechanism **2302** and a feature of the vehicle.

FIG. **23B** illustrates another embodiment of the energy recovery mechanism **2302** engaged with the drive shaft **2314B**. The drive shaft **2314B** includes an outer surface **2321** which contacts the one or more rollers **2301**. The outer surface **2321** includes a material and/or texture with a high coefficient of friction.

FIG. **23C** illustrates another embodiment of the energy recovery mechanism **2302** engaged with a drive shaft **2314C**. The drive shaft **2314C** includes a shoulder **2322** which has a larger diameter than the rest of the shaft. The shoulder **2322** contacts the one or more rollers **2301**. The shoulder **2322** optionally includes an outer surface having a material and/or texture with a high coefficient of friction.

FIG. **23D** illustrates another embodiment of the energy recovery mechanism **2302** engaged with a drive shaft **2314D**. The drive shaft **2314D** includes a groove **2324** which has a smaller diameter than the rest of the shaft. The groove **2324** contacts the one or more rollers **2301**. The groove **2324** optionally may include an outer surface having a material and/or texture with a high coefficient of friction.

FIG. **23E** illustrates another embodiment of the energy recovery mechanism **2302** engaged with a drive shaft **2314E**. The drive shaft **2314E** includes the groove **2324** which has a smaller diameter than the rest of the shaft. The groove **2324** contacts the one or more rollers **2301**. The groove **2324** may include an outer surface having a material and/or texture with a high coefficient of friction. Additionally, portions of the shaft immediately surrounding the groove **2324** may include this material. Here, the one or more rollers **2301** may take a cylindrical form that contacts the outer surface of the drive shaft **2314E** not including the groove **2324**. The one or more rollers **2301** may have a raised portion that extends radially and forms a roller bump. The roller bump will have a diameter that is larger than the roller diameter. The difference between the roller diameter and the diameter of the roller bump shall be substantially equal to the depth of the groove **2324** multiplied by a factor of two. The depth of the groove is substantially equal to the distance between the surface of the drive shaft **2314E** and the surface of the groove **2324**. The width of the roller bump may be less than or in some embodiments substantially equal to, but not in excess of, the groove **2324** width.

28

FIG. **24** illustrates a sideview of an example embodiment of an energy recovery mechanism **2402** and where the energy is recovered without physical contact with a drive shaft **2414**. The energy recovery mechanism **2402** includes a housing **2420** and one or more wires **2401** positioned within. A flexible arm **2417** connects the housing **2420** to a feature of the vehicle.

The drive shaft **2414** is magnetic and includes a magnetic north and magnetic south which creates a magnetic field. In some embodiments, the whole drive shaft **2414** is magnetic. In some embodiments, only a portion of the drive shaft **2414** is magnetic. In some embodiments, one or more magnets are fastened to the drive shaft. In some embodiments, one or more magnets are placed within the drive shaft. The magnets and/or drive shaft include permanent magnets. In some embodiments, the magnets and/or drive shaft include electromagnets.

A flexible arm **2417** is attached to the energy recovery mechanism **2402**. The flexible arm **2417** is configured to move the energy recovery mechanism **2402** to adjust the distance between the energy recovery mechanism **2402** and the drive shaft **2414** based on a signal received from a controller. The flexible arm **2417** may also electrically couple the **2420** to a feature of the vehicle such as an energy storage (capacitor and/or battery), generator, or motor.

The housing **2420** may be rectangular in shape. In some embodiments, the shape of the housing **2420** may be curved toward the drive shaft **2414**. In some embodiments, the housing **2420** may encompass the drive shaft **2414**.

In use, the magnetic drive shaft **2414** rotates in a clockwise or counterclockwise direction. This rotation causes the magnetic field of the shaft to rotate/alternate, thus inducing an electromagnetic force (EMF) in the one or more wires of the energy recovery mechanism **2402** and a current to flow in the wires. According to Faraday's law, the resulting EMF is proportional to the rate of change of the magnetic field and the number of turns in the coil of wire.

The one or more wires **2401** may be oriented to optimize the magnitude of the induced current. This may include one or more hoops, coils etc. The one or more wires may be connected to a resistor to convert the current to voltage. In some embodiments, the resistor may be located within the energy recovery mechanism **2402**. In some embodiments, the current is transferred out of the energy recovery mechanism **2402** before the current is converted to voltage.

A benefit of the energy recovery mechanism **2402** is that it does not require direct contact with the drive shaft **2414**. This increases the longevity of the energy recovery mechanism **2402** and drive shaft **2414** by eliminates friction and the associated wear from mechanical contact.

FIG. **25** illustrates a sideview of an example embodiment of an energy recovery mechanism **2502** positioned adjacent a magnetic drive shaft **2514**. The energy recovery mechanism **2502** includes a housing **2520** and one or more wires **2501** positioned within. A flexible arm **2517** connects the energy recovery mechanism **2502** to a feature of the vehicle. The one or more wires **2501** within the housing **2520** are oriented to form hoops positioned parallel to the drive shaft **2514**. In some embodiments, one or more of the energy recovery mechanisms may be positioned adjacent to the drive shaft **2514**.

ADDITIONAL EMBODIMENTS

As used herein, "system," "instrument," "apparatus," and "device" generally encompass both the hardware (for

example, mechanical and electronic) and, in some implementations, associated software (for example, specialized computer programs for graphics control) components.

It is to be understood that not necessarily all objects or advantages may be achieved in accordance with any particular embodiment described herein. Thus, for example, those skilled in the art will recognize that certain embodiments may be configured to operate in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein.

Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code modules executed by one or more computer systems or computer processors including computer hardware. The code modules may be stored on any type of non-transitory computer-readable medium or computer storage device, such as hard drives, solid state memory, optical disc, and/or the like. The systems and modules may also be transmitted as generated data signals (for example, as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (for example, as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The results of the disclosed processes and process steps may be stored, persistently or otherwise, in any type of non-transitory computer storage such as, for example, volatile or non-volatile storage.

Many other variations than those described herein will be apparent from this disclosure. For example, depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (for example, not all described acts or events are necessary for the practice of the algorithms). Moreover, in certain embodiments, acts or events can be performed concurrently, for example, through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. In addition, different tasks or processes can be performed by different machines and/or computing systems that can function together.

The various illustrative logical blocks, modules, and algorithm elements described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and elements have been described herein generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

The various features and processes described herein may be used independently of one another, or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be

performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

The various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a field programmable gate array ("FPGA") or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can include electrical circuitry configured to process computer-executable instructions. In another embodiment, a processor includes an FPGA or other programmable devices that performs logic operations without processing computer-executable instructions. A processor can also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Although described herein primarily with respect to digital technology, a processor may also include primarily analog components. For example, some, or all, of the signal processing algorithms described herein may be implemented in analog circuitry or mixed analog and digital circuitry. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

The elements of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module stored in one or more memory devices and executed by one or more processors, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of non-transitory computer-readable storage medium, media, or physical computer storage known in the art. An example storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The storage medium can be volatile or nonvolatile. The processor and the storage medium can reside in an ASIC. The ASIC can reside in a user terminal. In the alternative, the processor and the storage medium can reside as discrete components in a user terminal.

Conditional language, such as, among others, "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments

include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, and so forth, may be either X, Y, or Z, or any combination thereof (for example, X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

Any process descriptions, elements, or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of the embodiments described herein in which elements or functions may be deleted, executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those skilled in the art.

Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

All of the methods and processes described herein may be embodied in, and partially or fully automated via, software code modules executed by one or more general purpose computers. For example, the methods described herein may be performed by the computing system and/or any other suitable computing device. The methods may be executed on the computing devices in response to execution of software instructions or other executable code read from a tangible computer readable medium. A tangible computer readable medium is a data storage device that can store data that is readable by a computer system. Examples of computer readable mediums include read-only memory, random-access memory, other volatile or non-volatile memory devices, CD-ROMs, magnetic tape, flash drives, and optical data storage devices.

It should be emphasized that many variations and modifications may be made to the herein-described embodiments, the elements of which are to be understood as being among other acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. The section headings used herein are merely provided to enhance readability and are not intended to limit the scope of the embodiments disclosed in a particular section to the features or elements disclosed in that section. The foregoing description details certain embodiments. It will be appreciated, however, that no matter how detailed the foregoing appears in text, the systems and methods can be

practiced in many ways. As is also stated herein, it should be noted that the use of particular terminology when describing certain features or aspects of the systems and methods should not be taken to imply that the terminology is being re-defined herein to be restricted to including any specific characteristics of the features or aspects of the systems and methods with which that terminology is associated.

Those of skill in the art would understand that information, messages, and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

What is claimed is:

1. A system for generating energy in response to a movement of a vehicle, the system comprising:

an energy recovery mechanism configured to receive energy from the rotation of a component of the vehicle; an energy storage electrically coupled to the energy recovery mechanism and configured to store the energy as electrical energy;

a motor electrically coupled to the energy storage and configured to receive a portion of the electrical energy; a generator electrically coupled with the energy recovery mechanism and the energy storage; and

a controller in communication with the energy recovery mechanism, the energy storage, and the motor, wherein the controller further communicates with a driver of the vehicle, wherein the controller receives data from one or more of the energy recovery mechanism, energy storage, generator, and motor, wherein the controller transmits instructions to the energy recovery mechanism to change a total amount of energy received from the rotation of the vehicle component in response to the received data or an instruction from the driver, and wherein the controller transmits instructions to the energy recovery mechanism to control a ratio of angular velocity between the energy recovery mechanism and a rotational member of the generator in response to the received data or an instruction from the driver.

2. The system of claim 1, wherein the energy storage comprises one or more of a battery, a capacitor, or a combination thereof.

3. The system of claim 2, further comprising a generator electrically coupled with the energy recovery mechanism and the energy storage, wherein the controller further receives data from the generator.

4. The system of claim 1, wherein the data comprises a rotational velocity or acceleration of a roller of the energy recovery mechanism.

5. The system of claim 1, wherein the data comprises a magnitude of a voltage or current induced in the energy recovery mechanism.

6. The system of claim 3, wherein the data comprises an amount of energy or a rate of energy generated by the generator.

7. The system of claim 2, wherein the data comprises an amount of energy stored in the capacitor, a voltage of the capacitor, or a rate of charge or discharge of the capacitor.

8. The system of claim 2, wherein the data comprises an amount of energy stored in the battery, a voltage of the battery, or a rate of charge or discharge of the battery.

9. The system of claim 1, wherein the data comprises operational data of the motor.

33

10. The system of claim 1, wherein the data comprises the speed or acceleration of the vehicle.

11. The system of claim 3, wherein the controller transmits instructions to change a rate or an amount of energy generated by the generator in response to the received data or an instruction from the driver.

12. The system of claim 3, wherein the controller transmits instructions to control a flow of energy between two or more of the energy recovery mechanism, generator, capacitor, battery, and motor in response to the received data or an instruction from the driver.

13. The system of claim 1, wherein the controller transmits instructions to accelerate the vehicle based on the received data or an instruction from the driver.

14. The system of claim 1, wherein the controller transmits instructions to decelerate the vehicle based on received data or an instruction from the driver.

15. A method of generating energy in response to a movement of a vehicle, the method comprising:

rotating a component of a vehicle to generate energy;
receiving the energy in an energy recovery mechanism positioned adjacent the rotating component of the vehicle;

storing the energy in an energy storage;

alerting the driver when a portion of the energy storage is charged;

receiving, via the controller, data from one or more of the energy recovery mechanism, energy storage, and motor;

changing, via the controller, a total amount of energy received by the energy recovery mechanism in response to the received data or an instruction from the driver; and

changing, via the controller, a rate at which a generator rotates to generate energy.

16. The method of claim 15, wherein the energy storage comprises one or more of a battery, a capacitor, or a combination thereof.

17. The method of claim 15, further comprising receiving, via a controller, a signal to convey an electrical output of the energy from the energy storage to the motor;
conveying the electrical output to the motor; and
stopping the conveyance of the electrical output from the energy storage to the motor upon receipt of a second signal.

18. The method of claim 17, wherein the signal is manually sent by a driver of the vehicle.

19. The method of claim 17, where in the signal is sent automatically upon determining an acceleration of the vehicle or an energy demand of the motor.

20. The method of claim 15, wherein the data comprises a rotational velocity or acceleration of a roller of the energy recovery mechanism.

34

21. The method of claim 15, wherein the data comprises a magnitude of a voltage or current induced in the energy recovery mechanism.

22. A system for generating energy in response to a movement of a vehicle, the system comprising:

an energy recovery mechanism configured to receive energy from the rotation of a component of the vehicle, the energy recovery mechanism comprising a roller;

an energy storage electrically coupled to the energy recovery mechanism and configured to store the energy as electrical energy, wherein the energy storage comprises one or more of a battery, capacitor, or a combination thereof;

a motor electrically coupled to the energy storage and configured to receive a portion of the electrical energy;

a generator electrically coupled with the energy recovery mechanism and the energy storage; and

a controller in communication with the energy recovery mechanism, the energy storage, and the motor, wherein the controller receives data from one or more of the energy recovery mechanism, energy storage, and motor, the data comprising a rotational velocity or acceleration of the roller of the energy recovery mechanism; and wherein the controller transmits instructions to change a rate or an amount of energy generated by the generator in response to the received data.

23. The system of claim 22, wherein the controller transmits instructions to control a flow of energy between two or more of the energy recovery mechanism, generator, capacitor, battery, and motor in response to the received data or an instruction from the driver.

24. The system of claim 22, wherein the data further comprises an amount of energy or a rate of energy generated by the generator.

25. The system of claim 22, wherein the data comprises an amount of energy stored in the capacitor, a voltage of the capacitor, or a rate of charge or discharge of the capacitor.

26. The system of claim 22, wherein the data comprises an amount of energy stored in the battery, a voltage of the battery, or a rate of charge or discharge of the battery.

27. The system of claim 22, wherein the data comprises operational data of the motor.

28. The system of claim 22, wherein the data comprises the speed or acceleration of the vehicle.

29. The system of claim 22, wherein the controller transmits instructions to accelerate the vehicle based on the received data or an instruction from the driver.

30. The system of claim 22, wherein the controller transmits instructions to decelerate the vehicle based on received data or an instruction from the driver.

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